



RVS AGRICULTURAL COLLEGE
(Affli. Tamil Nadu Agricultural University)
THANJAVUR – 613 402



SAC - 101 FUNDAMENTALS OF SOIL SCIENCE (2+1)

Theory Notes

COURSE TEACHERS

Ms. C. RAMYA

Dr. L. VENKATAKRISHNAN

DEPARTMENT OF SOILSCIENCE AND AGRICULTURAL CHEMISTRY
RVS AGRICULTURAL COLLEGE
THANAJAVUR – 613 402
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1. SOIL – INTRODUCTION

Soil Science deals with soil as a natural resource on the surface of the earth. It includes soil formation, classification and mapping; physical, chemical, biological, and fertility properties of soils; and deals with these properties in relation to the use and management of soils.

The term SOIL was derived from the Latin Word “*SOLUM*” means FLOOR

- ❖ For a Layman soil is dirt or debris
- ❖ For an Agriculturist soil is a medium for plant growth
- ❖ For a Mining Engineer soil is a debris covering the rocks
- ❖ For a Civil Engineer soil is a material on which road bed or house bed is formed
- ❖ For a Home Owner soil is a mellow or loamy or hard material

Definitions

Whitney (1892) : Soil is a nutrient bin which supplies all the nutrients required for plant growth.

Hilgard (1892) : Soil is more or less a loose and friable material in which plants, by means of their roots, find a foothold for nourishment as well as for other conditions of growth”

Dokuchaiev (1900) : Russian scientist – regarded as **Father of Soil Science** - Soil is a natural body composed of mineral and organic constituents, having a definite genesis and a distinct nature of its own.

Joffee (1936) : “Soil is a natural body of mineral and organic constituents differentiated into horizons - usually unconsolidated - of variable depth which differs among themselves as well as from the underlying parent material in morphology, physical makeup, chemical properties and composition and biological characteristics”.

Jenny (1941) : Soil is a naturally occurring body that has been formed due to combined influence of climate and living organisms acting on parent material as conditioned by relief over a period of time

Soil Science Society of America (1970) :

- I. Soil is the unconsolidated mineral matter on the surface of the earth that has been subjected to and influenced by genetic and environmental factors of parent material, climate, macro and microorganisms and topography, all affecting over a period of time and producing a product, that is “SOIL” that differs from the material from which it is derived in physical, chemical, biological and morphological properties and characteristics.
- II. The unconsolidated mineral material on the immediate surface of the earth that serves as a natural medium for the growth of land plants.

As soil provides nutrients, water, air and anchorage and supports life on Earth,
it can be called as Soul of Infinite Life (SOIL) - J.S. Kanwar

2. COMPONENTS OF SOIL

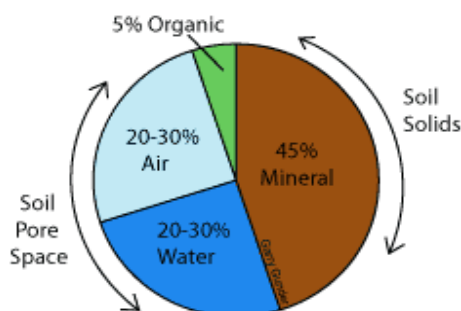
Soil as a three dimensional body

Soil is a three dimensional body having length, breadth and depth. They form a continuation over the land surface and differ in properties from place to place. Its upper boundary is air or water and lower boundary is the rock lithosphere.

Composition of soil on volume basis

Mineral matter	: 45%
Organic matter	: 5%
Soil water	: 25%
Soil air	: 25%

Soil Composition by Volume



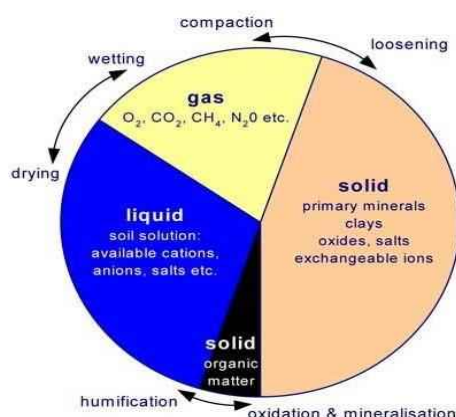
Phases of Soil

♣ Soil as a three phase system

♣ Solid, Liquid and gaseous phases

The diagram below shows the relative components of the three phases of soil: solids, liquids, and gasses. The solids may be mineral or organic. The pores are filled with either air or water.

Soil is a three-phase, porous media, composed of solids, liquids, and voids (empty spaces between the solids). The typical amount of total porosity (ratio of void volume to total volume) in a mineral soil ranges from about 40% to 60%. This means approximately 40 to 60% of the volume of a mineral soil is actually empty space between the solid particles (voids)



Mineral matter in soils

The size and composition of mineral matter in soils are variable due to nature of parent rock from which it has been derived. In general, the primary viz., Quartz, biotite, muscovite etc., dominate the coarser fractions of soil, on the other hand, the secondary minerals viz., silicate

clays and hydrous oxide clays of iron and aluminium etc., are present as very finer fraction, clay in the soils.

Organic matter in soils

Soil organic matter exists as partly decayed and partially synthesized plant and animal residues. Such organic residues are continually being broken down as a result of microbial activity in soil. Organic matter is a store house of nutrients soil. Besides these, organic matter is responsible for most desirable soil physical properties.

Soil water:

Soil water plays a very significant role in soil plant growth relationships. Water is held within the soil pores with varying degrees of forces depending on the amount of water present.

Soil water is also present along with dissolved salts and makes up the soil solution. This soil solution (water plus dissolved salts) acts as an important medium for supplying different essential nutrients for growing plants.

Soil Air

Air spaces or pore spaces (voids) in a soil consists of that portion of the soil volume not occupied by soil solids, either mineral or organic.

During rainy season, water replaces air from the soil pore spaces, but as soon as water disappeared by downward movement, surface evaporation and transpiration etc., air gradually replaces the water as it is lost from the pore spaces. Soil air influence the growth of the plants as well as the activity of different beneficial micro organisms present in soil.

Approaches of Soil Study

Two Concepts:

One treats soil as a natural body, weathered and synthesized product in nature (Pedology) while other treats soil as a medium for plant growth (Edaphology).

Pedological Approach:

- ❖ Study of the origin of the soil, its classification and its description is known as Pedology. (From Greek word *pedon*, means soil or earth).

- ❖ Pedology is the study of soil as a natural body and does not focus on the soil's immediate practical use. A pedologist studies, examines and classifies soil as they occur in their natural environment.

Edaphological Approach:

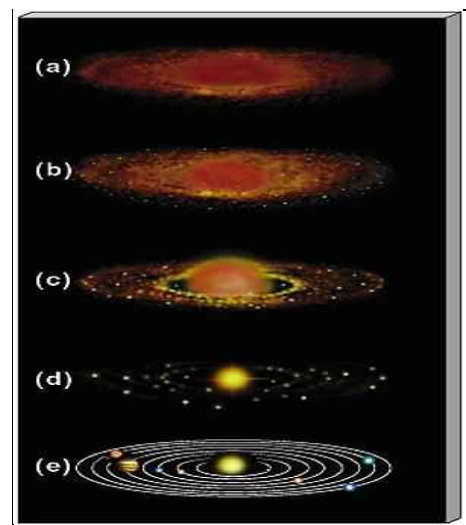
- ❖ Edaphology (from Greek word *edaphos*, means soil or ground) is the study of soil from the stand point of higher plants.
- ❖ Edaphologists consider the various properties of soil in relation to plant production. They are practical and have the production of food and fibre as their ultimate goal.
- ❖ They must determine the reasons for variation in the productivity of soils and find means for improvement.

Origin and Composition of Earth Crust

Theories on origin of earth : There are two important theories on the origin of earth

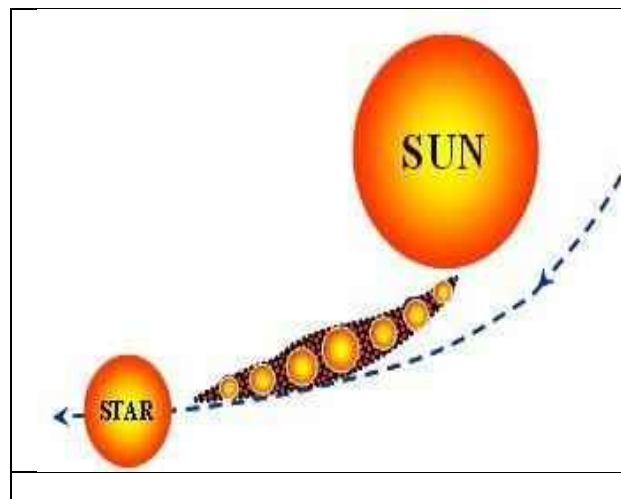
(i) Nebular hypothesis

- The exact knowledge of the way in which our earth was formed is based on an old theory known as the nebular hypothesis.
- According to this hypothesis, the whole solar system is supposed to have existed at one time as a nebula or a cloudy mass of matter in the form of vapour or gases.
- This cloudy mass was very hot and perpetually revolving. With the passage of time, the nebula began to cool.
- On cooling the outer layers of the vapours condensed and were thrown out in the form of rings.
- In this way several rings were thrown out at different times.
- These rings, on further condensation and disruption became what we know as the planets.
- The eight planets, in the order of distance from the sun are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.
- The asteroids, often called minor planets, number in thousands and lie between Mars and Jupiter.

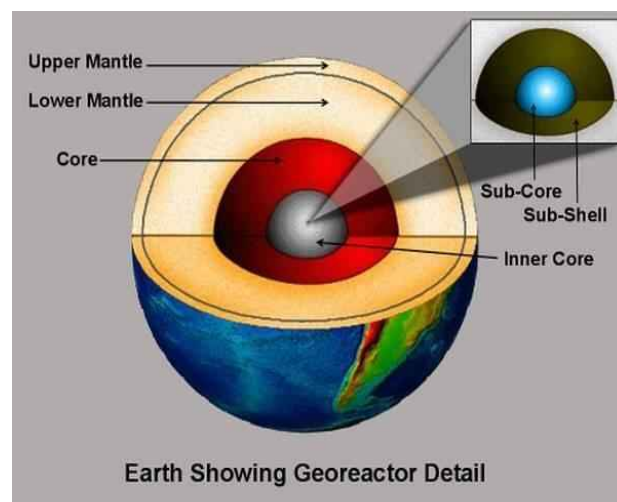


(ii) Planetesimal hypothesis

- This theory was postulated by Chamberlain and Moulton.
- The planets of the Solar System are seen to arise from an encounter between the Sun and another star. In this scenario, the gravity of the passing star tears a succession of bolts from the solar surface.
- Bolts coming from the side nearer the star are thrown out to distances comparable with those of the giant planets, while those from the far side of the Sun are ejected less violently to the distances of the terrestrial planets.
- A lot of heat was generated in this process. The gas particles attracted from the floating materials and from the volcanic eruptions resulted in the formation of atmosphere around the earth.
- This theory explains the fact that the total mass of the planet is $\frac{1}{700}$ times of the solar system.



Interior of Earth



The structure of the earth's interior is three layered and are named as crust, mantle and the core.

Crust

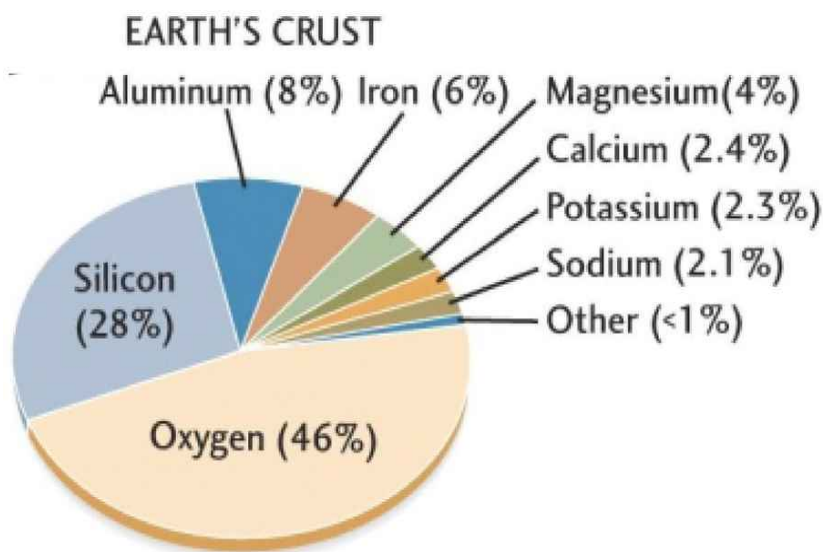
Crust is the outer thin layer with a total thickness of around 100 km. It forms 0.5% of the earth's volume. The outer covering of the crust is of sedimentary material and below that lies crystalline, igneous and metamorphic rocks.

Composition of earth crust

The earth's crust is principally composed of mineral matter. The mineral matter is made up of various elements combined together to form compounds. Some elements exist as such without forming compounds. Out of 106 elements known, 8 are sufficiently abundant and constitute about 99 percent by weight of the earth's crust.

Elements present in the Earth's crust (Composition)

	Element	Amount	
Non-metallic	Oxygen	46.60	75 % of earth crust
	Silicon	27.72	
Metals	Aluminium	8.16	25% of earth's crust
	Iron	5.00	
	Calcium	3.50	
	Sodium	2.83	
	Magnesium	2.09	
	Potassium	2.59	



3. SOIL GENESIS - SOIL FORMING ROCKS

Igneous, Sedimentary and Metamorphic Rocks

Rocks are the materials that form the essential part of the Earth's solid crust. A rock may be defined as a hard mass of mineral matter comprising two or more rock forming minerals. **Petrology** (in Greek, *petra* means rock, *logos* means science) deals with science of rocks. It consists of

- a. **Petrography** which deals with the description of rocks.
- b. **Petrogenesis** which is the study of the origin of rocks.

Formation of rocks:

- Cooling and consolidation of molten magma within or on the surface of earth results in the formation of **Igneous** or **Primary rocks**
- Disintegration and decomposition lead to the breaking down of pre-existing rocks. Transportation and cementation of primary rocks results in the formation of **Sedimentary** or **Secondary rocks**
- The primary and the secondary rocks when subjected to earth's movement and to high temperature and pressure are altered to new rocks called **Metamorphic rocks**

Classification of rocks

According to the **mode of formation** the rocks are divided into the following three main classes.

1. Igneous or Primary rocks
2. Sedimentary or Secondary rocks
3. Metamorphic rocks

1. Igneous rocks (primary or massive rocks)

These are first formed in the earth crust due to the solidification of molten magma. They are the **source of parent material** for other rocks and ultimately for soils.

Based on the mode of origin Igneous rocks are classified as

- Extrusive rocks (or volcanic rocks)

These rocks are formed due to the consolidation of magma on the **surface of the earth**. The magma, when flows on the Earth surface are called LAVA. eg. Basalt

➤ Intrusive rocks (or plutonic rocks)

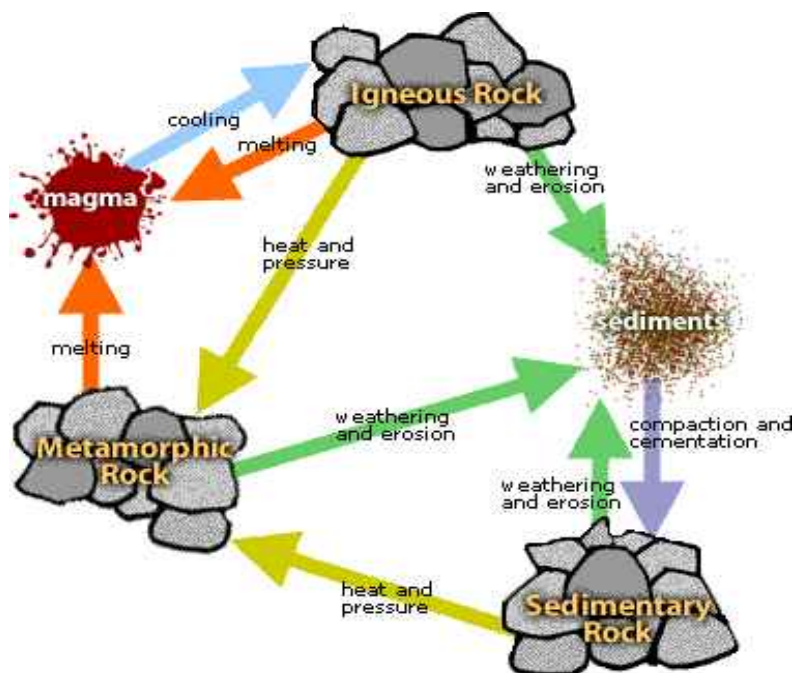
These rocks are produced due to solidification of magma **below the surface of the earth**. These intrusive rocks solidifies at greater depths. eg. Granite.

Based on chemical composition Igneous rocks may be divided into

- ❖ Acid rocks : > 65% silica (Granite, Rhyolite)
- ❖ Sub acid rocks : 60-65% silica (Syenite and Trachyte)
- ❖ Sub basic rocks : 55-60% silica (Diorite and Andesite)
- ❖ Basic rocks : 45-55% silica (Gabbro, Basalt)

2. Sedimentary rocks (Clastic or stratified rocks)

The sedimentary rocks are formed from sediments, derived from the breaking down of pre-existing rocks. The sediments are transported to new places and deposited in new arrangements and cemented to form secondary rocks. Sediments may contain various size particles cemented together by substances like SiO_2 , Fe_2O_3 or lime. These rocks are called as clastic rocks. **Stratification** is the most common feature of these rocks and are also termed as **stratified rocks**.



The sedimentary rocks are classified a) based on origin and b) based on grain size

a) Based on the origin the sedimentary rocks are classified as

1. Residual : Laterite



Laterite is well known in Asian countries as a building material for more than 1000 years. It was excavated from the soil and cut in form of large blocks; temples at Angkor are famous examples for this early use. At beginning of the 19.century it obtained scientific interest when the English surgeon Francis Buchanan travelled along the western coast of southern India and published his manifold observations and results (1807). He coined the term laterite when he wrote: “What I have called indurated clay ...is one of the most valuable materials for building. It is diffused in immense masses, without any appearance of stratification and is placed over the granite that forms the basis of Malayala. It is full of cavities and pores, and contains a very large quantity of iron in the form of yellow and red ochers. In the mass, while excluded from the air, it is so soft, that any iron instrument readily cuts it, and is dug up in square masses with a pick-axe, and immediately cut into the desired shape with a trowel, or large knife. It very so after becomes as hard as brick, and resists the air and water much better than any brick that I have seen in India. The most proper English name would be laterite, from laterites, the appellation that may be given to it in science”. (The Latin word `later` means brick)

2. Transported :

- a. Deposited as solids in suspension : Sandstone, shale
- b. Deposited by chemical precipitation : Limestone, ironstone
- c. Deposited through agency of organic matter : Peat, Phosphatic deposits

b) Based on the grain size sedimentary rocks are classified as

- 1. Rocks with boulder pebbles sized minerals (Rudaceous) : Conglomerate
- 2. Rocks with sand size particles (Arenaceous) : Sandstone

- | | |
|--|-------------|
| 3. Rocks with silt size particles (silt rocks) | : Siltstone |
| 4. Rocks with clay size particles (Argillaceous) | : Shale |

3. Metamorphic rocks

These are formed from igneous and sedimentary rocks under the influence of heat, pressure, chemically active liquids and gases. Changes may occur in mineral composition of rocks or texture or both. The change due to water is called hydrometamorphism, due to heat is thermometamorphism and due to pressure is called dynamometamorphism.

Sand stone	→	: Quartzite
Shale	→	: Slate/mica, schist
Lime stone	→	: Marble
Granite	→	: granite gneiss
Dolerite	→	: Hornblende gneiss

Relative abundance of rocks in earth's crust

(i) Composition of Earth's crust as a whole :

♦ Igneous Rocks	-	95%
♦ Sedimentary Rocks	-	5%
- Shales	-	(4.0%)
- Sandstones	-	(0.75%)
- Limestones	-	(0.025%)

(ii) Composition of the upper 5 km of the Earth's crust :

♦ Sedimentary Rocks

Shales	-	52%	} (74%)
Sandstones	-	15%	
Limestones and dolomite	-	7%	

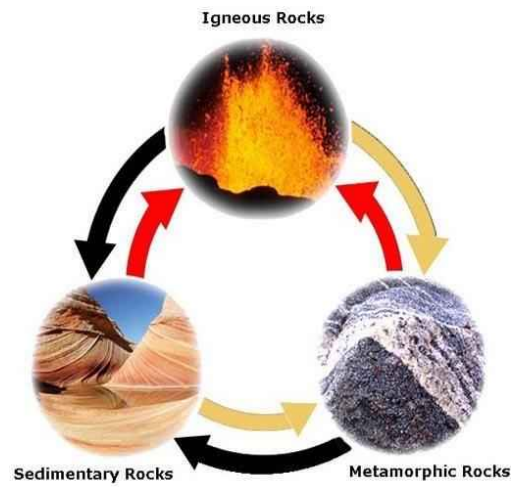
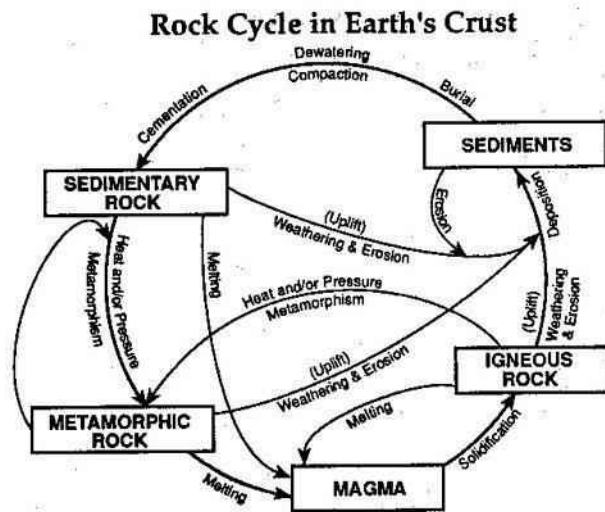
♦ Igneous Rocks

Granite	-	15%	} (18%)
Basalt	-	3%	

♦ Others	-	8%	(8%)
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♦ Total		100	100
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Rock Cycle



4. BRIEF DESCRIPTION OF IMPORTANT ROCKS

♣ Mineralogical Composition

Igneous Rocks

S. No	Rocks	Origin	Essential minerals	Common minerals	Average specific gravity	Remarks
i.	Granite	Plutonic holocrystalline	Quartz (20 to 30%)	Hornblende, magnetite, mica	2.64	Light coloured white or reddish
ii.	Syenite	Plutonic Holocrystalline	Quartz, orthoclase	Hornblende, magnetite, biotite	2.80	Light coloured white or reddish
iii.	Diorite	Plutonic Holocrystalline	Quartz	Hornblende, magnetite, biotite	2.85	Darker
iv	Gabbro	Plutonic Holocrystalline	Labradorite, augite, olivine	Hornblende, ilmenite	3.0	Blackish
v.	Dolerite	Hypabasal	Labradorite, augite, olivine	Hornblende, ilmenite	3.0	Blackish
vi.	Basalt	Volcanic crystalline with glassy mass	Labradorite, augite, olivine	Hornblende, ilmenite	3.0	Blackish

Sedimentary rocks

S. No	Rock	Mineral composition	Colour and structure
1.	Sandstone	Mainly quartz with some CaCO ₃ , iron oxides and clay	Light to red, granular
2.	Shale	Clay minerals, quartz and some organic matter	Light to dark thinly laminated
3.	Limestone	Mainly calcite with some dolomite, iron oxides, clay, phosphate and organic matter	Light grey to yellow, fine grained and compact

Metamorphic rocks

S. No.	Rock	Mineral composition	Colour and structure
1.	Gneiss	Formed from granite	Alternating light and dark colours, banded and foliated
2.	Schist	Formed from basalt or shale	As original rock, foliated
3.	Quartzite	Formed from sandstone	Light or brown, compact and uniform texture, foliated structure
4.	Slate	Formed from shale	Grey to black, compact and uniform texture, foliated structure
5.	Marble	Formed from lime stone	Light red, green, black, compact fine to coarse texture, foliated structure

Igneous Rock



5. MINERALS

♣ Definition, occurrence

♣ Classification of primary minerals

Mineral:

A mineral is a naturally occurring inorganic solid with a definite chemical composition and a crystalline structure. More than 90% of all of the minerals in the Earth's Crust are made up of compounds containing Silicon and Oxygen, the two most abundant elements on Earth. There are over 2000 minerals on Earth, but only 100 are commonly found. 30 minerals make up the majority of the rocks on Earth.

A **mineral**, by definition, must satisfy five conditions:

- It must be naturally occurring.
- It must be inorganic.
- It must be a solid element or compound.
- It must have a definite composition.
- It must have a regular internal crystal structure

Occurrence:

Of the 2000 known minerals, only few occur in abundance in the Earth crust.

The relative abundance of important rock forming minerals is given below:

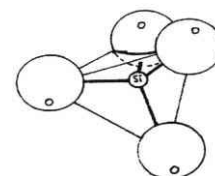
Minerals (arranged in the order of their crystallization)	Important constituents	Percent distribution
Primary minerals		
Ferro Magnesium minerals		
♦ Ortho-ino silicates		16.8
♦ Olivine	Fe, Mg	
♦ Pyroxenes	Ca, Na, Fe, Mg	
♦ Amphiboles	Ca, Na, Fe, Mg, Al, OH	
♦ Phyllo Silicates		3.6
♦ Biotite	K, Fe, Mg, Al, OH	
♦ Muscovite	K, Al, OH	

Non-Ferro Magnesium minerals		
Tecto Silicates		
♦ Feldspars		61.0
Anorthite	Ca, Al	
Albite	Na, Al	
Orthoclase	K, Al	
♦ Quartz		
Secondary minerals		
♦ Clay minerals	Na, K, Ca	11.6
♦ Others	Mg, Fe, Al, OH	6.0

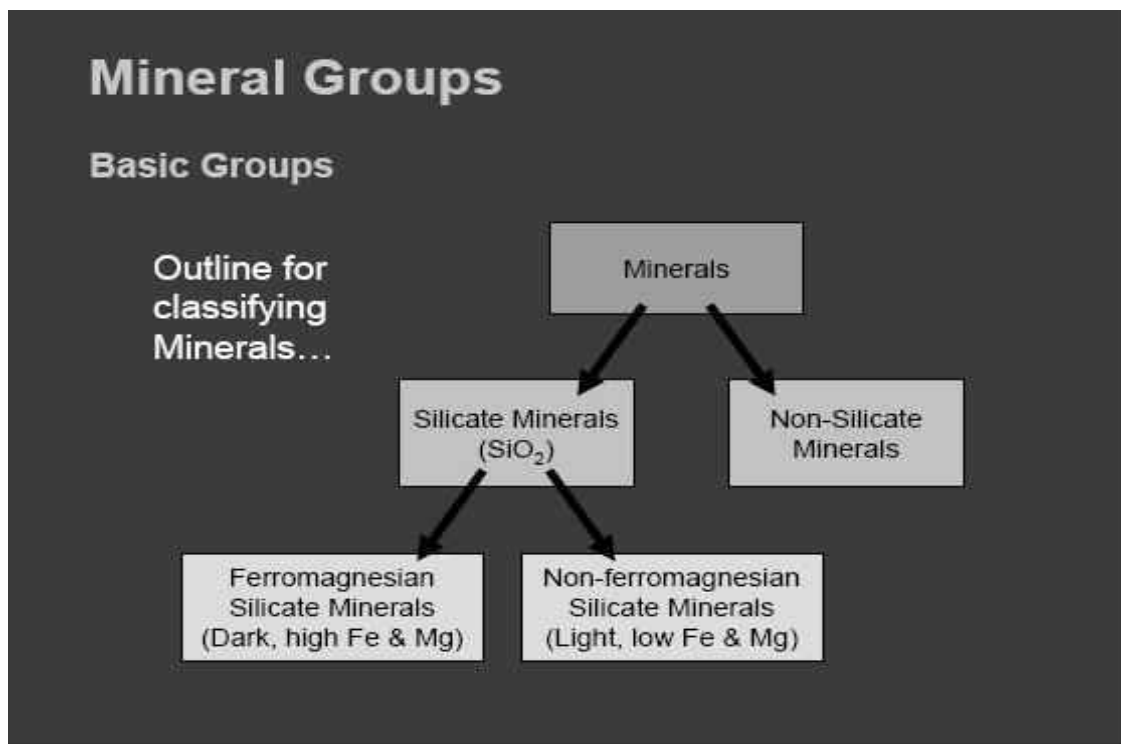
Formation of minerals

- ♦ When the molten magma solidifies, the different elements present in it freely arrange themselves in accordance with attractive forces and geometric form.
- ♦ The earth's crust contains dominant amount of oxygen (46.60%) followed by silicon (27.72%).
- ♦ In order to achieve neutrality between the negatively – charged oxygen and the positively – charged silicon, there would be a greater tendency for silicon and oxygen to combine to form the basic compound, called the silicon – oxygen tetrahedron (SiO_4^{4-}).
- ♦ This explains the dominance (>90%) of silicate minerals in the earth's crust.
- ♦ Geometrically, it is possible to arrange only 4 oxygen anions around a central silicon cation so that all are touching each other. This is the arrangement of a tetrahedron.
- ♦ The amount of charge carried by silicon ion is 4^+ and by oxygen is 2^- . In order to attain neutrality, one silicon (4^+) ion would combine with two oxygen ion ($2 \times 2^-$) to form SiO_2 but geometrically stable structure is formed when 1 silicon combines with 4 oxygen ions to form tetrahedron (SiO_4^{4-}) which carries a net negative charge of 4^- .

The silicate tetrahedron (SiO_4^{4-}) is the fundamental building block of all the silicate minerals



The Si-O bonds are very strong with mixed covalent and ionic character.



Classification of minerals

i. Based on the mode of origin

Primary mineral: A mineral that forms the original component of the rock is known as the primary mineral eg. Feldspar, Hornblende, Mica, Quartz etc;

Secondary mineral: A mineral that has been formed, deposited or introduced as a result of subsequent changes in the rock is known as secondary mineral. Eg: Limonite, Gibbsite etc., and clay minerals like Kaolinite, Montmorillonite etc.,

ii. Based on its importance or amount

Essential mineral: Those minerals which form the chief constituents of a rock, and are known as essential minerals. They are present in large quantities varying from 95-98% eg. Calcite and Silicate minerals.

Accessory minerals: Those minerals which occur only in small quantities and whose presence / absence is of no consequence as far as the character of the rock is concerned is called accessory minerals. Eg. Tourmaline, magnetite, pyrites etc

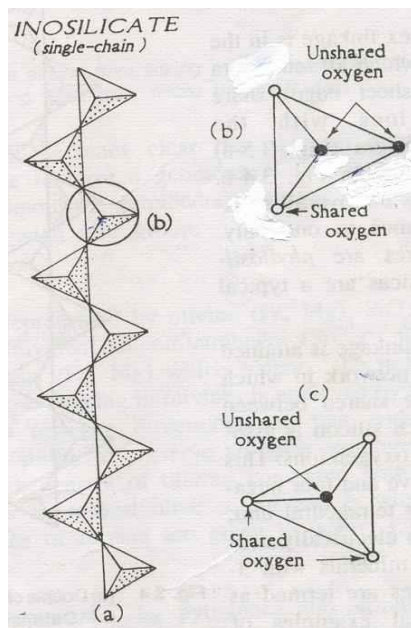
♦ Silicate minerals

(A) Ferro magnesian silicate minerals

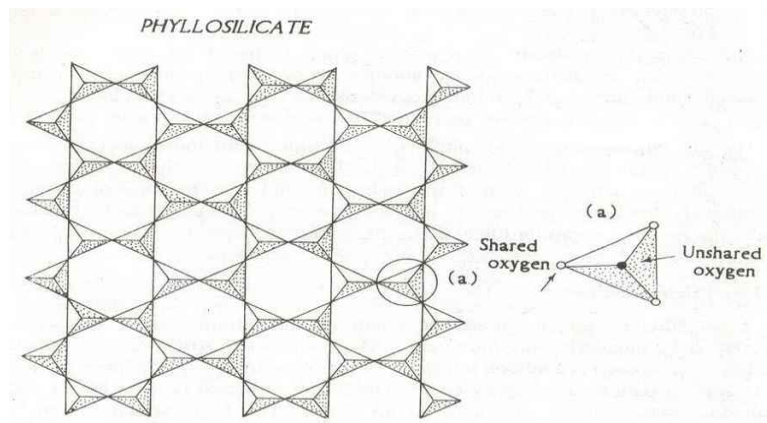
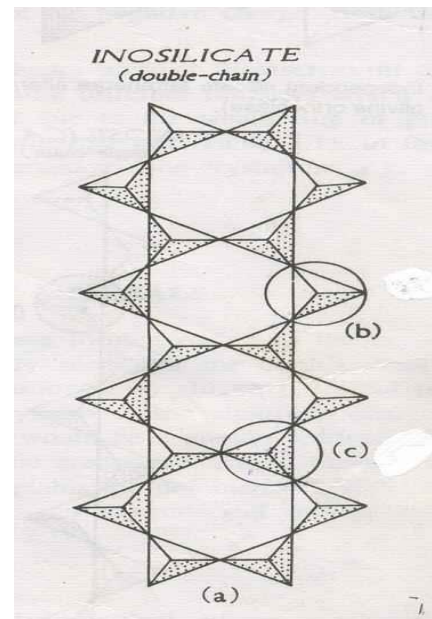
(i) Inosilicates (Pyroxenes and amphiboles)

The pyroxenes and amphiboles are two groups of ferromagnesian minerals. The structure consists of long chains of linked silica tetrahedra. The pyroxenes consist of a single chain (2 oxygen shared in each tetrahedron) whereas amphiboles consist of double chains (alternately 2 and 3 oxygen atoms share the successive tetrahedra). These chain silicates are sometimes referred to **inosilicates**.

Pyroxene



Amphibole

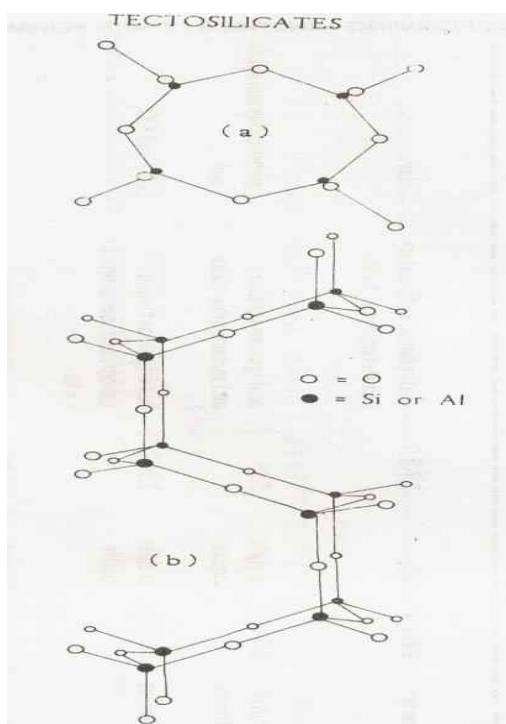


(ii) Phyllosilicates

The phyllosilicates are an important group of soil – forming minerals and are represented by micas (biotite, muscovite). They have sheet structure of tetrahedra where each silicon ion shares three oxygen ions with adjacent silicon ion to form a honey – comb like pattern. The fourth unshared oxygen ion of each tetrahedron stands above the plane of all others. The basic structural unit of phyllosilicates is formed by the condensation of two sheets of silicon-tetrahedra with one sheet of aluminum or magnesium octahedron.

(B) Non Ferro magnesian minerals

(i) **Tectosilicates:** The most common minerals of this group are feldspars and quartz.



Feldspars:

- Feldspars are aluminosilicates of K, Na and Ca. The feldspar structure consists of tetrahedral which are attracted by sharing each oxygen atom between neighbouring tetrahedra.
- The tetrahedra contain mainly Si ions with sufficient Al substitution. It belongs to the group of minerals that are light in weight.
- There are two groups of feldspars: (i) Potassium feldspars (KAlSi_3O_8) include orthoclase and microcline. Orthoclase and microcline are more common in the plutonic and metamorphic rocks.

- The potassium feldspars occur commonly in the silts and sands of soils and also abundant in clay-size fractions of soil (ii) plagioclase feldspars- a series consisting of solid solution of albite ($\text{NaAlSi}_3\text{O}_8$) high in sodium and anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$) high in calcium.
- Plagioclase weathers more rapidly than orthoclase.

Quartz

It is very densely packed and occurs in a high degree of purity. It is strongly resistant to weathering as the structure is densely packed, and free from any substitution. **It is the most abundant mineral next to feldspars.**

Non-silicate minerals

Oxides: Haematite (Fe_2O_3)
 Limonite ($\text{Fe}_2\text{O}_3, 3\text{H}_2\text{O}$)
 Goethite ($\text{FeO}(\text{OH}) \cdot \text{H}_2\text{O}$)
 Gibbsite ($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$)

The red, yellow or brown colours in soils are due to the presence of goethite and haematite, which occur as coatings on the surface of soil particles.

Carbonates : Calcite (CaCO_3)
 Dolomite (CaMgCO_3)

Sulphates : Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)

Phosphates : Apatite (Rock phosphate - $\text{Ca}_3(\text{PO}_4)_2$ - primary source of phosphorus)

6. SECONDARY MINERALS

♣ Clay minerals

♣ Amorphous minerals

Formation of secondary mineral

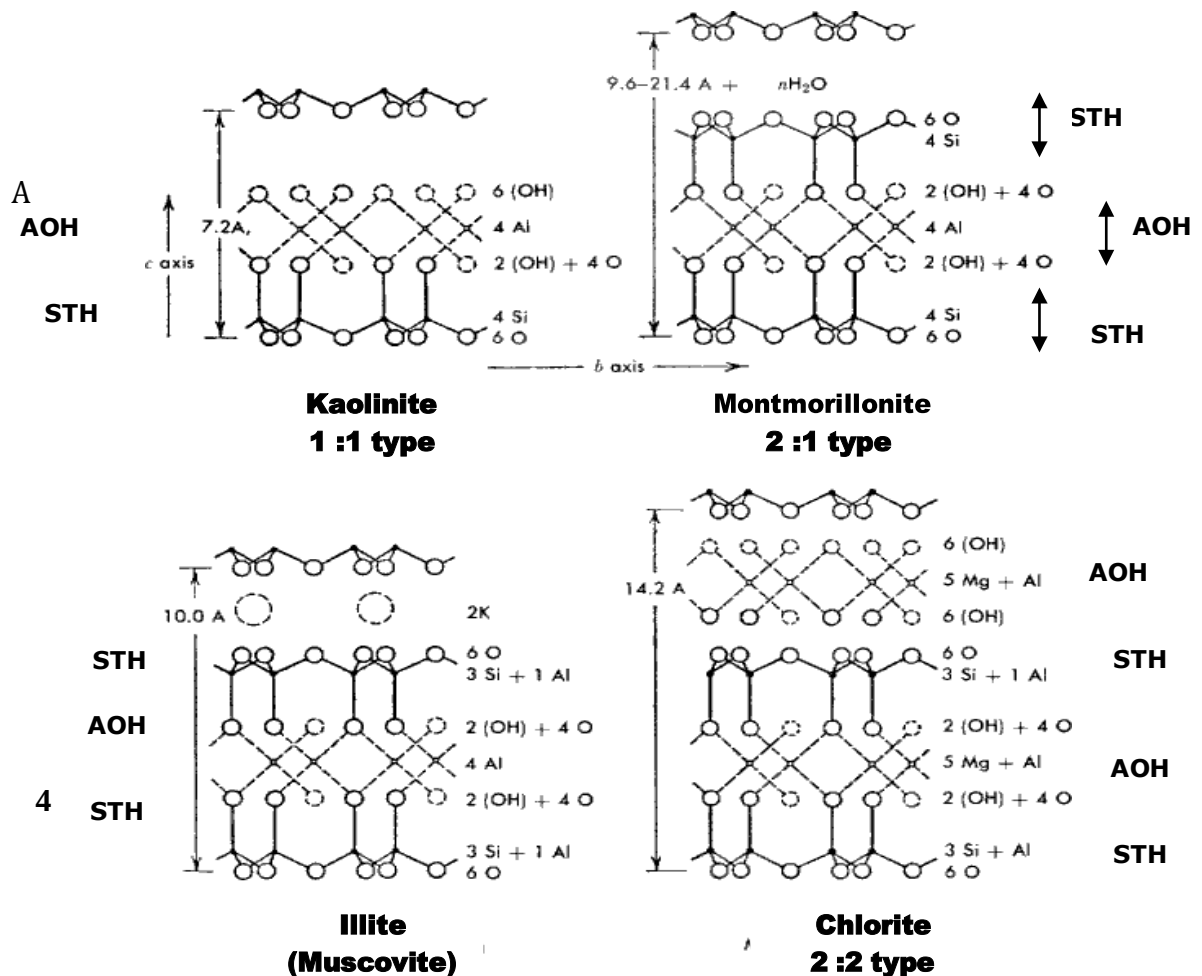
The secondary minerals are formed at the earth's surface by weathering of the pre-existing primary minerals under variable conditions of temperature and pressure. These clay minerals are of size <0.002 mm and **are considered to be the most reactive part of soil**. Important soil properties like nutrient and water holding capacity are controlled by clay minerals. These minerals are layered silicates consisting of silica tetrahedron and aluminium octahedron. (Kaolinite)

- 1) 1:1 clay mineral (1 silicon tetrahedron + 1 aluminium octahedron) : (Kaolinite)
- 2) 2: 1 non-expanding clay mineral :
 - i. Black mica (Biotite)
 - ii. White mica (Muscovite)
 - iii. Weathered mica (illite)
- 3) 2: 1 expanding clay mineral :
 - i. partially expanding (Vermiculite)
 - ii. Fully expanding (Montmorillonite)
- 4) 2: 2 clay mineral :
 - i. Chlorite

The common minerals (in descending order) as observed in the dominant soil groups of India are

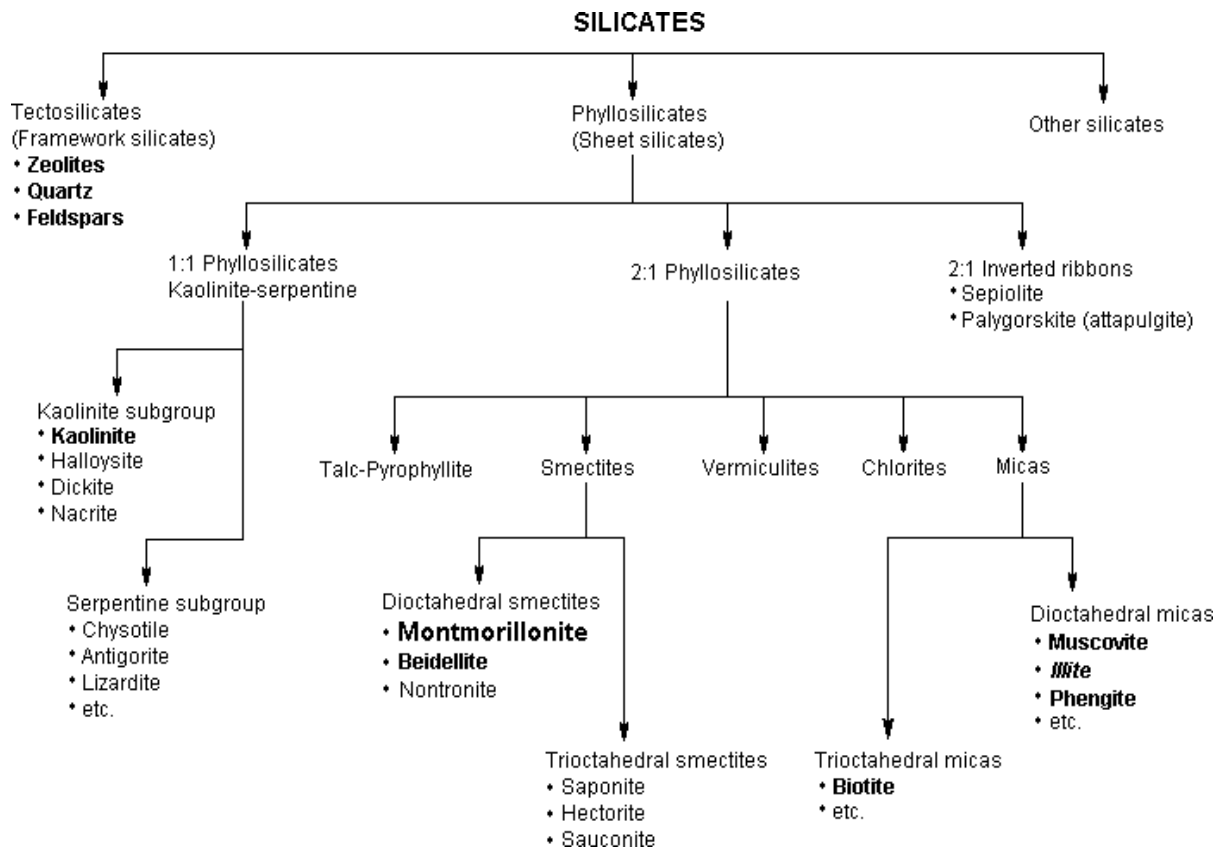
1.	Alluvium derived soils	:	Illite, chlorite (intergrades) Montmorillonite (Flood plains and basins), Kaolinite.
2.	Forest soils	:	Illite, chlorite (Vermiculite), Montmorillonite, Kaolinite
3.	Black (cotton) soils	:	Smectite, Illite, Chlorite / Chloritized Intergrades
4.	Desert soils	:	Illite, Attapulgitite, Vermiculite
5.	Red and lateritic soils	:	Kaolinite (gibbsite), Chlorite, Illite

Structure of Clay Minerals



Amorphous minerals

- A **mineral** is a solid material, made of one substance that occurs naturally on Earth. Most of the common minerals are made of crystals.
- A Crystal is a solid formed by a repeating, three-dimensional pattern of atoms, ions, or molecules and having fixed distances between the different parts.
- Minerals that do not grow in these regular or crystalline patterns are called Amorphous solids. Eg. Allophane.



7. WEATHERING OF ROCKS AND MINERALS

♣ Weathering

♣ Physical, chemical and biological weathering

Weathering is the breakdown and alteration of rocks and minerals at or near the Earth's surface by a set of physical, chemical and biological processes. Rock and soil is altered physically by disintegrating and chemically by decomposing. Because weather and climate occur at the earth's surface, the intensity of weathering decreases with depth and **most of it occur within less than a meter of the surface of soil and rock.**

Weathering results in the formation of Regolith or Parent material.

Regolith is an unconsolidated residual mass of the weathering rock on the earth's surface or above the solid rocks.

Parent material: It is the regolith or at least its upper portion. May be defined as the unconsolidated and more or less chemically weathered mineral material from which soils are developed.

Weathering involves three processes

Physical (or) mechanical - disintegration

Chemical - decomposition

Biological - disintegration and decomposition

Different agents of weathering

Physical/ Mechanical (disintegration)	Chemical (decomposition)	Biological (disintegration + decomposition)
1. Physical condition of rock	1. Hydration	1. Man & animals
2. Change in temperature	2. Hydrolysis	2. Higher plants & their roots
3. Action of H ₂ O	3. Solution	3. Micro organisms
- fragmentation & transportation	4. Carbonation	
- action of freezing	5. Oxidation	
- alternate wetting & drying	6. Reduction	
- action of glaciers		
4. Action of wind		
5. Atmospheric electric phenomenon		

The sequence of processes in the formation of soils is:

- ◆ Weathering of rocks and minerals → formation of regolith or parent material → formation of true soil from regolith.
- ◆ Rock → Weathering → Regolith → soil forming factors and processes → True soil

Two processes involved in the formation of soil are

- i) Formation of regolith by breaking down (weathering) of the bed rock.
- ii) The addition of organic matter through the decomposition of plant and animal tissues, and reorganization of these components by soil forming processes to form soil.

1. Physical weathering

Physical weathering:

Physical weathering refers to the disintegration and breaking down of rocks to comparatively smaller pieces, without producing any new substances

Factors influencing Physical weathering

Physical condition of rocks:

- The permeability of rocks is the most important single factor. Coarse textured (porous) sandstone weathers more readily than a fine textured (almost solid) basalt.
- Unconsolidated volcanic ash weathers quickly as compared to unconsolidated coarse deposits such as gravels.

Action of Temperature:

- The variations in temperature exert great influence on the disintegration of rocks. During day time, the rocks get heated up by the sun and expand.
- At night, the temperature falls and the rocks get cooled and contract. This alternate expansion and contraction weakens the surface of the rock and crumbles it.
- The minerals within the rock also vary in their rate of expansion and contraction. The cubical expansion of quartz is twice as feldspar.
- Dark coloured rocks are subjected to fast changes in temperature as compared to light coloured rocks.
- The differential expansion of minerals in a rock surface generates stress between the heated surface and cooled un-expanded parts resulting in fragmentation of rocks.
- This process causes the surface layer to peel off from the parent mass and the rock ultimately disintegrates. This process is called **Exfoliation**

Action of Water:

Water acts as a disintegrating, transporting and depositing agent.

i) Fragmentation and transport:

Water beats over the surface of the rock when the rain occurs and starts flowing towards the ocean

- ◆ Moving water has the great cutting and carrying force.
- ◆ It forms gullies and ravines and carries the suspended soil material of variable sizes with it.
- ◆ Transporting power of water varies. It is estimated that the transporting power of stream varies as the sixth power of its velocity i.e the greater the speed of water, more is the transporting power and carrying capacity

Speed/Sec	Carrying capacity
15 cm	fine sand
30 cm	gravel
1.2 m	stones (1kg)
9.0 m	boulders (several tons)

The disintegration is greater near the source of river than its mouth

ii) Action of freezing:

Frost is much more effective than heat in producing physical weathering

- ❖ In cold regions, the water in the cracks and crevices freezes into ice and the volume increases to one tenth
- ❖ As the freezing starts from the top there is no possibility of its upward expansion. Hence, the increase in volume creates enormous out ward pressure which breaks apart the rocks

iii) Alternate wetting and Drying:

- ❖ Some natural substances increase considerably in volume on wetting and shrink on drying. (e.g.) smectite, montmorillonite
- ❖ During dry summer/ dry weather - these clays shrink considerably forming deep cracks or wide cracks.
- ❖ On subsequent wetting, it swells.
- ❖ This alternate swelling and shrinking / wetting or drying of clay enriched rocks make them loose and eventually breaks

iv) Action of glaciers:

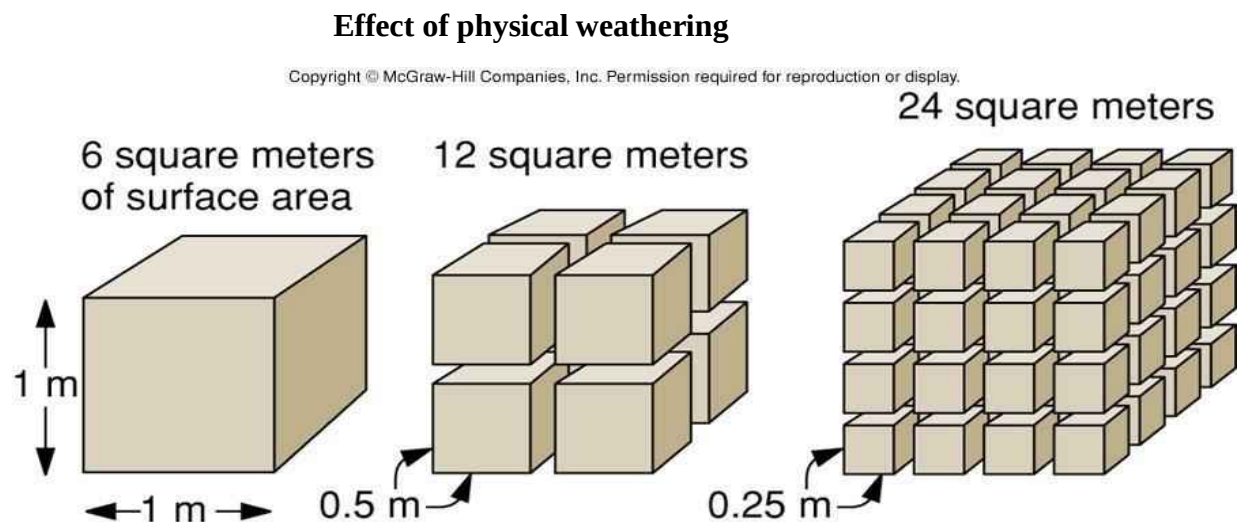
- ❖ In cold regions, when snow falls, it accumulates and changes into an ice sheet.
- ❖ These big glaciers start moving owing to the change in temperature and/or gradient.
- ❖ On moving, these exert tremendous pressure over the rock on which they pass and carry the loose materials
- ❖ These materials get deposited on reaching the warmer regions, where its movement stops with the melting of ice

v) Action of wind:

- ❖ Wind has an erosive and transporting effect. Often when the wind is laden with fine material viz., fine sand, silt or clay particles, it has a serious abrasive effect and the sand laden winds itch the rocks and ultimately breaks down under its force
- ❖ The dust storm may transport tons of material from one place to another. The shifting of soil causes serious wind erosion problem and may make cultivated land as degraded (e.g) Rajasthan deserts

vi) Atmospheric electrical phenomenon

It is an important factor causing break down during rainy season and lightning breaks up rocks and or widens cracks



* Increase in the surface area of the rocks

II. Chemical Weathering

Decomposition of rocks and minerals by various chemical processes is called chemical weathering. It is the most important process for soil formation.

Chemical weathering takes place mainly at the surface of rocks and minerals with disappearance of certain minerals and the formation of secondary products (new materials). This is called chemical transformation.

Feldspar + water $\rightarrow$ clay mineral + soluble cations and anions

Chemical weathering becomes more effective as the surface area of the rock increases. Since the chemical reactions occur largely on the surface of the rocks, smaller the fragments, the greater the surface area per unit volume available for reaction. The effectiveness of chemical weathering is closely related to the mineral composition of rocks.

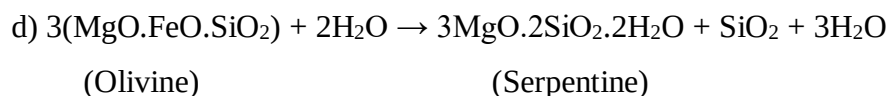
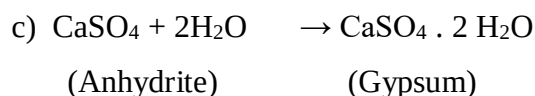
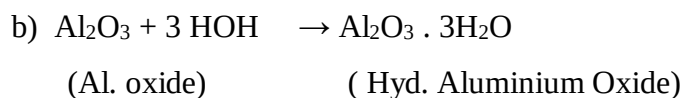
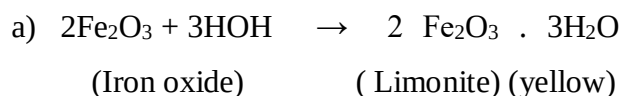
(e.g) quartz responds far slowly to the chemical attack than olivine or pyroxene.

Chemical Processes of weathering:

Hydration:

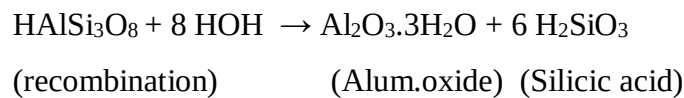
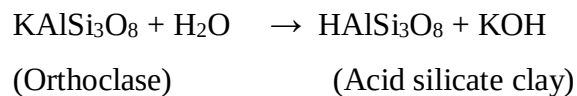
It refers to the chemical combination of water molecules with a particular substance or mineral leading to a change in structure.

Soil forming minerals in rocks do not contain any water and undergo hydration when exposed to humid conditions. Upon hydration there is swelling and increase in volume of minerals. The minerals lose their luster and become soft. It is one of the most common processes in nature and works with secondary minerals, such as aluminium oxide and iron oxide minerals and gypsum.



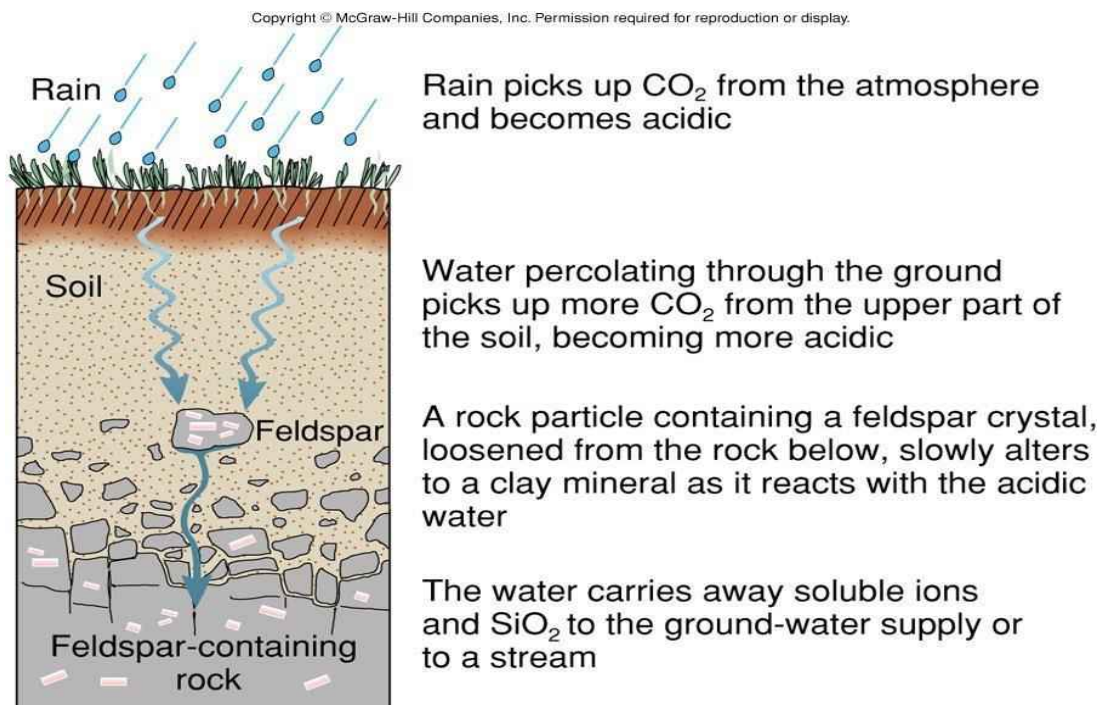
Hydrolysis:

It is the most important process in chemical weathering. It is due to the dissociation of H_2O into H^+ and OH^- ions which chemically combine with minerals and bring about changes, such as exchange, decomposition of crystalline structure and formation of new compounds. Water acts as a weak acid on silicate minerals.



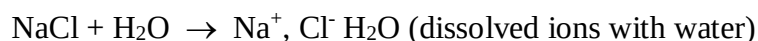
This reaction is important because of two reasons.

- i) Clay, bases and silicic acid - the substances formed in these reactions - are available to plants
- ii) Water often containing CO_2 (absorbed from atmosphere), reacts with the minerals directly to produce insoluble clay minerals, positively charged metal ions (Ca^{++} , Mg^{++} , Na^+ , K^+) and negatively charged ions (OH^- , HCO_3^-) and some soluble silica – all these ions are made available for plant growth.



Solution:

Some substances present in the rocks are directly soluble in water. The soluble substances are removed by the continuous action of water and the rock no longer remains solid and form holes, rills or rough surface and ultimately falls into pieces or decomposes. The action is considerably increased when the water is acidified by the dissolution of organic and inorganic acids. (e.g) halides, NaCl

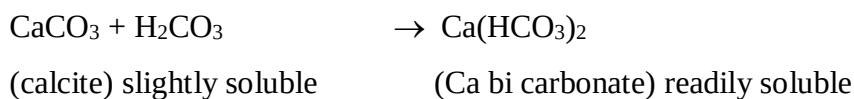


Carbonation:

Carbon di oxide when dissolved in water it forms carbonic acid.

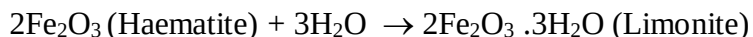
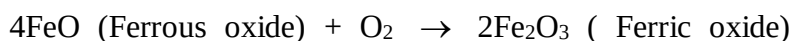


This carbonic acid attacks many rocks and minerals and brings them into solution. The carbonated water has an etching effect upon some rocks, especially lime stone. The removal of cement that holds sand particles together leads to their disintegration.



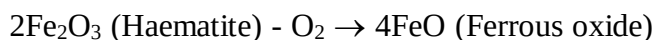
Oxidation:

The process of addition and combination of oxygen to minerals is called oxidation. The absorption is usually from O_2 dissolved in soil water and that present in atmosphere. The oxidation is more active in the presence of moisture and results in hydrated oxides.(e.g.) minerals containing Fe and Mg.



Reduction:

The process of removal of oxygen and the reverse of oxidation is called reduction. It is equally important in changing soil colour to grey, blue or green as ferric iron is converted to ferrous iron compounds. Under the conditions of excess water or water logged condition (less or no oxygen), reduction takes place.



Oxidized form

Reduced form

Weathering sequence:

It refers to the generalized order of weathering of minerals. Quartz (most resistant) > muscovite > K feldspar > Na and Ca feldspar > biotite, hornblende, augite > olivine > dolomite and calcite > gypsum (least resistant)

- In **conclusion**, during chemical weathering **igneous and metamorphic rocks** are involved in the destruction of primary minerals and the production of secondary minerals.
- In **sedimentary rocks**, which are made up of primary and secondary minerals, weathering acts initially to destroy relatively weak bonding agents and the particles are freed. These particles are then individually subjected to weathering.

III. Biological Weathering

Unlike physical and chemical weathering, the biological or living agents are responsible for both decomposition and disintegration of rocks and minerals. The biological life is mainly controlled by the prevailing environment.

Man and Animals

- ❖ The action of man in disintegration of rocks is well known as he cuts rocks to build dams, channels and construct roads and buildings. All these activities result in increasing the surface area of the rocks for attack of chemical agents and accelerate the process of rock decomposition.
- ❖ A large number of animals, birds, insects and worms, by their activities make holes in them and thus help in weathering.
- ❖ In tropical and sub tropical regions, ants and termites build galleries and passages and carry materials from lower to upper surface and excrete acids. The oxygen and water with many dissolved substances, reach every part of the rock through the cracks, holes and galleries and thus brings about speedy disintegration.
- ❖ Rabbits, by burrowing in to the ground, destroy soft rocks.
- ❖ Moles, ants and bodies of the dead animals, provides substances which react with minerals and aid in decaying process.
- ❖ The earthworms pass the soil through the alimentary canal and thus brings about physical and chemical changes in soil material.

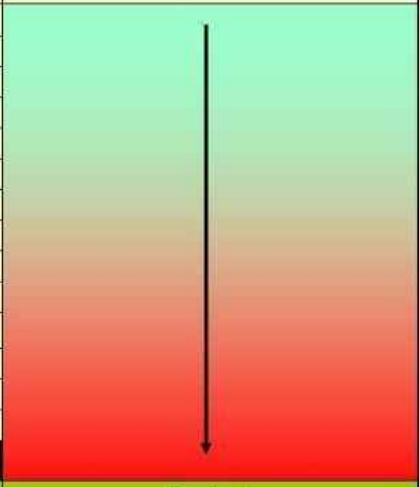
Higher Plants and Roots

- The roots of trees and other plants penetrate into the joints and crevices of the rocks. As they grew, they exert a great disruptive force and the hard rock may break apart. (e.g.) peepal tree growing on walls/ rocks.
- The grass roots form a sponge like mass, prevent erosion and conserve moisture and thus allow moisture and air to enter in to the rock for further action.
- Some roots penetrate deep into the soil and may open some sort of drainage channel. The roots running in crevices in lime stone and marble produces acids. These acids have a solvent action on carbonates
- The dead roots and plant residues decompose and produce carbon dioxide which is of great importance in weathering

Micro- organisms

- In early stages of mineral decomposition and soil formation, the lower forms of plants and animals like, mosses, bacteria and fungi and actinomycetes play an important role.
- They extract nutrients from the rock and N from air and live with a small quantity of water. In due course of time, the soil develops under the cluster of these micro-organisms.
- These organisms are closely associated with the decay of plant and animal remains and thus liberate nutrients for the next generation plants and also produce CO₂ and organic compounds which aid in mineral decomposition.

Relative stabilities of common minerals under weathering

Stability of minerals	Rate of weathering
MOST STABLE	Slowest
Iron oxide (hematite)	
Aluminum hydroxides (gibbsite)	
Quartz	
Clay minerals	
Muscovite mica	
Potassium feldspar	
Biotite	
Albite (Na-rich feldspar)	
Amphiboles	
Pyroxene	
Anorthite (Ca-rich feldspar)	
Olivine	
Calcite	
Halite	
LEAST STABLE	Fastest

8. FACTORS OF SOIL FORMATION

♣ Active and Passive soil forming factors

The soil formation is the process of two consecutive stages.

1. The weathering of rock (R) into Regolith
2. The formation of true soil from Regolith

The formation of true soil from regolith takes place by the combined action of soil forming factors and processes.

1. The first step is accomplished by weathering (disintegration & decomposition)
2. The second step is associated with the action of Soil Forming Factors

Factors of soil formation

Dokuchaiev (1889) established that the soils develop as a result of the action of soil forming factors viz., $S = f(p, cl, o)$

Further, Jenny (1941) formulated the following equation

$$S = f(cl, o, r, p, t, \dots)$$

Where,

cl - Environmental climate

o - Organisms and vegetation (biosphere)

r - Relief or topography

p - Parent material

t - Time

... - additional unspecified factors

The five soil forming factors act simultaneously at any point on the surface of the earth to produce soil

Active soil forming factors:

- i) Climate, ii) organism (vegetation or biosphere)

These factors supply energy and act upon other soil forming factors

Passive soil forming factors

- i) Parent material ii) Relief iii) Time

The passive soil forming factors are those which represent the source of soil forming mass and conditions affecting it. These provide a base on which the active soil forming factors work or act for the development of soil.

A. Passive soil forming factors

1. Parent Material:

It is that mass (consolidated material) from which the soil has formed.

i) Sedentary: Formed in original place. It is the residual parent material. The parent material differ as widely as the rocks

ii) Transported: the parent materials are transported from their place of origin. They are named according to the main force responsible for the transport and redeposition.

- a) by gravity - Colluvial
- b) by water - Alluvial
- c) by glacial ice - Moraine
- d) by lake water - Lacustrine
- e) by wind - Aeolian (sandy texture)
- Loess (silty texture)

The soils developed on such transported parent materials bear the name of the parent material; viz. Alluvial soils from alluvium, colluvial soils from colluvium *etc.* In the initial stages, however, the soil properties are mainly determined by the kind of parent material.

Endodynamomorphic soils: Soils wherein the composition of parent material subdues the effects of climate and vegetation. These soils develop under the **influence of parent material**.

Ectodynamomorphic soils: Development of normal profile under the **influence of climate and vegetation**.

Soil properties as influenced by parent material:

Different parent materials affect profile development and produce different soils, especially in the initial stages.

- ❖ Acid igneous rocks (like granite, rhyolite) produce light-textured soils (Alfisols).
- ❖ Basic igneous rocks (basalt), alluvium or colluvium derived from limestone or basalt, produce fine-textured cracking-clay soils (Vertisols).
- ❖ Basic alluvium or aeolian materials produce fine to coarse-textured soils (Entisols or Inceptisols).
- ❖ The nature of the elements released during the decaying of rocks has a specific role in soil formation. (e.g.) Si and Al forms the skeleton for the production of secondary clay minerals.
- ❖ Iron and Manganese are important for imparting red colour to soils and for oxidation and

reduction phenomena.

- ❖ Sodium and Potassium are important dispersing agents for clay and humus colloids.
- ❖ Calcium and Magnesium have a flocculating effect and result in favorable and stable soil structure for plant growth.

2. Relief or Topography:

The relief and topography sometimes are used as synonymous terms. They denote the configuration of the land surface. The topography refers to the differences in elevation of the land surface on a broad scale.

The prominent types of topography designations, as given in FAO Guidelines (1990) are:

Land surface	Slope
1. Flat to almost flat	0 – 2 %
2. Gently undulating	2 - 5 %
3. Undulating	5 – 10 %
4. Rolling	10 – 15 %
5. Hilly	15 – 30 %
6. Steeply dissect	> 30 % with moderate range of elevation (<300 m)
7. Mountainous	> 30% with great range of elevation (>300 m)

Soil formation on flat to almost flat position:

On level topographic positions, almost the entire water received through rain percolates through the soil. Under such conditions, the soils formed may be considered as representative of the regional climate. They have normal solum with distinct horizons

Soil formation on undulating topography:

The soils on steep slopes are generally shallow, stony and have weakly- developed profiles with less distinct horizonation. It is due to accelerated erosion, which removes surface material before it has the time to develop.

Soil formation in depression:

The depression areas in semi-arid and sub humid regions reflect more moist conditions than actually observed on level topographic positions due to the additional water received as runoff. Such conditions (as in the Tarai region of the Uttar Pradesh) favour more vegetative growth and slower rate of decay of organic remains. This results in the formation of comparatively dark- coloured soils rich in organic matter (Mollisols).

Soil formation and Exposure / Aspect:

Topography affects soil formation by affecting temperature and vegetative growth through slope exposures (aspect). The southern exposures (facing the sun) are warmer and subject to marked fluctuations in temperature and moisture. The northern exposures, on the other hand are cooler and more humid. The eastern and western exposures occupy intermediate position in this respect.

3. Time:

Soil formation is a very slow process requiring thousands of years to develop a mature pedon. The period taken by a given soil from the stage of weathered rock (i.e. regolith) up to the stage of maturity is considered as **time**. The matured soils mean the soils with fully developed horizons (A, B, C). It takes hundreds of years to develop an inch of soil. The time that nature devotes to the formation of soils is termed as **Pedologic Time**.

It has been observed that rocks and minerals disintegrate and/or decompose at different rates; the coarse particles of limestone are more resistant to disintegration than those of sandstone. However, in general, limestone decomposes more readily than sandstone (by chemical weathering).

Weathering stages in soil formation

Stages	Characteristic
1. Initial	Un weathered parent material
2. Juvenile	Weathering started but much of the original material still un weathered
3. Virile	Easily weatherable minerals fairly decomposed; clay content increased, slowly weatherable minerals still appreciable
4. Senile	Decomposition reaches at a final stage; only most resistant minerals survive
5. Final	Soil development completed under prevailing environments

❖	The soil properties also change with time, for instance nitrogen and organic matter contents increase with time provided the soil temperature is not high.
❖	CaCO ₃ content may decrease or even lost with time provided the climatic conditions are not arid
❖	In humid regions, the H ⁺ concentration increases with time because of chemical weathering.

B. Active soil forming factors

The active soil forming factors are those which supply energy that acts on the mass for the purpose of soil formation. These factors are climate and vegetation (biosphere).

1. Climate

Climate is the most significant factor controlling the type and rate of soil formation. The dominant climates recognized are:

- ❖ **Arid climate:** The precipitation here is very less than the water-need. Hence the soils remain dry for most of the time in a year.
- ❖ **Humid climate:** The precipitation here is much more than the water need. The excess water results in leaching of salt and bases followed by translocation of clay colloids.
- ❖ **Oceanic climate:** Moderate seasonal variation of rainfall and temperature.
- ❖ **Mediterranean climate:** Moderate precipitation in winters and summers are dry and hot.
- ❖ **Continental climate:** Warm summers and extremely cool or cold winters.
- ❖ **Temperate climate:** Cold humid conditions with warm summers.
- ❖ **Tropical and subtropical climate:** Warm to hot humid with isothermal conditions in the tropical zone.

Climate affects the soil formation directly and indirectly.

- ◆ **Directly**, climate affects the soil formation by supplying water and heat to react with parent material.
- ◆ **Indirectly**, it determines the fauna and flora activities which serve a source of energy in the form of organic matter. This energy acts on the rocks and minerals in the form of acids and salts are released.
- ◆ Precipitation and temperature are the two major climatic elements which contribute most to soil formation.

Precipitation:

- ◆ Precipitation is the most important among the climatic factors. The rain water percolates and moves from one part of the parent material to another.
- ◆ It carries with it substances in solution and suspension.
- ◆ The substances so carried are redeposited in another part or completely removed from the material.

- ◆ The soluble substances move with it and are translocated to the upper layer.
- ◆ Thus rainfall brings about a redistribution of soluble substances in the soil body.

Temperature

- ❖ Temperature is another climatic agent influencing the process of soil formation.
- ❖ High temperature hinders the process of leaching and causes an upward movement of soluble salts.
- ❖ High temperature favors rapid decomposition of organic matter and increase microbial activities in soil while low temperatures favour the accumulation of organic matter by slowing down the process of decomposition.
- ❖ Temperature thus controls the rate of chemical and biological reactions taking place in the parent material.

Jenny (1941) computed that in the **tropical regions** the rate of **weathering proceeds three times faster** than in temperate regions and nine times faster than in arctic.

2. Organism & Vegetation

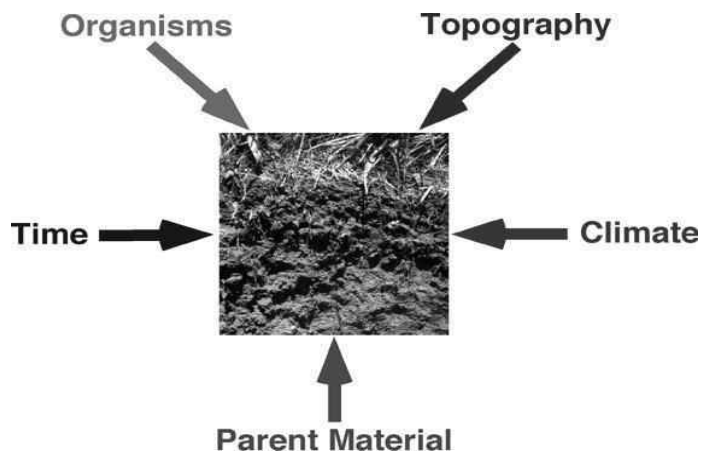
Organism

- ❖ The active components of soil ecosystem are plants, animals, microorganisms and man.
- ❖ The role of microorganisms in soil formation is related to the humification and mineralization of vegetation
- ❖ The action of animals especially burrowing animals is to dig and mix-up the soil mass
- ❖ Man influences the soil formation through his manipulation of natural vegetation, agricultural practices *etc.*
- ❖ Compaction by traffic of man and animals decrease the rate of water infiltration into the soil and thereby increase the rate of runoff and erosion.

Vegetation

- ❖ The roots of the plants penetrate into the parent material and act both mechanically and chemically.
- ❖ They facilitate percolation and drainage and bring about greater dissolution of minerals through the action of CO₂ and acidic substances secreted by them.
- ❖ The decomposition and humification of the materials further adds to the solubilization of minerals.
- ❖ Forests – reduces temperature, increases humidity, reduce evaporation and increases precipitation.

- ❖ Grasses reduce runoff and result greater penetration of water in to the parent material.



9. FUNDAMENTAL SOIL FORMING PROCESSES

- ❖ **Eluviation,**
- ❖ Illuviation
- ❖ **Humification**
- ❖ Horizonation:

The basic soil forming processes involved in soil formation (Simonson, 1959) includes the following consequences.

- ❖ **Gains** or Additions of water, mostly as rainfall, organic and mineral matter to the soil.
- ❖ **Losses** of the above materials from the soil.
- ❖ **Transformation** of mineral and organic substances within the soil.
- ❖ **Translocation** of soil materials from one point to another within the soil. It is usually divided into two aspects.
 - 1) Movement in solution (leaching) and
 - 2) Movement in suspension (eluviation) of clay, organic matter and hydrous oxides

A. Fundamental Soil Forming Processes

1. Eluviation:

- ❖ It is the mobilization and translocation of certain constituents viz., Clay, Fe_2O_3 , Al_2O_3 , SiO_2 , humus, CaCO_3 , other salts etc. from one point of soil body to another.
- ❖ Eluviation means **washing out**. It is the process of removal of constituents in suspension or solution by the percolating water from the upper to lower layers.
- ❖ The eluviation process involves mobilization and translocation of soil constituents resulting in textural differences.
- ❖ The horizon formed by the process of eluviation is termed as **eluvial horizon (A₂ or E horizon)**. Translocation depends upon relative mobility of elements and depth of percolation.

2. Illuviation:

The process of **deposition** of soil materials (removed from the eluvial horizon) in the lower layer is termed as Illuviation.

- The horizon formed by this process is termed as illuvial horizon (B-horizon, especially Bt).

This is the **horizon of gains** having the property of stabilizing translocated clay materials. **1.**

3. Humification:

- ❖ Humus accumulation on the surface and formation of the O horizon. It is the process of decomposition of organic matter and synthesis of new organic substances.
- ❖ It is the process of transformation of raw organic matter into humus. It is extremely a complex process involving various organisms.
- ❖ First, simple compounds such as sugars and starches are attacked followed by proteins and cellulose and finally very resistant compounds such as tannins, are decomposed and the dark coloured substance, known as humus, is formed.

4. Horizonation:

It is the process of differentiation of soil in different horizons along the depth of the soil body. The differentiation is due to the fundamental processes of eluviation and illuviation.

10. SPECIFIC SOIL FORMING PROCESSES

- ♣ **Podzolization and Laterization**
- ♣ Salinization and Alkalinization
- ♣ **Calcification and Decalcification**
- ♣ Carbonation
- ♣ **Gleization**
- ♣ Pedoturbation

The basic pedologic processes provide a framework for later operation of more specific processes

Podzolization:

It is a process of soil formation resulting in the formation of Podzols and Podzolic soils. The process operates under favorable combination of the following environments.

- i) **Climate:** A cold and humid climate is most favorable for podzolization
- ii) **Parent material:** Siliceous (Sandy) material, having poor reserves of weatherable minerals, favor the operation of podzolization as it helps in easy percolation of water.
- iii) **Vegetation:** Acid producing vegetation such as coniferous pines is essential
- iv) **Leaching and Translocation of Sesquioxides:**

In the process of decomposition of organic matter various organic acids are produced. The organic acids thus formed act with Sesquioxide and the remaining clay minerals, forming organic- sesquioxide and organic clay complexes, which are soluble and move with the percolating water to the lower horizons.

As **iron and aluminium are removed**, the A horizon gives a bleached grey or ashy appearance. The Russians used the term Podzols (*pod* means under, the *zola* means ash like i.e. ash-like horizon appearing beneath the surface horizon) for such soils.

Laterization:

- The term laterite is derived from the word *later* meaning brick or tile and was originally applied to a group of high clay Indian soils found in Malabar hills of Kerala, Tamil Nadu, Karnataka and Maharashtra.

- It refers specifically to a particular cemented horizon in certain soils which when dried, become very hard, like a brick.
- Such soils (in tropics) when massively mixed with sesquioxides (iron and aluminium oxides) to an extent of 70 to 80 per cent of the total mass, are called laterites or latosols (Oxisols).
- The soil forming process is called Laterization or Latozation.
- Laterization is the process that **removes silica, instead of sesquioxides** from the upper layers and thereby leaving sesquioxides to concentrate in the solum. The process operates under the following conditions.

i) **Climate:** Unlike podzolization, the process of laterization operates most favorable in warm and humid (tropical) climate with 2000 to 2500 mm rainfall and continuous high temperature (25°C) throughout the year.

ii) **Natural vegetation:** The rain forests of tropical areas are favorable for the process.

iii) **Parent Material:** Basic parent materials having sufficient iron bearing ferromagnesian minerals (Pyroxene, amphiboles, biotite and chlorite), which on weathering release iron that are congenial for the development of laterites.

Salinization

- ◆ It is the process of accumulation of salts, such as sulphates and chlorides of calcium, magnesium, sodium and potassium in soils in the form of a salty (salic) horizon.
- ◆ It is quite common in arid and semi arid regions.
- ◆ It may also take place through capillary rise of saline ground water and by inundation with seawater.
- ◆ Salt accumulation may also result from irrigation or seepage in areas of impeded drainage.

Desalinization

- ◆ It is the process of removal of excess soluble salts from horizons that contained enough soluble salts to impair the plant growth.
- ◆ Leaching is done by ponding water and improving the drainage conditions by installing artificial drainage network.

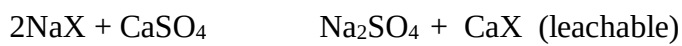
Solonization or Alkalization

- ◆ The process involves the accumulation of sodium ions on the exchange complex of the clay resulting in the formation of sodic soils (Solonetz).
- ◆ All cations in solution are engaged in a reversible reaction with the exchange sites on the clay and organic matter particles.
- ◆ The reaction can be represented as:
$$\text{Ca.Mg.2NaX} \rightarrow \text{Ca}^{++} + \text{Mg}^{++} + 2\text{Na}^+ + \text{X} \quad (3\text{CO}_3^{2-} \rightarrow \text{Na}_2\text{CO}_3 + \text{MgCO}_3 + \text{CaCO}_3)$$

(Where X represents clay or organic matter exchange sites)

Solodization or Dealkalization

- ◆ The process refers to the removal of Na^+ from the exchange sites. This process involves dispersion of clay. Dispersion occurs when Na^+ ions become hydrated.
- ◆ Much of the dispersion can be eliminated if Ca^{++} and or Mg^{++} ions are concentrated in the water. These Ca and Mg ions can replace the Na on exchange complex, and the salts of sodium are leached out as:

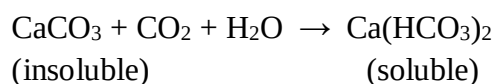


Calcification

- ◆ It is the process of precipitation and accumulation of calcium carbonate (CaCO_3) in some part of the profile. The accumulation of CaCO_3 may result in the development of a calcic horizon.
- ◆ Calcium is readily soluble in acidic soil water and/or when CO_2 concentration is high in root zone as:
 - $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$
 - $\text{H}_2\text{CO}_3 + \text{Ca} \rightarrow \text{Ca}(\text{HCO}_3)_2$ (soluble)
 - $\text{Ca}(\text{HCO}_3)_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2$ (precipitates)
- ◆ The process of precipitation after mobilization under these conditions is called calcification.

Decalcification

It is the reverse of calcification that is the process of removal of CaCO_3 or calcium ions from the soil by leaching



Carbonation

- It occurs when carbon dioxide interacts chemically with minerals. When carbon dioxide is dissolved in water, it forms weak carbonic acid.
- When carbonic acid comes in contact with the surface of the earth it dissolves large masses of limestone, creating caves and caverns.
- Other common terms associated with carbonation are sink holes, karst topography, stalactites and stalagmites.

Gleization:

- The term *glei* is of Russian origin means blue, grey or green clay.
- The Gleization is a process of soil formation resulting in the development of a gley horizon in the lower part of the soil profile above the parent material due to poor drainage condition and where waterlogged conditions prevail. Such soils are called hydromorphic soils.
- The process is not particularly dependent on climate (high rainfall as in humid regions) but often on drainage conditions.
- Under such conditions, iron compounds are reduced to soluble ferrous forms.
- The solubility of Ca, Mg, Fe, and Mn is increased and most of the iron exists as Fe^{++} organo - complexes in solution or as mixed precipitate of ferric and ferrous hydroxides.
- This is responsible for the production of typical bluish to greyish horizon with mottling of yellow and or reddish brown colors.

Pedoturbation:

Another process that may be operative in soils is pedoturbation. It is the process of mixing of the soil. Mixing to a certain extent takes place in all soils. The most common types of pedoturbation are:

- ❖ **Faunal pedoturbation:** It is the mixing of soil by animals such as ants, earthworms, moles, rodents, and man himself
- ❖ **Floral pedoturbation :** It is the mixing of soil by plants as in tree tipping that forms pits and mounds
- ❖ **Argillic pedoturbation:** It is the mixing of materials in the solum by the churning process caused by swell-shrink clays as observed in deep Black Cotton Soils.

* * * * *

11. SOIL PROFILE

♣ Profile description

♣ Master horizons

♣ Pedon and polypedon

Definition of soil profile:

The vertical section of the soil showing the various layers from the surface to the unaffected parent material is known as a soil profile.

The various layers are known as horizons.

Soil Horizons (layers):

Soil is made up of distinct horizontal layers; these layers are called horizons. They range from rich, organic upper layers (humus and topsoil) to underlying rocky layers (subsoil, regolith and bedrock).

Master horizons :

Most soils have a distinct profile or sequence of horizontal layers. Generally, these horizons result from the processes of chemical weathering, eluviation, illuviation, and organic decomposition. Up to five layers can be present in a typical soil viz., O, A, B, C, and R horizons which are called the Master horizons.

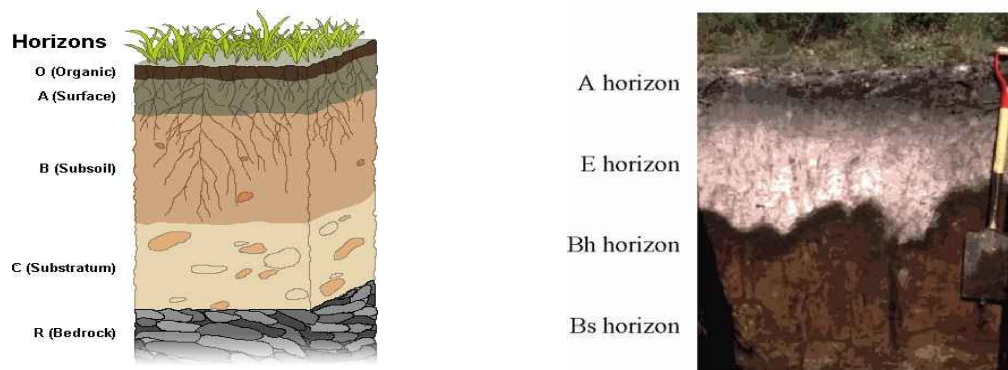


Figure: Typical layers found in soil profiles.

The O horizon is the topmost layer of most soils. It is composed mainly of plant litter at various levels of decomposition and humus.

A horizon is found below the O layer. This layer is composed primarily of mineral particles which has two characteristics: it is the layer in which humus and other organic materials are mixed with mineral particles, and it is a **zone of translocation** from which

eluviation has removed finer particles and soluble substances, both of which may be deposited at a lower layer. Thus the A horizon is dark in color and usually light in texture and porous. The A horizon is commonly differentiated into a darker upper horizon or organic accumulation, and a lower horizon showing loss of material by eluviation.

The B horizon is a mineral soil layer which is strongly influenced by illuviation. Consequently, this layer receives material eluviated from the A horizon. The B horizon also has a higher bulk density than the A horizon due to its enrichment of clay particles. The B horizon may be colored by oxides of iron and aluminum or by calcium carbonate illuviated from the A horizon.

The E horizon is normally found in between the the A and B layers. This layer is a leached / eluviated layer with pale / lighter colour compared to the upper and lower layers due to eluviation of clays, OM, Fe and Al oxides; and accumulation of silica, sand and silt sized particles. The presence or absence of this layer in a profile is decided by the soil forming processes.

The C horizon is composed of weathered parent material. The texture of this material can be quite variable with particles ranging in size from clay to boulders. The C horizon has also not been significantly influenced by the pedogenic processes, translocation, and/or organic modification.

The final layer in a typical soil profile is called the R horizon. This soil layer simply consists of unweathered bedrock.

Sub ordinate horizons:

Lowercase letters used to designate subordinate distinction within master horizons

Letter	Distinction	Letter	Distinction
a	Highly decomposed Organic material	q	Accumulation of silica
b	Buried soil horizon	r	Weathered or soft bed rock
c	Concretions or nodules	s	Illuvial accumulation of organic matter and Fe and Al oxides
d	Dense unconsolidated material / Physical root restriction	t	Accumulation of silicate clays
e	Organic material of intermediate decomposition	v	Plinthite (high iron, red material)
f	Frozen soil and water	w	Distinctive colour or structure
g	Strong gleying (mottling)	x	Fragipan (high bulk density)

h	Illuvial accumulation of organic matter	y	Accumulation of gypsum
i	Slightly decomposed organic material	z	Accumulation of soluble salts
j	Accumulation of jarosite (K or FeSO ₄ minerals)	co	Coprogenous earth
k	Accumulation of carbonates	di	Diatomaceous earth
m	Cementation	ff	Dry permafrost
n	Accumulation of sodium	jj	Evidence of cryoturbation
o	Accumulation of Fe and Al oxides	ma	marl
p	Ploughing or other disturbance	ss	Presence of slicken sides

Special Features

Pedon: A pedon is the smallest volume that can be recognized as a soil individual. It has three dimensions and its area ranges from 1 to 10 square meters, depending on the variability in the horizons. The shape of the pedon is roughly hexagonal. A soil volume that consists of more than one pedon is termed a polypedon

Polypedon: A polypedon is defined as a continuous similar pedons bounded on all sides by "not-soil" or by pedons of unlike characters. It is a real physical soil body which has a minimum area of more than 1 sq. km and an unspecified maximum area.

Regolith: The unconsolidated materials above the parent rock including A, B and C horizons are called the weathered soil material including O, A and B horizons are called **solum**.

Control Section: The Control section is the vertical section of soil or the part of the soil upon which classification is based. It provides a uniform basis for soil classification. It is that part of the soil profile between depths of 25 cm and 100 cm.

Lithological discontinuity : When two or more genetically unrelated (contrasting) materials are present in a profile as in the case of alluvial or colluvial soils, then the phenomenon is known as lithological discontinuity.

12. SOIL PHYSICAL PROPERTIES – SOIL TEXTURE

♣ Soil texture

♣ Particle size distribution

♣ Textural classes and textural triangular diagram

♣ Significance of soil texture

The physical properties of a soil are the result of soil parent materials being acted upon by climatic factors (such as rainfall and temperature), and being affected by relief (slope and direction or aspect), and by vegetation, with time. A change in any one of these soil-forming factors usually results in a difference in the physical properties of the resulting soil.

- Physical properties (mechanical behaviour) of a soil greatly influence its use and behaviour towards plant growth.
- The plant support, root penetration, drainage, aeration, retention of moisture, and plant nutrients are linked with the physical condition of the soil.
- Physical properties also influence the chemical and biological behaviour of soil.
- The physical properties of a soil depend on the amount, size, shape, arrangement and mineral composition of its particles. These properties also depend on organic matter content and pore spaces.

Important physical properties of soils

1. Soil texture
2. Soil structure
3. Surface area
4. Soil density
5. Soil porosity
6. Soil colour
7. Soil consistence

Soil Texture

Definition:

Soil texture refers to the relative proportion of the three soil separates viz., sand, silt and clay or simply refers to the size of soil particles.

Knowledge on Texture is important since it is a guide to the value of the land. Land use capability and methods of soil management depends on Texture

The proportion of each size group in a given soil (the texture) cannot be easily altered and it is considered as a basic property of a soil. The soil separates are defined in terms of diameter (in millimeters) of the particles. Soil particles >2 mm in diameter are excluded from soil textural determinations.

Particles less than 2 mm are called fine earth (soil), normally considered in chemical and mechanical analysis.

Sand:

- ❖ Usually consists of quartz but may also contain fragments of feldspar, mica and occasionally heavy minerals viz., zircon, Tourmaline and hornblende.
- ❖ Has uniform dimensions
- ❖ Can be represented as spherical
- ❖ Not necessarily smooth and has jagged surface

Silt:

- ❖ Particle size intermediate between sand and clay
- ❖ Since the size is smaller, the surface area is more
- ❖ Coated with clay
- ❖ Has the physico- chemical properties as that of clay to a limited extent
- ❖ Sand and Silt forms the SKELETON

Clay:

- ❖ Particle size less than 0.002 mm
- ❖ Plate like or needle like in shape
- ❖ Belong to alumino silicate group of minerals
- ❖ Some times considerable concentration of fine particles which does not belong to alumino silicates. (eg). iron oxide and CaCO_3
- ❖ These are secondary minerals derived from primary minerals in the rock
- ❖ FLESH of the soil

The size limits of these fractions have been established by various organizations. There are a number of systems of naming soil separates.

- ❖ The American system developed by United States Department of Agriculture(USDA)
- ❖ The English system or British system developed by British Standard Institute (BSI)
- ❖ The International system developed by International Society of Soil Science (ISSS)
- ❖ European system

i) USDA

Soil Separates	Diameter (mm)
Clay	< 0.002
Silt	0.002 – 0.05
Very Fine Sand	0.05 – 0.10
Fine Sand	0.10 – 0.25
Medium Sand	0.25 - 0.50
Coarse Sand	0.50 - 1.00
Very Coarse Sand	1.00 – 2.00

iii) ISSS

Soil separates	Diameter (mm)
Clay	< 0.002
Silt	0.002 – 0.02
Fine sand	0.02 – 0.2
Coarse sand	0.2 – 2.0

ii) BSI

Soil separates	Diameter (mm)
Clay	< 0.002
Fine Silt	0.002 – 0.01
Medium Silt	0.01 – 0.04
Coarse Silt	0.04 – 0.06
Fine Sand	0.06 - 0.20
Medium Sand	0.20 - 1.00
Coarse Sand	1.00 – 2.00

(iv) European System

Soil separates	Diameter (mm)
Fine clay	< 0.0002
Medium clay	0.0002 – 0.0006
Coarse clay	0.0006 – 0.002
Fine silt	0.002 - 0.006
Medium silt	0.006 - 0.02
Coarse silt	0.02 - 0.06
Fine sand	0.06 - 0.20
Medium sand	0.20 - 0.60
Coarse sand	0.60 - 2.00

Particle size distribution/ determination

The determination of relative distribution of the ultimate or individual soil particles below 2 mm diameter is called as Particle size analysis or Mechanical analysis.

Two steps are involved

- i) Separation of all the particles from each other ie., Complete dispersion into ultimate particles
- ii) Measuring the amount of each group

Separation

S.No	Aggregating agents	Dispersion method
1	Lime and Oxides of Fe & Al	Dissolving in HCl
2	Organic matter	Oxidizes with H_2O_2
3	High concentration of electrolytes (soluble salts)	Precipitate and decant or filter with suction
4	Surface tension	Elimination of air by stirring with water or boiling

After removing the cementing agents, disperse by adding NaOH

Measurement

Once the soil particles are dispersed into ultimate particles, measurement can be done

- i) **Coarser fractions** – sieving – sieves used in the mechanical analysis corresponds to the desired particle size separation
 - For 2 mm, 1 mm and 0.5 mm – sieves with circular holes
 - For smaller sizes, wire mesh screens are used (screening)
- ii) **Finer fractions** – by settling in a medium

The settling or the velocity of the fall of particles is influenced by

 - Viscosity of the medium
 - Difference in density between the medium and falling particles
 - Size and shape of object

Stokes' Law:

Particle size analysis is based on a simple principle i.e. "when soil particles are suspended in water they tend to sink. Because there is little variation in the density of most soil particles, their velocity (V) of settling is proportional to the square of the radius 'r' of each particles.

Thus $V = kr^2$, where k is a constant. This equation is referred to as **Stokes' law**.

Stokes (1851) was the first to suggest the relationship between the radius of the particles and its rate of fall in a liquid. He stated that "the velocity of a falling particle is proportional to the square of the radius and not to its surface. The relation between the diameter of a particle and its settling velocity is governed by Stokes' Law :

$$V = \frac{2}{9} gr^2 \frac{(ds - dw)}{\eta}$$

where

- V - velocity of settling particle (cm/sec.)
- g - acceleration due to gravity cm/sec^2 (981)
- ds - density of soil particle (2.65)
- dw - density of water (1)
- η - coefficient of viscosity of water (0.0015 at 4°C)
- r - radius of spherical particles (cm).

Methods of Textural determination

Numerous methods for lab and field use have been developed

- i) Elutriation method – Water & Air
- ii) Pipette method
- iii) Decantation/ beaker method
- iv) Test tube shaking method
- v) Feel method – Applicable to the field – quick method – by feeling the soil between thumb and fingers

Feel Method

Evaluated by attempting to squeeze the moistened soil into a thin ribbon as it is pressed with rolling motion between thumb and pre finger or alternately to roll the soil into a thin wire.

Four aspects to be seen

- i) Feel by fingers
- ii) Ball formation
- iii) Stickiness
- iv) Ribbon formation

Soil Textural Classes

To convey an idea of the textural make up of soils and to give an indication of their physical properties, soil textural class names are used. These are grouped into three main fractions viz., Sand, Silt and Clay.

According to the proportion of these three fractions a soil is given a name to indicate its textural composition. Such a name gives an idea not only of the textural composition of a soil but also of its various properties in general.

On this basis soils are classified into various textural classes like sands, clays, silts, loams *etc.*

Sands:

The sand group includes all soils in which the sand separates make up at least 70% and the clay separate 15% or less of the material by weight. The properties of such soils are therefore characteristically those of sand in contrast to the stickier nature of clays. Two specific textural classes are recognized in this group sandy and loamy sand.

Silt:

The silt group includes soils with at least 80% silt and 12% or less clay. Naturally the properties of this group are dominated by those of silt. Only one textural class - Silt is included in this group.

Clays:

To be designated clay a soil must contain at least 35% of the clay separate and in most cases not less than 40%. In such soils the characteristics of the clay separates are distinctly dominant, and the class names are clay, sandy clay and silty clay. Sandy clays may contain more sand than clay. Likewise, the silt content of silty clays usually exceeds clay fraction

Loams:

The loam group, which contains many subdivisions, is a more complicated soil textural class. An ideal loam may be defined as a mixture of sand, silt and clay particles that exhibits the properties of those separates in about equal proportions. Loam soils do not exhibit dominant physical properties of sand, silt or clay. Loam does not contain equal percentage of sand, silt and clay. However, exhibit approximately equal properties of sand, silt and clay.

Textural Classes :

In the American system as developed by the United State Department of Agriculture twelve textural classes are proposed. They are :

Clay
Silty Clay
Sandy clay

Silt

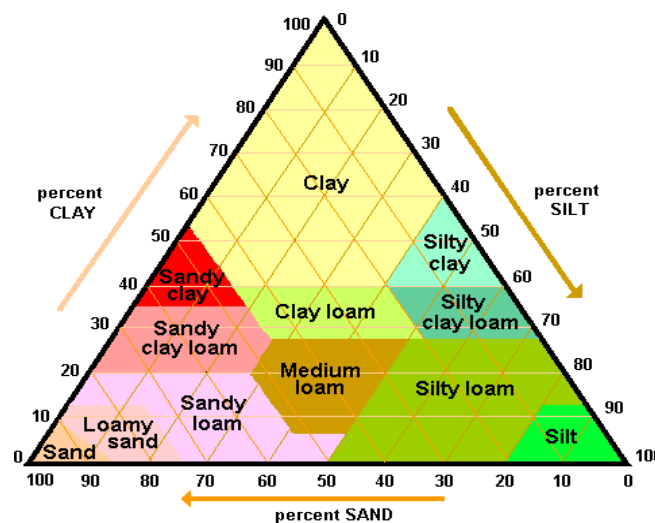
Loam (Medium loam)
Clay loam
Silty loam
Silty clay loam
Sandy clay loam
Sandy loam

Sand
Loamy sand

The textural triangle:

It is used to determine the soil textural name after the percentages of sand, silt, and clay are determined from a laboratory analysis. Since the soil's textural classification includes only mineral particles and those of less than 2mm diameter, the sand plus silt plus clay percentages equal 100 percent. (note that organic matter is not included.)

Using the Soil Texture Triangle to Determine Soil Texture (Whitney) :



Procedure:

First, you will need to know the percentages of sand, silt, and clay in your soil, as determined by laboratory particle size analysis.

1. Locate the percentage of clay on the left side of the triangle and move inward horizontally, parallel to the base of the triangle.
2. Follow the same procedure for sand, moving along the base of the triangle to locate your sand percentage
3. Then, move up and to the left until you intersect the line corresponding to your clay percentage value.
4. At this point, read the *textural class* written within the bold boundary on the triangle. For example: a soil with 40% sand, 30% silt, and 30% clay will be a clay loam. With a

moderate amount of practice, soil textural class can also be reliably determined in the field

Significance / Importance of Soil Texture

- Presence of each type of soil particles makes its contribution to the nature and properties of soil as a whole
- Texture has good effect on management and productivity of soil. Sandy soils are of open character usually loose and friable.
- Such type of texture is easy to handle in tillage operations.
- **Sand** facilitates drainage and aeration. It allows rapid evaporation and percolation.
- Sandy soils have very little water holding capacity. Such soils can not stand drought and unsuitable for dry farming.
- Sandy soils are poor store house of plant nutrients
- Contain low organic matter
- Leaching of applied nutrients is very high.
- In sandy soil, few crops can be grown such as potato, groundnut and cucumbers.
- **Clay** particles play a very important role in soil fertility.
- Clayey soils are difficult to till and require much skill in handling. When moist clayey soils are exceedingly sticky and when dry, become very hard and difficult to break.
- They have fine pores, and are poor in drainage and aeration.
- They have a high water holding capacity and poor percolation, which usually results in water logging.
- They are generally very fertile soils, in respect of plant nutrient content. Rice, jute, sugarcane can be grown very successfully in these soils.
- **Loam and Silt loam soils** are highly desirable for cultivation
- Generally, **the best agriculture soils are** those that contain 10 – 20 per cent clay, 5 – 10 per cent organic matter and the rest equally shared by silt and sand

13. SOIL STRUCTURE

♣ Definition

♣ Genesis and Classification

♣ Factors influencing soil structural stability.

Definition

The arrangement and organization of primary and secondary particles in a soil mass is known as soil structure. (or) The grouping and arrangement of individual soil particles is known as soil structure.

Genesis of soil structure

- Soil particles are present as single individual grains and as aggregates *i.e.* group of particles bound together into granules or compound particles. These granules or compound particles are known as secondary particles.
- Particles in sandy and silty soils are present as single individual grains while in clayey soil they are present in granulated condition. The individual particles are usually solid, while the aggregates are not solid but they possess a porous or spongy character.
- Most soils are mixture of single grain and compound particle. Soils, which predominate with single grains are said to be structure-less, while those possess majority of secondary particles are said to be aggregate, granulated or crumb structure.

Mechanism of Aggregate Formation

- The bonding of the soil particles into structural unit is called as the genesis of soil structure. The bonding between individual particles in the structural units is generally considered to be stronger than the structural units themselves.
- In aggregate formation, a number of primary particles such as sand, silt and clay are brought together by the cementing or binding effect of soil colloids.
- The cementing materials taking part in aggregate formation are **colloidal clay, iron and aluminium hydroxides** and **decomposing organic matter**. Whatever may be the cementing material, it is ultimately the dehydration of colloidal matter accompanied with pressure that completes the process of aggregation.

Classification

The primary particles - sand, silt and clay - usually occur grouped together in the form of aggregates.

Natural aggregates are called **peds** whereas **clod** is an artificially formed soil mass. Structure is studied in the field under natural conditions and it is described under three categories.

- 1) Type - Shape or form and arrangement pattern of peds
- 2) Class - Size of Peds
- 3) Grade - Degree of distinctness of peds

Types of Structure: There are four principal forms of soil structure

1) Plate-like (Platy)

In this type, the aggregates are arranged in relatively thin horizontal plates or leaflets. The horizontal axis or dimensions are larger than the vertical axis.

- ❖ When the units/ layers are thick they are called platy
- ❖ When they are thin then it is laminar.

Platy structure is most noticeable in the surface layers of virgin soils but may be present in the subsoil. This type is inherited from the parent material, especially by the action of water or ice.

2) Prism-like:

The vertical axis is more developed than horizontal, giving a pillar like shape.

- ❖ Vary in length from 1- 10 cm.
- ❖ Commonly occur in sub soil horizons of Arid and Semi arid regions.
- ❖ When the tops are rounded, the structure is termed as columnar when the tops are flat / plane, level and clear cut - prismatic.

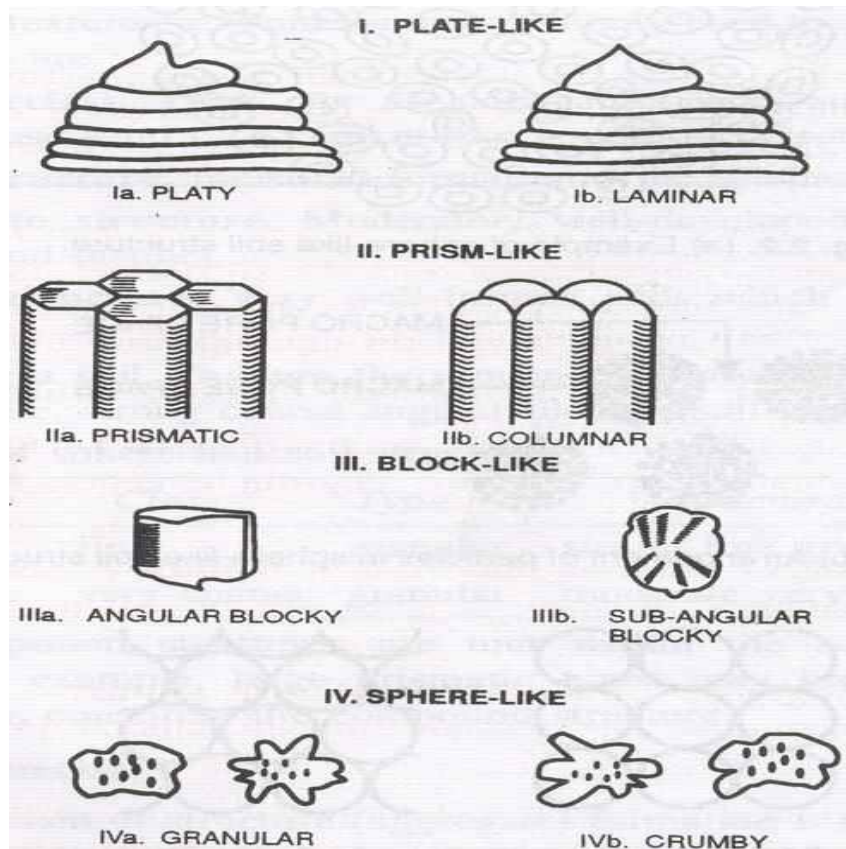
3) Block like:

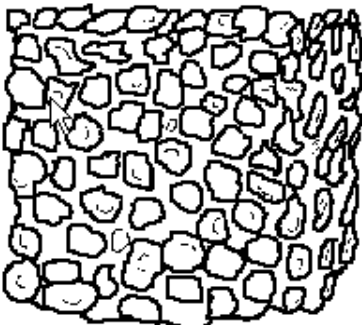
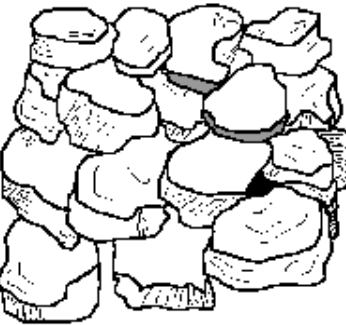
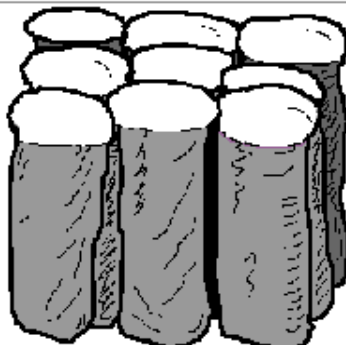
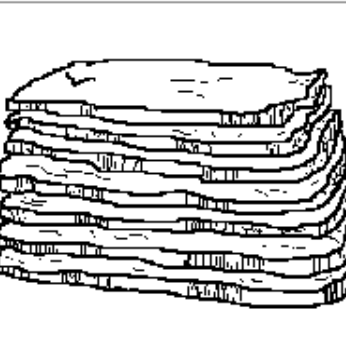
- ❖ All three dimensions are about the same size.
- ❖ The aggregates have been reduced to blocks.
- ❖ Irregularly six faced with their three dimensions more or less equal.
- ❖ When the faces are flat and distinct and the edges are sharp angular, the structure is named as angular blocky.

- ❖ When the faces and edges are mainly rounded it is called sub angular blocky.
- ❖ These types usually are confined to the sub soil and characteristics have much to do with soil drainage, aeration and root penetration.

4) Spheroidal (Sphere like):

- ❖ All rounded aggregates (peds) may be placed in this category.
- ❖ Not exceeding an inch in diameter.
- ❖ These rounded complexes usually loosely arranged and readily separated.
- ❖ When wetted, the intervening spaces generally are not closed so readily by swelling as may be the case with a blocky structural condition.
- ❖ Therefore in sphere-like structure, infiltration, percolation and aeration are not affected by wetting of soil.
- ❖ The aggregates of this group are usually termed as granular which are relatively less porous.
- ❖ When the granules are very porous, it is termed as crumb.
- ❖ This is specific to surface soil particularly high in organic matter/ grass land soils.



		
Granular: Resembles cookie crumbs and is usually less than 0.5 cm in diameter. Commonly found in surface horizons where roots have been growing.	Blocky: Irregular blocks that are usually 1.5 - 5.0 cm in diameter.	Prismatic: Vertical columns of soil that might be a number of cm long. Usually found in lower horizons.
		
Columnar: Vertical columns of soil that have a salt "cap" at the top. Found in soils of arid climates.	Platy: Thin, flat plates of soil that lie horizontally. Usually found in compacted soil.	Single Grained: Soil is broken into individual particles that do not stick together. Always accompanies a loose consistence. Commonly found in sandy soils.

Classes of Structure

Each primary structural type of soil is differentiated into 5 size classes depending upon the size of the individual peds.

The terms commonly used for the size classes are:

1. Very fine or very thin
2. Fine or thin
3. Medium
4. Coarse or thick
5. Very Coarse or very thick

The terms thin and thick are used for platy types, while the terms fine and coarse are used for other structural types.

Grades of Structure

Grades indicate the degree of distinctness of the individual peds. It is determined by the stability of the aggregates. Grade of structure is influenced by the moisture content of the soil. Grade also depends on organic matter, texture etc. Four terms commonly used to describe the grade of soil structure are:

- 1. Structure-less:** There is no noticeable aggregation, such as conditions exhibited by loose sand.
- 2. Weak Structure:** Poorly formed, indistinct formation of peds, which are not durable and much un-aggregated material.
- 3. Moderate structure:** Moderately well developed peds, which are fairly durable and distinct.
- 4. Strong structure:** Very well formed peds, which are quite durable and distinct.

Structure naming

For naming a soil structure the sequence followed is grade, class and type; for example strong coarse angular blocky, moderate thin platy, weak fine prismatic.

Factors Affecting Soil Structure

The development of structure in arable soil depends on the following factors:

1. Climate:

Climate has considerable influence on the degree of aggregation as well as on the type of structure. In arid regions there is very little aggregation of primary particles. In semi arid regions, the degree of aggregation is greater.

2. Organic matter:

Organic matter improves the structure of a sandy soil as well as of a clay soil. In case of a sandy soil, the sticky and slimy material produced by the decomposing organic matter and the associated microorganism cement the sand particles together to form aggregates.

3. Tillage:

Cultivation implements break down the large clods into smaller fragments and aggregates. For obtaining good granular and crumbly structure, optimum moisture content in the soil is necessary. If the moisture content is too high, it will form large clods on drying. If it is too low, some of the existing aggregates will be broken down.

4. Plants, Roots and Residues

- ❖ Excretion of gelatinous organic compounds and exudates from roots serve as a link
- ❖ Root hairs make soil particles to cling together. Grass and cereal roots Vs other roots
- ❖ Pressure exerted by the roots also held the particles together
- ❖ Dehydration of soil - strains the soil due to shrinkage result in cracks lead to aggregation
- ❖ Plant tops and residues – shade the soil – prevent it from extreme and sudden temperature and moisture changes and also from rain drop impedance. Plant residues – serve as a food to microbes which are the prime aggregate builders.

5. Animals:

Among the soil fauna small animals like earthworms, moles and insects etc., that burrow in the soil are the chief agents that take part in the aggregation of finer particles.

6. Microbes:

Algae, fungi and actinomycetes keep the soil particles together by the products of decomposition.

7. Fertilizers:

Fertilizer like Sodium Nitrate destroys granulation by reducing the stability of aggregates. Few fertilizers for example Calcium Ammonium Nitrate (CAN) help in development of good structures.

8. Wetting and drying:

When a dry soil is wetted, the soil colloids swell on absorbing water. On drying, shrinkage produces strains in the soil mass gives rise to cracks, which break it up into clods and granules of various sizes.

9. Exchangeable cations: Ca, Mg → Flocculating; Good structure
H, Na → Deflocculating; Poor structure

10. Inorganic cements: CaCO₃ and Sesquioxides

11. Clay and Water : The quantity and quality of the clay minerals and the water content of the soil play a major role in the formation and stability of the soil structure.

Significance / Importance of Soil Structure

Soil structure influences rather indirectly by the formation of an array of pores of various shapes and sizes. These pores are controlling factors governing water, air and temperature in soil.

The role of soil structure in relation to plant growth

- ❖ Soil structure influences the amount and nature of porosity.
- ❖ Structure controls the amount of water and air present in the soil. Not only the amount of water and air dependent on soil structure, but their movement and circulation are also controlled by soil structure.
- ❖ Structure can be modified by cultivation and tillage operations while texture is an inherent property of soil and cannot be modified within short period of time .It affects tillage practices.
- ❖ Structure controls runoff and erosion.
- ❖ Platy structure normally hinders free drainage whereas sphere like structure (granular and crumby) helps in drainage.
- ❖ Crumby and granular structure provides optimum infiltration, water holding capacity, aeration and drainage. It also provides good habitat for microorganisms and supply of nutrients.

14. DENSITIES AND POROSITY OF SOIL

Density is the weight per unit volume of a substance. It is expressed in gram per cubic centimeter or pound per cubic foot. Two density measurements particle density and bulk density are common for soils.

$$\text{Density (D)} = \frac{\text{Mass (M)}}{\text{Volume (V)}} \quad \text{gm / cc}$$

Bulk Density (Db)

Soil bulk density, like all density measurements, is an expression of the mass to volume relationship for a given material. Soil bulk density measures total soil volume.

- ◆ Thus, bulk density takes into account solid space as well as pore space.
- ◆ Soils that are loose, porous, or well-aggregated will have lower bulk densities than soils that are compacted or nonaggregated.
- ◆ This is because pore space weighs less than solid space (soil particles). Sandy soils have less total pore than clayey soils, so generally they have higher bulk densities.
- ◆ Bulk densities of sandy soils vary between 1.2 to 1.8 Mg m⁻³. Fine-textured soil, such as Clays, silty clays, or clay loams, have bulk densities between 1.0 and 1.6 Mg m⁻³.

Factors Affecting Bulk Density

- Bulk density is an indirect measure of pore space and is affected primarily by texture and structure.
- As aggregation and clay content increase, bulk density decreases .
- Tillage operations do not affect texture, but they do alter structure (soil particle aggregation). Primary tillage operations, such as plowing, generally decrease bulk density and increase pore space, which is beneficial. Secondary tillage (cultivation) generally increases bulk density and decreases pore space.
- The compaction resulting from cultivation can be detrimental to plant growth.

- Cropped soils generally have higher bulk densities than uncropped soils.
- The movement of machinery over the field forces solid particles into spaces once occupied by water or air, resulting in less pore space and increased bulk density.

Clayey	Loamy	Sandy
• Well-aggregated	• Moderately aggregated	• Nonaggregated
• High organic	• Moderate organic	• Low organic matter
• Matter content	• matter content	• content

- The density of water is 1.0 Mg m^{-3} and mineral soils are usually heavier than water. However, organic soils generally have a bulk density less than water.
- As the organic matter content of mineral soils increases, the bulk density decreases. Manure additions in large amounts tend to lower the surface bulk density of mineral soils because of the addition of low bulk density material and the consequent promotion of soil aggregation.
- Soil bulk density increases with soil depth primarily because of less organic matter and decreased aggregation.

Particle Density (Dp)

Soil particle density is a measure of the mass per unit volume of the soil solids only.

- Texture and structure do not affect particle density.
- However, organic matter, which is a soil solid, readily influences particle density. Organic matter weighs much less per unit volume than soil minerals.
- Soils high in organic matter have lower particle densities than soils similar in texture that are low in organic matter.
- Soil particle density generally increases with soil depth because of the concurrent decrease in organic matter
- Particle density varies with the type of soil minerals present as well as the amount of organic matter.
- The particle density of most mineral soils is in the range of 2.60 to 2.75 Mg m^{-3} . Particle density is used in the calculation of pore space and bulk density on a coarse fragment free basis.

- When unknown, particle density of mineral soils is assumed to be 2.65 Mg m^{-3} . Generally quartz, feldspars, and colloidal silicates dominate the mineral fraction of soils.
- The particle density of these minerals averages about 2.65 Mg m^{-3} . When large amounts of heavy minerals, such as hornblende or magnetite, are present, the soil particle density is greater than 2.65 Mg m^{-3} .
- Soils formed in volcanic parent materials, such as pumice or ash, generally have particle densities less than 2.65 Mg m^{-3} .
- Soil particle density is determined using Archimede's Principle.

POROSITY OF SOIL

Pore-spaces (also called voids) in a soil consists of that portion of the soil volume not occupied by solids, either mineral or organic. The pore-space under field conditions, are occupied at all times by air and water. Pore-spaces directly control the amount of water and air in the soil and indirectly influence the plant growth and crop production. In general there are broadly two types of pores in soils (i) Macro pores and (ii) Micro or capillary pores.

(i) Macro pores. Dimension is larger than about 0.08 mm. Large-sized pores are referred to as macro-pores which allow air and water movement readily. Sands and sandy soils have a large number of macro-pores. It is found in between the granules.

(ii) Micro or Capillary pores. Dimension is smaller than about 0.08 mm. Smaller sized pores are generally referred to as a micro or capillary pores in which movement of air and water is restricted to some extent. Clays and clayey soils have a greater number of micro or capillary pores. It has got more importance in the plant growth relationship. It is found within the granules.

Biopores

Macropores created by roots, earthworms and other organisms constitute a very important type of pores termed biopores.

Porosity refers to the percentage of soil volume occupied by pore spaces. Size of individual pores, rather than total pore space in a soil, is more significant in its plant growth relationship. For optimum growth of the plant, the existence of approximately equal amount of macro and micro-pores which influence aeration, permeability, drainage and water retention favourably is essential. Porosity of a soil can be easily changed.

Factors Affecting Porosity of Soil

(i) Soil structure: A soil having granular and crumb structure contains more pore spaces than that of prismatic and platy soil structure. So well aggregated soil structure has greater pore space as compared to structureless or single grain soil.

(ii) Soil texture: In Sandy soils the total pore space is small whereas in fine textured clay and clay loam soils total pore space is high and there is a possibility of more granulation in clay soils.

(iii) Arrangement of soil particles: When the sphere like particles are arranged in columnar form (*i.e.* one after another on the surface forming column like shape) it gives the most open packing system resulting very low amount of pore spaces. When such particles are arranged in the pyramidal form it gives the most close packing system resulting high amount of pore spaces.

(iv) Organic matter: Soil containing high organic matter possesses high porosity because of good aggregate formation.

(v) Macro-organisms: Macro-organisms like earthworms, rodents and insects etc. increase macro-pores in the soil.

(vi) Depth of soil: With the increase in depth of soil, the porosity will decrease because of compactness in the sub-soil.

(vii) Cropping: Intensive crop cultivation tends to lower the porosity of soil as compared to fallow soils. The decrease in porosity may be due to reduction in organic matter content.

(vi) Depth of soil: With the increase in depth of soil, the porosity will decrease because of compactness in the sub-soil.

(vii) Cropping: Intensive crop cultivation tends to lower the porosity of soil as compared to fallow soils. The decrease in porosity may be due to reduction in organic matter content.

(viii) Puddling: Due to puddling under sufficient soil moisture, the soil surface layer is made dense and compact. Eventually, the porosity of this surface soil is reduced by the infiltration of muddy surface materials.

Relationship between Porosity and Densities of Soil

$$\% \text{ solid space} = \frac{\text{Bulk density}}{\text{Particle density}} \times 100$$

Since, % pore space + % solid space = 100

$$\% \text{ pore space} = 100 - \% \text{ solid space}$$

$$\% \text{ pore space} = 100 - \frac{\text{Bulk density}}{\text{Particle density}} \times 100$$

$$\% \text{ pore space} = \left(100 - \frac{\text{Bulk density}}{\text{Particle density}} \times 100 \right)$$

15. SOIL COLOUR

♣ Importance of soil colour, kinds of soil colour

♣ Determination of soil colour

♣ Munsell colour chart

Soil colour is the most obvious soil property and gives an immediate indication of the condition of the soil system.

Importance of soil colour

- Soil colour provides an indirect measure of important soil characteristics including drainage, aeration, organic matter content and fertility.
- However, in general, soil colour is controlled by soil forming processes. Organic matter is brown/black and is found predominantly in the A-horizon of the soil profile.
- Iron compounds, in various states of oxidation and reduction are major colouring agents of subsoil horizons. The colour of soil containing iron oxides varies from **red and rust brown to yellow** depending on the degree of hydration. **Reduced iron** normally displays a **green-blue tinge**.
- Soils in anaerobic conditions, such as those in **poorly drained** depressions, will normally have **dull, grey B-horizons**. Alternatively, **aerobic soils** have **bright reddish-brownish colours**

Kinds of Soil Colour

Soil colour is inherited from its parent material and that is referred to as *litho chromic*, e.g. red soils developed from red sandstone. Besides soil colour also develops during soil formation through different soil forming processes and that is referred to as *acquired* or *pedochromic* colour e.g. red soils developed from granite gneiss or schist.

Factors Affecting Soil colour

There are various factors or soil constituents that influence the soil colour which are as follows:

(i) Organic matter: Soils containing high amount of organic matter show the colour variation from black to dark brown.

(ii) Iron compounds: Soils containing higher amount of iron compounds generally impart red, brown and yellow tinge colour.

(iii) Silica, lime and other salts: Sometimes soils contain either large amounts of silica and lime or both. Due to presence of such materials in the soil the colour of the soil appears like *white* or light coloured.

(iv) Mixture of organic matter and iron oxides:

Very often soils contain a certain amount of organic matter and iron oxides. As a result of their existence in soil, the most common soil colour is *brown*.

(v) Alternate wetting and drying condition:

- ◆ During monsoon period due to heavy rain the reduction of soil occurs and during dry period the oxidation of soil also takes place.
- ◆ Due to development of such alternating oxidation and reduction condition, the colour of soil in different horizons of the soil profile is *variegated* or *mottled*.
- ◆ This mottled colour is due to residual products of this process especially iron and manganese compounds.

(vi) Oxidation-reduction conditions:

When soils are waterlogged for a longer period, the permanent reduced condition will develop. The presence of ferrous compounds resulting from the reducing condition in waterlogged soils impart *bluish* and *greenish* colour.

Therefore, it may be concluded that the soil colour is indirectly indicative of many other important soil properties. Besides soil colour directly modify the soil temperature *e.g.* dark coloured soils absorb more heat than light coloured soils.

Determination of soil colour

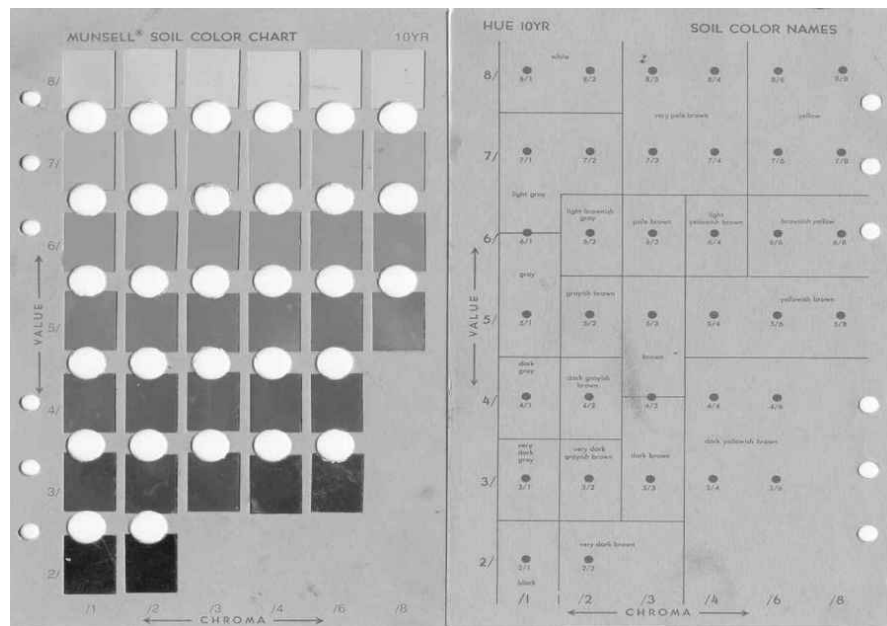
- ◆ The soil colours are best determined by the comparison with the **Munsell colour chart**.
- ◆ This colour chart is commonly used for this purpose. The colour of the soil is a result of the light reflected from the soil. Soil colour notation is divided into three parts.

- Hue** - it denotes the dominant spectral colour (red, yellow, blue and green).
- Value** - it denotes the lightness or darkness of a colour (the amount of reflected light).
- Chroma** - it represents the purity of the colour (strength of the colour).

The Munsell colour notations are systematic, numerical with letter designations of each of these three variables (Hue, value and chroma). For example, the numerical notation 2.5 YR 5/6 suggests a hue of 2.5 YR, value of 5 and chroma of 6. The equivalent or parallel soil colour name for this Munsell notation is 'red'.



Munsell Colour Chart



16. SOIL CONSISTENCE AND PLASTICITY

♣ Cohesion and Adhesion

♣ Plasticity - Atterberg's constant

♣ Upper and lower plastic limits, plasticity number

Soil consistence is defined as “the resistance of a soil at various moisture contents to mechanical stresses or manipulations”.

Soil consistency is the strength with which soil materials are held together or the resistance of soils to deformation and rupture.

It combines both the „**cohesive**“ and „**adhesive**“ forces, which determine the ease with which a soil can be reshaped or ruptures.

Adhesion	Cohesion
Molecular attraction that holds the surfaces of two substances (eg. Water and soil particles) in contact	Force holding a solid or liquid together, owing to attraction between like molecules. Decreases with rise in temperature

Soil consistence is described at three moisture levels namely „wet“, „moist“ and „dry“.

1. **Wet soils:** Consistency is denoted by terms stickiness and plasticity

Stickiness is grouped into four categories namely i) non sticky, ii) slightly sticky, iii) sticky and iv) very sticky

Plasticity of a soil is its capacity to be moulded (to change its shape depending on stress) and to retain the shape even when the stress is removed. Soils containing more than about 15% clay exhibit plasticity – pliability and the capacity of being molded. There are four degrees in plasticity namely i) non plastic, ii) slightly plastic, iii) plastic and iv) very plastic.

2. Moist soil:

Moist soil with least coherence adheres very strongly and resists crushing between the thumb and forefinger. The different categories are

- Loose-non coherent
- Very friable - coherent, but very easily crushed

- iii. Friable - easily crushed
- iv. Firm - crushable with moderate pressure
- v. Very firm - crushable only under strong pressure
- vi. Extremely firm - completely resistant to crushing. (type and amount of clay and humus influence this consistency)

3. Dry soil:

In the absence of moisture, the degree of resistance is related to the attraction of particles for each other. The different categories are

- i) Loose - non coherent
- ii) Soft - breaks with slight pressure and becomes powder
- iii) Slightly hard - break under moderate pressure
- iv) Hard - breaks with difficulty with pressure
- v) Very hard - very resistant to pressure
- vi) Extremely hard - extreme resistance and cannot be broken.

Atterberg's Constants

Atterberg (1912) studied plasticity from the point of view of moisture range over which plasticity range is maintained. The constants are also known as Atterberg's limits. There are three limits namely i) shrinkage limit or lower plastic limit, ii) plastic limit and iii) liquid limit or upper plastic limit. From the upper and lower limits, plasticity number or index is calculated. Plasticity number or index is an indirect measure of the force required to mould the soil.

i) Shrinkage limit (or) lower plastic limit (SL)

It is the soil moisture content below which the soil becomes friable. The maximum water content at which a reduction in moisture will not cause a decrease in the volume of the soils. This defines the arbitrary limit between solid and semi-solid states.

ii) Plastic limit (PL)

Moisture content at which a soil cannot be deformed without cracking. Water content corresponding to an arbitrary limit between the plastic and semisolid states. It is the moisture content at which the soil begins to exhibit plasticity. Soil cannot be deformed without cracking. Soils should not be ploughed at moisture contents above the plastic limit

iii) Liquid limit (or) upper plastic limit (LL)

The moisture content at which soil ceases to be plastic, becomes semi-fluid and tends to flow like a liquid

iv) Plasticity number

The difference between the moisture content (consistency) of upper and lower plastic limits.

Factors affecting Atterberg's constants

1. **Clay content:** Plasticity is a function of finer soil fractions and moisture content. Because of the plate like shape of clay particles and the lubricating effect of water, the fine soil fractions tend to slide over each other. High clay content increases the moisture contents of different plastic limits and increases the plasticity index / number. (Higher clay content requires more water to exhibit plasticity because of higher surface area)
2. **Nature of clay minerals:** Quartz and feldspars are non plastic. Kaolinite, illite, talc, muscovite, biotite, vermiculite, montmorillonite clays are plastic.
3. **Nature of exchangeable cations:** Sodium saturated soils have lower plastic limits than potassium, calcium and magnesium saturated soils.
4. **Organic matter content:** Organic matter favours plasticity.

Sticky point:

The moisture content at which the attractive power of the soil for water is satisfied. The sticky point moisture percentage is near the liquid limit (slightly higher for less plastic soils and slightly lower for more plastic soils).

Swelling and shrinkage:

Some soils develop cracks - sometimes few feet deep – when dried and the cracks closes when wet. Eg. Black soils. This is because of the penetration of water molecules between clay crystals. This property is observed in smectite type clays (eg. Montmorillonite) which allows water molecules to penetrate between two layers of clay crystals and expands itself – causing swelling of soil. When moisture evaporates, the expanded clay crystals shrink, reducing their volume and thereby causing cracks in soil.

17. SOIL WATER

♣ Importance, classification, measurement of soil moisture

♣ Soil water potentials

♣ Methods of expression and pF scale

♣ Soil moisture constants

Soil water

Water contained in soil is called soil moisture. The water is held within the soil pores. Soil water is an important component of the soil which influences soil organisms and plant growth. Unlike open source water like ponds and lakes, soil water exists in different forms. Some forms are available to plants while others are not. Soil water with dissolved salts is called soil solution. Soil solution supplies nutrients to growing plants.

Importance of Soil Water

- ❖ Water is essential for life including soil organisms and plants.
- ❖ Soil water serves as a solvent and carrier of food nutrients for plant growth.
- ❖ Yield of crop is first determined by the amount of water available in soil followed by the availability of other food nutrients.
- ❖ Soil water itself is a nutrient.
- ❖ Soil water regulates soil temperature
- ❖ Soil water is essential for soil forming processes and weathering
- ❖ Microorganisms require water for their metabolic activities
- ❖ Soil water helps in chemical and biological activities of soil
- ❖ It is a principal constituent of the growing plant
- ❖ Water is essential for photosynthesis

Classification of soil water

Soil water has been classified from a physical and biological point of view as Physical classification of soil water, and biological classification of soil water.

Physical classification of soil water

- i) Gravitational water ii) Capillary water and iii) Hygroscopic water

1. Gravitational water :

- ◆ Gravitational water occupies the larger soil pores (macro pores) and moves down readily under the force of gravity.

- ◆ Water in excess of the field capacity is termed gravitational water. Gravitational water is of no use to plants because it occupies the larger pores.
- ◆ It reduces aeration in the soil. Thus, its removal from soil is a requisite for optimum plant growth. **Soil moisture tension at gravitational state is zero or less than 1/3 atmosphere.**

2. Capillary water:

- ◆ Capillary water is held in the capillary pores (micro pores). Capillary water is retained on the soil particles by surface forces.
- ◆ It is held so strongly that gravity cannot remove it from the soil particles. The molecules of capillary water are free and mobile and are present in a liquid state.
- ◆ Due to this reason, it evaporates easily at ordinary temperature though it is held firmly by the soil particle; plant roots are able to absorb it. Capillary water is, therefore, known as **available water. The capillary water is held between 1/3 and 31 atmosphere pressure.**

3. Hygroscopic water:

- ◆ The water that held tightly on the surface of soil colloidal particle is known as hygroscopic water.
- ◆ It is essentially non-liquid and moves primarily in the vapour form.
- ◆ Hygroscopic water held so tightly (**31 to 10000 atmosphere**) by soil particles that plants can not absorb it. Some microorganism may utilize hygroscopic water.

B. Biological Classification of Soil Water

There is a definite relationship between moisture retention and its utilization by plants. This classification based on the availability of water to the plant. Soil moisture can be divided into three parts.

i. Available water: The water which lies between wilting coefficient and field capacity. It is obtained by subtracting wilting coefficient from moisture equivalent.

ii. Unavailable water: This includes the whole of the hygroscopic water plus a part of the capillary water below the wilting point.

iii. Super available or superfluous water:

The water beyond the field capacity stage is said to be super available. It includes gravitational water plus a part of the capillary water removed from larger interstices. This water

is unavailable for the use of plants. The presence of super-available water in a soil for any extended period is harmful to plant growth because of the lack of air.

Measuring Soil Moisture

- i) Direct measurement of moisture content
- ii) Measurement of soil moisture potential (tension or suction)

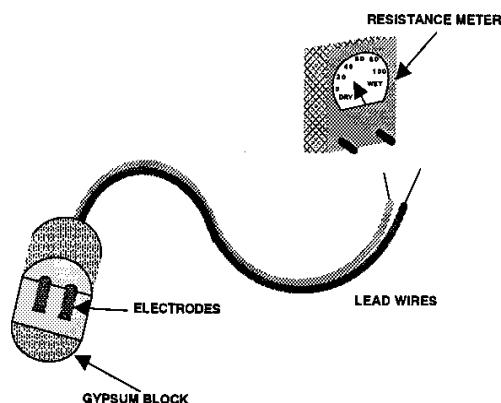
Measuring soil moisture content in laboratory

1. Gravimetric method:

- ◆ This consists of obtaining a moist sample, drying it in an oven at 105°C until it loses no more weight and then determining the percentage of moisture.
- ◆ The gravimetric method is time consuming and involves laborious processes of sampling, weighing and drying in laboratory.

2. Electrical conductivity method:

- ◆ This method is based upon the changes in electrical conductivity with changes in soil moisture.
- ◆ **Gypsum blocks** inside of which two electrodes at a definite distance are apart are used in this method. These blocks require previous calibration for uniformity.
- ◆ The blocks are buried in the soil at desired depths and the conductivity across the electrodes is measured with a modified Wheatstone bridge.
- ◆ These electrical measurements are affected by salt concentration in the soil solution and are not very helpful in soils with high salt contents.



3. Pressure plate and pressure membrane apparatus:

- ◆ The pressure plate apparatus is used for moisture measurements up to 1 atm. and pressure membrane apparatus is used for moisture measurements up to 20 atm.



4. Neutron scattering method:

- ◆ Fast neutrons are emitted by neutron emitters like radium-beryllium or americium-beryllium.

- ♦ The fast moving neutrons interact with hydrogen atoms and get slowed down. The number of slow moving neutrons returned back after colliding with hydrogen is proportional to the quantity of water present in soil as water is considered to be the sole source of hydrogen atoms.
- ♦ The slow moving neutrons are detected by proportional or scintillation counters and interpreted in terms of moisture percentage at different depths.



5. Profile probe and theta probe:

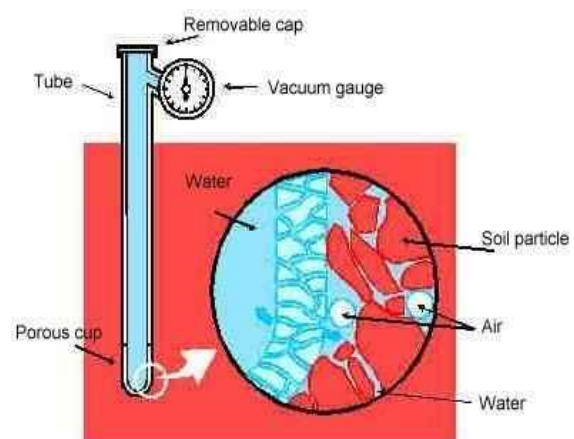
This new technology uses electro-magnetic radiation. The emitted radiation is intercepted by soil water and the radiation reflected is detected and interpreted in terms of per cent moisture content in different layers.



6. Measuring soil moisture potential *in-situ* (field):

- Suction method or equilibrium tension method:

- Field tensiometers measure the tension with which water is held in the soils. They are used in determining the need for irrigation.
- The tensiometer is a porous cup attached to a glass tube, which is connected to a mercury monometer.
- The tube and cup are filled with water and cup inserted in the soil. The water flows through the porous cup into the soil until equilibrium is established.
- These tension readings in monometer, expressed in terms of cm or atmosphere, measures the tension or suction of the soil.
- If the soil is dry, water moves through the porous cup, setting up a negative tension (or greater is the suction).
- The tensiometers are more useful in sandy soils than in fine textured soils and can be used only up to a tension of 1 bar.
- At tensions above one bar the water column is broken and air enters the ceramic cup. Once the air gets entrapped in the tensiometer, the reliability of readings is questionable.



18. SOIL WATER POTENTIALS

♣ **Matric, Osmotic and gravitational potential**

♣ **Methods of expression and pF scale**

♣ **Soil moisture constants**

- The retention and movement of water in soils, its uptake and translocation in plants and its loss to the atmosphere are all energy related phenomenon.
- The more strongly water is held in the soil, the greater is the heat (energy) required.
- In other words, if water is to be removed from a moist soil, work has to be done against adsorptive forces.
- Conversely, when water is adsorbed by the soil, a negative amount of work is done. The movement is from a zone where the free energy of water is high (standing water table) to one where the free energy is low (a dry soil). This is called soil water energy concept.

Free energy of soil solids for water is affected by:

- i) **Matric (solid) force** i.e., the attraction of the soil solids for water (adsorption) which markedly reduces the free energy (movement) of the adsorbed water molecules.
- ii) **Osmotic force** i.e., the attraction of ions and other solutes for water to reduce the free energy of soil solution. Matric and Osmotic potentials are negative and reduce the free energy level of the soil water. These negative potentials are referred as suction or tension.
- iii) **Force of gravity:** This acts on soil water, the attraction is towards the earth's center, which tends to pull the water down ward. This force is always positive. The difference between the energy states of soil water and pure free water is known as soil water potential.

Total water potential (Pt) is the sum of **gravitational potential (Pg)**, **matric potential (Pm)** and the **Osmotic potential** or **solute potential (Po)**.

$$P_t = P_g + P_m + P_o$$

Potential represents the difference in free energy levels of pure water and of soil water. The soil water is affected by the force of gravity, presence of soil solid (matric) and of solutes.

Methods of expressing soil water potential

There are two units to express differences in energy levels of soil water.

- i) **pF Scale:** The free energy is measured in terms of the height of a column of water required to produce necessary suction or pressure difference at a particular soil moisture level. The pF, therefore, represents the logarithm (to the base 10) of the height of water column (cm) to give the necessary suction.
- ii) **Atmospheres or Bars:** It is another common mean of expressing suction. Atmosphere is the average air pressure at sea level.
 - ◆ If the suction is very low as occurs in the case of a wet soil containing the maximum amount of water that it can hold, the pressure difference is of the order of about 0.01 atmosphere or 1 pF equivalent to a column of water 10 cm in height.
 - ◆ Similarly, if the pressure difference is 0.1 atmosphere the pF will be 20. Soil moisture constants can be expressed in term of pF values.
 - ◆ A soil that is saturated with water has pF : 0 while an oven dry soil has a pF : 7.
 - ◆ pF values are related to pressure and height of column of water

Height of water column	Pressure (atmosphere)	pF value
1	0.001	0
10	0.01	1
10^2	0.1	2
10^3	1.0	3
10^4	10.0	4
10^5	100.0	5
10^6	1000.0	6
10^7	10,000.0	7

Soil Moisture Constants

- Earlier classification divided soil water into gravitational, capillary and hygroscopic water. The hygroscopic and capillary waters are in equilibrium with the soil under given condition.
- The hygroscopic coefficient and the maximum capillary capacity are the two equilibrium points when the soil contains the maximum amount of hygroscopic and capillary waters, respectively.
- The amount of water that a soil contains at each of these equilibrium points is known as soil moisture constant.
- The soil moisture constant, therefore, represents definite soil moisture relationship and retention of soil moisture in the field.

- The three classes of water (gravitational, capillary and hygroscopic) are however very broad and do not represent accurately the soil - water relationships that exists under field conditions.
- Though the maximum capillary capacity represents the maximum amount of capillary water that a soil holds, the whole of capillary water is not available for the use of the plants.
- A part of it, at its lower limit approaching the hygroscopic coefficient is not utilized by the plants. Similarly a part of the capillary water at its upper limit is also not available for the use of plants.
- Hence two more soil constants, viz., field capacity and wilting coefficient have been introduced to express the soil-plant-water relationships as it is found to exist under field conditions.

1. Field capacity:

- ◆ Assume that water is applied to the surface of a soil. With the downward movement of water all macro and micro pores are filled up.
- ◆ The soil is said to be saturated with respect to water and is at maximum water holding capacity or maximum retentive capacity.
- ◆ It is the amount of water held in the soil when all pores are filled.
- ◆ Sometimes, after application of water in the soil all the gravitational water is drained away, and then the wet soil is almost uniformly moist.
- ◆ The amount of water held by the soil at this stage is known as the field capacity or normal moisture capacity of that soil.
- ◆ It is the capacity of the soil to retain water against the downward pull of the force of gravity. At this stage only micropores or capillary pores are filled with water and plants absorb water for their use.
- ◆ At field capacity water is held with a **force of 1/3 atmosphere**. Water at field capacity is **readily available** to plants and microorganism.

2. Wilting coefficient:

- ◆ As the moisture content falls, a point is reached when the water is so firmly held by the soil particles that plant roots are unable to draw it.
- ◆ **The plant begins to wilt.** At this stage even if the plant is kept in a saturated atmosphere it does not regain its turgidity and wilts **unless water is applied to the soil**.

- ◆ The stage at which this occurs is termed the **Wilting point** and the percentage amount of water held by the soil at this stage is known as the **Wilting Coefficient**.
- ◆ It represents the point at which the soil is unable to supply water to the plant. Water at wilting coefficient is held with a **force of 15 atmospheres**.

3. Hygroscopic coefficient:

- ◆ The hygroscopic coefficient is the **maximum amount of hygroscopic water absorbed** by 100 g of dry soil under standard conditions of humidity (50% relative humidity) and temperature (15°C).
- ◆ This tension is equal to a **force of 31 atmospheres**. Water at this tension is **not available to plant** but may be available to certain bacteria.

4. Available water capacity:

- ◆ The amount of water required to apply to a soil at the wilting point to reach the field capacity is called the "available" water.
- ◆ The water supplying power of soils is related to the amount of available water a soil can hold.
- ◆ The available water is the difference in the amount of water at field capacity (- 0.3 bar) and the amount of water at the permanent wilting point (- 15 bars).

5. Maximum water holding capacity:

- ◆ It is also known as **maximum retentive capacity**. It is the amount of moisture in a soil when its pore spaces both micro and macro capillary are completely filled with water.
- ◆ It is a rough measure of total pore space of soil. Soil moisture tension is very low between **1/100th to 1/1000th of an atmosphere** or pF 1 to 0.

6. Sticky point moisture:

- ◆ It represents the moisture content of soil at which it no longer sticks to a foreign object.
- ◆ The sticky point represents the maximum moisture content at which a soil remains friable.
- ◆ Sticky point moisture values vary nearly approximate to the moisture equivalent of soils.
- ◆ Summary of the soil moisture constants, type of water and force with which it held is given in following table.

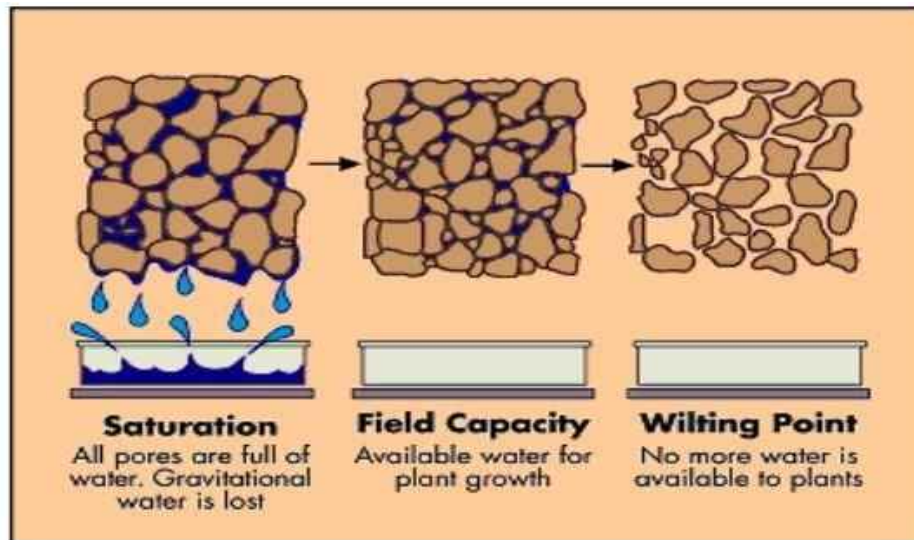
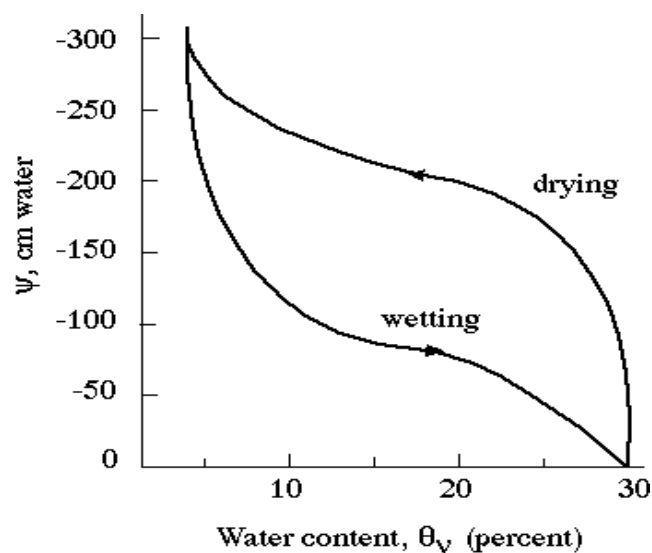


Fig: Soil moisture constants

Hysterisis

- The moisture content at different tensions during wetting of soil varies from the moisture content at same tensions during drying. This effect is called as **hysterisis**.
- This is due to the presence of capillary and non capillary pores and the effect is also called ink-bottle effect.
- The moisture content is always low during sorption (wetting) and high during desorption (drying).



Hysterisis Curve in a normal soil

19. SOIL WATER MOVEMENT

♣ Saturated Flow and Unsaturated Flow

♣ Entry of water into the soil

Saturated flow

- This flow occurs when the soil pores are completely filled with water. This water moves at water potentials larger than – 33 kPa.
- Saturated flow is water flow caused by gravity's pull. It begins with *infiltration*, which is water movement into soil when rain or irrigation water is on the soil surface. When the soil profile is wetted, the movement of more water flowing through the wetted soil is termed *percolation*.

Hydraulic conductivity can be expressed mathematically as

$$V = kf$$

Where,

V = Total volume of water moved per unit time

f = Water moving force

k = Hydraulic conductivity of soil

Factors affecting movement of water

1. Texture, 2. Structure, 3. Amount of organic matter, 4. Depth of soil to hard pan,
5. Amount of water in the soil, 6. temperature and 7. Pressure

Vertical water flow

The vertical water flow rate through soil is given by *Darcy's law*. The law states that the rate of flow of liquid or flux through a porous medium is proportional to the hydraulic gradient in the direction of flow of the liquid.

Unsaturated flow

Unsaturated flow occurs along soil surfaces, not through large pores. Unsaturated flow is driven by matric forces that are much stronger than gravity.

Gravity is not sufficiently strong to exert a significant influence on unsaturated flow because when unsaturated, much of the soil water is affixed to solid surfaces. Unsaturated flow is slow.

- Even though the driving force is usually greater than for saturated flow, the resistance to flow is enormous.

- Water will flow toward a lower (more negative) potential regardless of direction. In other words it will flow towards: drier medium small pores and finer texture .

Entry of Water into Soil

1. Infiltration:

Infiltration refers to the downward entry or movement of water into the soil surface

- ❖ It is a surface characteristic and hence primarily influenced by the condition of the surface soil.
- ❖ Soil surface with vegetative cover has more infiltration rate than bare soil
- ❖ Warm soils absorb more water than colder ones
- ❖ Coarse surface texture, granular structure and high organic matter content in surface soil, all help to increase infiltration
- ❖ Infiltration rate is comparatively lower in wet soils than dry soils

2. Percolation:

The movement of water through a column of soil is called percolation. It is important for two reasons.

- ❖ This is the only source of recharge of ground water which can be used through wells for irrigation
- ❖ Percolating waters carry plant nutrients down and often out of reach of plant roots (leaching)
- ❖ In dry region it is negligible and under high rainfall it is high
- ❖ Sandy soils have greater percolation than clayey soil
- ❖ Vegetation and high water table reduce the percolation loss

3. Hydraulic conductivity

- ❖ Symbolically represented as K , is a property of vascular plants, soil or rock, that describes the ease with which water can move through pore spaces or fractures.
- ❖ It depends on the intrinsic permeability of the material and on the degree of saturation. The term hydraulic conductivity refers to the readiness with which a soil transmits fluids through it

4. Permeability:

It indicates the relative ease of movement of water with in the soil. The characteristics that determine how fast air and water move through the soil is known as permeability. Usually,

the finer the soil texture, the slower the permeability and the more permeable the soil, the greater the seepage. The size of the soil pores is of great importance with regard to the rate of **infiltration** (movement of water into the soil) and to the rate of **percolation** (movement of water through the soil). Pore size and the number of pores closely relate to soil texture and structure, and also influence soil permeability.

5. Drainage:

It is the natural or artificial removal of surface and sub-surface water from a given area. It refers to the soil's natural ability to allow water to pass through it. Dense / heavy textured soil will hold water, while loose / light textured soil will allow water to pass through quickly. Soil drainage may determine which types of plants grow well in it. Many agricultural soils need drainage to improve production or to manage water supply.

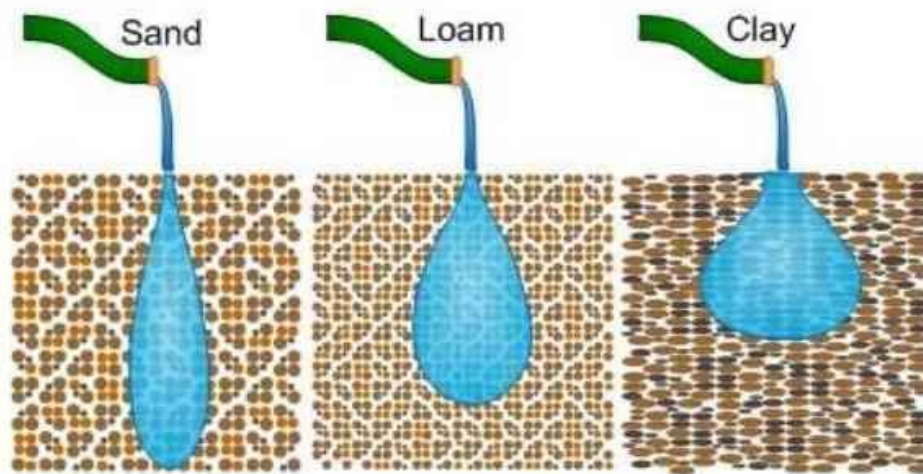


Fig: Response of water to different soils

20. SOIL AIR

♣ Composition

♣ Gaseous exchange

♣ Effect on crop growth

Composition of soil air :

Soil air/gases are the gases found in the air space between soil components. The primary natural soil gases include nitrogen, carbon dioxide and oxygen. The oxygen is critical because it allows for respiration of both plant roots and soil organisms.

Air spaces or pore spaces (voids) in a soil consists of that portion of the soil volume not occupied by soil solids, either mineral or organic. During rainy season, water replaces air from the soil pore spaces, but as soon as water disappeared by downward movement, surface evaporation and transpiration etc., air gradually replaces the water as it is lost from the pore spaces. Soil air influence the growth of the plants as well as the activity of different beneficial micro organisms present in soil. Soil air is a continuation of the atmospheric air. Unlike the other components, it is constant state of motion from the soil pores into the atmosphere and from the atmosphere into the pore space. This constant movement or circulation of air in the soil mass resulting in the renewal of its component gases is known as soil aeration.

Composition of Soil Air : The soil air contains a number of gases of which nitrogen, oxygen, carbon dioxide and water vapour are the most important. Soil air and atmospheric air differ in their compositions. Soil air contains a much greater proportion of carbon dioxide and a lesser amount of oxygen than atmospheric air. At the same time, soil air contains a far great amount of water vapour than atmospheric air. The amount of nitrogen in soil air is almost the same as in the atmosphere. The average relative composition of the soil and atm. air is as follows:

Composition (%)	O ₂	CO ₂	N
Atm. air	20.97	0.03	79.9
Soil air	20.60	0.30	79.2

Factors Affecting the Composition of Soil Air :

1. Nature and condition of soil: The quantity of oxygen in soil air is less than that in atmospheric air. The amount of oxygen also depends upon the soil depth. The oxygen content of the air in lower layer is usually less than that of the surface soil. This is possibly due to more readily diffusion of the oxygen from the atmosphere into the surface soil than in the subsoil.

Light texture soil or sandy soil contains much higher percentage than heavy soil. The concentration of CO₂ is usually greater in subsoil probably due to more sluggish aeration in lower layer than in the surface soil.

2. Type of crop: Plant roots require oxygen, which they take from the soil air and deplete the concentration of oxygen in the soil air. Soils on which crops are grown contain more CO₂ than fallow lands. The amount of CO₂ is usually much greater near the roots of plants than further away. It may be due to respiration by roots.

3. Microbial activity: The microorganisms in soil require oxygen for respiration and they take it from the soil air and thus deplete its concentration in the soil air. Decomposition of organic matter produces CO₂ because of increased microbial activity. Hence, soils rich in organic matter contain higher percentage of CO₂.

4. Seasonal variation: The quantity of oxygen is usually higher in dry season than during the monsoon. Because soils are normally drier during the summer months, opportunity for gaseous exchange is greater during this period. This results in relatively high O₂ and low CO₂ levels. Temperature also influences the CO₂ content in the soil air. High temperature during summer season encourages microorganism activity which results in higher production of CO₂.

Importance of soil air / soil aeration and its effect on crop growth:

1. Plant and root growth: Soil aeration is an important factor in the normal growth of plants. The supply of oxygen to roots in adequate quantities and the removal of CO₂ from the soil atmosphere are very essential for healthy plant growth. When the supply of oxygen is inadequate, the plant growth either retards or ceases completely as the accumulated CO₂ hampers the growth of plant roots, most noticeable on the root crops.

2. Soil micro and macroorganism population and activity: Animals that live in the soil (like worms) and many types of bacteria that are important for other plants and animals need air for respiration and metabolism. Some of the important microbial activities such as the decomposition of organic matter, nitrification, sulphur oxidation etc depend upon oxygen present in the soil air. The deficiency of air (oxygen) in soil slows down the rate of microbial activity.

3. Formation of toxic material: Poor aeration results in the development of toxin and other injurious substances such as ferrous oxide, H₂S gas, CO₂ gas etc in the soil. Heavy rainfall, excessive irrigation or puddling together with flooding brings about a reduced condition of the soil and subsequently reduction of elements like, Fe and Mn and slow rate or incomplete of decomposition of organic matter will take place in the soil leading to the formation of various

organic toxic acids like, lactic, butyric and citric acids etc. and also toxic concentrations of Fe and Mn are found in the anaerobic soil condition. Due to production of such toxic substances plant roots get injury and thereby lost their ability to absorb nutrients and water for their growth.

4. Water and nutrient absorption: A deficiency of oxygen has been found to check the nutrient and water absorption by plants. The energy of respiration is utilized in absorption of water and nutrients. Under poor aeration condition (this condition may arise when soil is water logged), plants exhibit water and nutrient deficiency. The availability of nutrient elements like phosphorus, iron and manganese are increased by poor aeration due to flooding while Cu and Zn show deficiency under poor aeration (waterlogged) conditions.

5. Development of plant diseases: Insufficient aeration of the soil leads to the development of diseases (eg.: wilt of gram and dieback of citrus and peach). Changes in aeration may modify the susceptibility of the plant, the virulence of the organism or both, so that disease incidence may vary with conditions of soil aeration. The effect of aeration has to do mainly with the sensitivity of the disease organism and not the plant. Cereals are moderately tolerant to poor aeration.

Gaseous exchange in soils :

The exchange of gases between the soil and the atmosphere is facilitated by two mechanisms

i) Mass flow: With every rain or irrigation, a part of the soil air moves out into the atmosphere as it is displaced by the incoming water. As and when moisture is lost by evaporation and transpiration, the atmospheric air enters the soil pores. The variations in soil temperature cause changes in the temperature of soil air. As the soil air gets heated during the day, it expands and the expanded air moves out into the atmosphere. On the other hand, when the soil begins to cool, the soil air contracts and the atmospheric air is drawn in.

ii) Diffusion: Most of the gaseous interchange in soils occurs by diffusion. Atmospheric and soil air contains a number of gases such as nitrogen, oxygen, carbon dioxide etc., each of which exerts its own partial pressure in proportion to its concentration.

The movement of each gas is regulated by the partial pressure under which it exists. If the partial pressure on one of the gases (i.e. carbon dioxide) is greater in the soil air than in the atmospheric air, it (CO_2) moves out into the atmosphere. On the other hand, partial pressure of oxygen is low in the soil air, as oxygen present in soil air is consumed as a result of biological activities. The oxygen present in the atmospheric air (partial pressure of O_2 is greater) therefore, diffuses into the soil air till equilibrium is established. Thus, diffusion allows extensive movement and continual change of gases between the soil air and the atmospheric air. Oxygen and carbon dioxide are the two important gases that take in diffusion

21. SOIL TEMPERATURE

❖ Source, amount and flow of heat in soil

❖ Soil temperature and crop growth

Sources of soil heat :

There are various sources of soil heat namely solar radiation, bio chemical reactions, conduction, precipitation, exposure, vegetation etc.

1. **Solar radiation:** Radiant energy from the sun is the power source that determines the thermal regime of the soil and the growth of Plants. Much of the radiation from the sun is dissipated and distributed by components of the earth's atmosphere as it passed downward towards the earth. The heat absorbed by the surface of the earth from the solar radiation is affected by several parameters like latitude, distribution of land and water, slope of the land etc., The southern sloped exposure of Landscape receive more Solar radiation per unit area and hence the temperature as well. The presence of Soil water also favours larger absorption of heat because of its high specific heat.
2. **Bio Chemical reaction:** Larger amount of heat is liberated during the various biochemical reactions in soil. Besides, decomposition of soil org. matter and other microbial processes Liberate heat in the soil and thus contributes to soil heat.
3. **Conduction:** The inner core of the earth is very hot and the conduction of the heat to the outer earth crust / soil body is very slow. Generally during night, the uppermost surface soil becomes cooler flows from the regions of sub soil to the region of surface soil (cooler soil layer)
4. **Precipitation:** During winter season precipitation increases soil heat because of the high specific heat of water.
5. **Exposure:** It is of significance in the middle latitude where the elevation is lower and it is of little importance in the tropics because of high elevation of the sun.
6. **Vegetation:** The insulating property of the Plant cover plays a significant role on lowering the soil heat. The major impact of vegetation are associated with (i) the “albedo effect” (absorptivity of soil), (ii) decreasing the depth of penetration of global radiation thro’ the crop canopy (iii) increasing the latent heat in evapo transpiration and (iv) decreasing the rate of heat loss from the soil thro’ its insulating influence.

Soil Temperature regimes:

- **Pergelic** – Mean annual soil temperature is $< 0^{\circ}\text{C}$
- **Cryic** – Mean annual soil temperature is 0 to 8°C with summer temperature $< 15^{\circ}\text{C}$
- **Iso** – When used as a prefix on the following temperature regimes, iso refers to soils in which the average soil temperature for the 3 warmest months differs by less than 5°C from the average temperature of 3 coldest months.

Frigid and Isofrigid	:	0°C to $< 8^{\circ}\text{C}$
Mesic and Isomesic	:	8°C to $< 15^{\circ}\text{C}$
Thermic and Isothermic	:	15°C to $< 22^{\circ}\text{C}$
Hyperthermic and Isohyperthermic	:	$> 22^{\circ}\text{C}$

Factors influencing soil temperature:

The soil temperature mainly depends on the factors that influence the supply of and its dissipation in soil environment.

- Composition of the soil:** The thermal regimes of the soil depend on its Volumetric composition, size and arrangement of soil particles and the interface relationship between the phases of soil. Soils containing much more mineral matter (sp. heat 0.10-0.20) get heated very easily than soils containing higher amount of org. matter (Sp. heat 0.44).
- Soil structure:** Open or close system of packing of soil have a significant influence on the pore spaces and therefore on the soil temperature as well. Soil having steroidal type of structure warm up more quickly because of better drainage.
- Soil texture:** A light textured sandy soil absorb heat very quickly than heavy textured clayey soils which carry a greater quantity of water hence warming up slowly.
- Soil moisture:** Most soils have high specific heat than the dry soils, consequently a moist soil has a lower soil temperature than a dry soil.
- Soil Colour:** Darker soils, due to the presence of humus,.....
- Vegetation:** Soils covered by Vegetation absorb less heat and thereby reduces the thermal regime of the soil.
- Irrigation and drainage:** Irrigation raises the humidity of soil air as well the atmospheric air near the soil surface thereby reduces the daily soil temp. Variations. Drainage favours warming up of the soil.
- Topography:** The ground slope influences the soil temperature largely wherein a ridged surface of a soil will be warmer than the levelled.
- Compactness of the soil:** Compactness of the soil increases the density and the thermal conductivity of the soil. Tillage creates surface mulch which reduces the heat flux from the surface to the sub surface layers.

- j) **Climate:** Soils in tropical climate are warmer than those in temperate climate.
- k) **Season:** Summer months in the northern hemisphere results in the peak of global radiation and hence maximum soil temperature. Temperature of the surface soil is always higher than the air temperature.

Some important heat constants / Thermal characteristics of soil are

- a) **Specific heat (c) :** Amount of heat required to raise the temperature of one gram of a substance by 1°C. Specific heat of dry soil (0.2 cal/g) is only one fifth that of water (1.0 cal/g)
- b) **Heat capacity/Thermal capacity(C):** It is the specific heat (c) of a material multiplied by its mass / density (P)

$$C = Pc$$
- c) **Thermal conductivity (K):** It is a measure of the ability of a substance to conduct heat. It is the amount of heat passing in unit time through a unit cross sectioned area of the soil under a unit temperature gradient.
- d) **Thermal diffusivity (E) :** The ratio of the thermal conductivity to its heat (thermal) capacity is called as thermal diffusivity

$$E = K/C$$

A high value of the thermal diffusivity implies a capability for rapid and considerable changes in soil temperature. Soil thermal diffusivity grew with sand content and bulk density while it got declined with clay and organic carbon content of the soils.

Impact of soil Temperature on Plant growth and nutrition:

Soil temperature has a profound influence on the soil fertility by changing the soil environment which in turn affect the plant growth.

- i) **Germination of seeds:** Germinating seeds require a specified optimum temperature. The Germination will fail or the rate will be slow at below the optimum temperature of 10-35° C and on the other hand the seeds will get injured if the soil temperature is high. The temperature
- ii) **Physical properties of Soil:** Soil temperature greatly influence the soil physical properties particularly the soil structure. Crumb / granular structured friable soils favour greater germination.
- iii) **Microbial activity:** The growth and activity of thermophilic and thermophobic microbes are greatly influenced by soil temperature (25-35° C) which in turn influence important microbial processes like mineralisation of nutrients, nitrogen fixation, pesticides degradation etc.,

- iv) **Decomposition of organic matter in soil:** The decomposition products vary with the rate of decomposition of soil org. matter which in turn depends on soil temperature vis – a- vis microbial activity. Higher soil temperature favours faster degradation of SOM and the release of nutrients thereby influencing the plant growth.
- v) **Absorption of water:** The absorption of water by Plant roots is a function of soil temperature and the root activity for which an optimum soil temperature is necessary.
- vi) **Availability of nutrients:** Soil temperature greatly influences the physic-chemical and chemical reactions in soil controlling the transformation and quail ability of nutrients to Plants.
- vii) **Root growth:** Low temperature encourage white succulent roots with little branching while high temperature encourage a browner, finer and much more branching root system. The temperature optima for root growth is 20-25° C for most crops which is often lower than for shoot growth.
- viii) **Plant diseases:** At low temperature, the soil contains many parasitic fungai which will grow actively and rapidly, leading to higher mortality of seedlings.

Management of soil temperature:

- i) **Water Management:** draining the excess water or irrigating a dry soil helps to maintain soil temperature favouring Plant growth.
- ii) **Soil colour:** Dark coloured surface soils (by addition of charcoal powder or organic matter) maintain a higher temperature while a light coloured surface soil (by addition of chalk or other light substances) shows low soil temperature.
- iii) **Providing mulches:** Mulching the surface soil will help to keep the soil warmer during winter and cooler during summer.
- iv) **Tillage:** A tilled surface soil is generally warmer during the day time because of its higher porosity and low thermal conductivity than a compacted untilled soil. Ridges will have fluctuation in soil temperature compared to furrows which will be relatively cooler.
- v) **Soil Compaction:** Loose soils have abrupt changes in soil temperature while a compacted soil will have a stabilised temperature due to its high thermal conductivity.

Technique of measuring the soil temperature

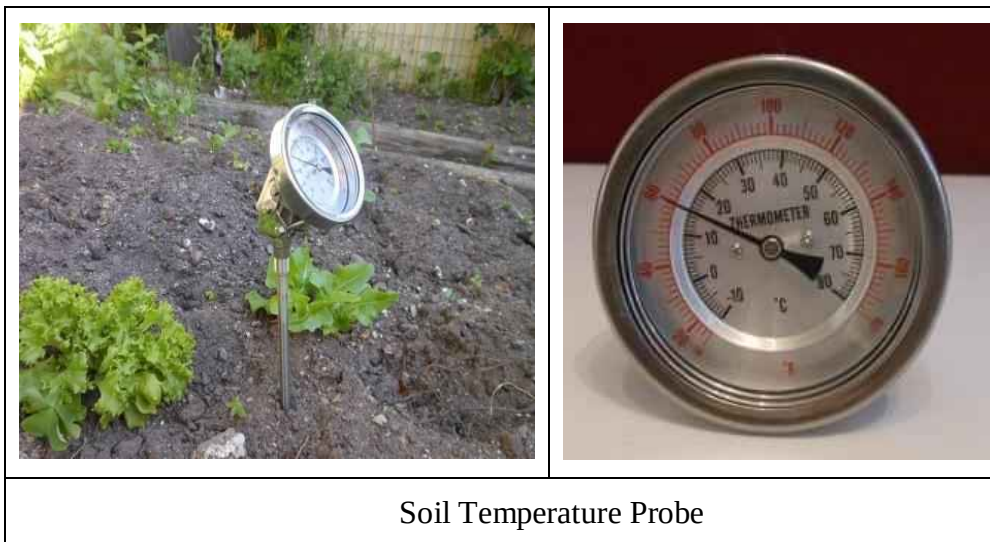
Soil temperature gauges or thermometers are the common way to take the measurements. There are special soil temperature gauges used by farmers and soil sample companies, but it can be easily measured simply by just the use of a soil thermometer.

- **Measure at the right depth:** If you are planting seeds or new plants, take your measurement at the recommended planting depth. Soil



readings for seed beds are done in 1 to 2 inches of soil. Sample at least 4 to 6 inches deep for transplants or for a mixed garden.

- **Make a pilot hole:** Use a screwdriver to make a pilot hole so that you don't break your thermometer by pushing it into hard soil. Insert the thermometer to the hilt, or maximum depth, and hold it for a minute. Do this for three consecutive days.
- **Follow directions:** Refer to your thermometer package for specific instructions. With most glass bulb thermometers, make sure it is firmly touching the soil, and allow a few minutes for the temperature to register.
- **Provide shade:** If the sun is bright, shade the thermometer with your hand to keep the reading accurate.
- **Multiple measurements:** Take a reading in the morning and late afternoon, then average the two numbers. If you're seeding a lawn, take readings on all four sides of your house, since some areas warm more quickly than others.



Soil Temperature Probe

Soil Temperature and its Importance.

The **soil** layer of the earth is critical in maintaining plant life, offering mechanical support, and supplying water and nutrients. ... Germination of various seeds requires different **soil temperature** ranges. For example, maize starts to germinate at **soil temperatures** from 7 to 10°C.

Soil functions as a major storage mechanism of heat, collecting energy throughout the day and releasing heat to the surface during the night. Over the course of a year, soil retains energy during warmer seasons and releases heat to the air throughout colder seasons.

Why is soil temperature important?

1. Soil temperature directly affects plant growth. In other words, nearly every crop slows down its growth when soil temperatures are below 9°C and above 50°C.

2. Germination of various seeds requires different soil temperature ranges. For example, maize starts to germinate at soil temperatures from 7 to 10°C.
3. Most of the soil organisms function best at an optimum soil temperature of 25 to 35°C
4. Nitrification requires a soil temperature of 32°C.
5. It also influences soil moisture content, aeration and availability of plant nutrients.

Is the ground warmer than the air ?

During sunny days, **ground** temperatures can be much **warmer than** the **air** temperature (10-40 F is not unusual), and on a cold, clear winter nights, when the **ground** radiates heat to space, the **ground** temperature can be 1-8 F cooler **than air** temperature

What is the temperature of the ground about 10 feet deep ?

In a cold winter day, the outside air **temperature** is 30 °F, but the **temperature of the ground 10 feet** down is around 50 °F. By putting pipes in the **ground**, we can exchange the heat from the **ground** to the house. A fluid is pumped through a closed loop of piping into the earth where it warms up.

Ideal Soil Temperatures for Planting

The perfect temperature for planting varies dependent upon the variety of vegetable or fruit. Planting before it is time can reduce fruit set, stunt plant growth and prevent or reduce seed germination. Plants such as tomatoes, cucumbers and snap peas benefit from soils at least 60 F (16 C.). Sweet corn, lima beans and some greens need 65 degrees F. (18 C.) Warmer temperatures into the 70s (20s C.) are required for watermelon, peppers, squash, and at the higher end, okra, cantaloupe and sweet potatoes. If you are in doubt, check your seed packet for ideal soil temperatures for planting. Most will list the month for your USDA zone.

Realistic Soil Temperatures :

Somewhere between the minimum soil temperature for plant growth and the optimum temperature is the realistic soil temperature. For instance, plants with higher temperature needs, such as okra, have an optimum temperature of 90 F. (32 C.). However, healthy growth can be achieved when they are transplanted into soils of 75 F. (24 C). This happy medium is suitable for beginning plant growth with the assumption that optimum temperatures will occur as the season progresses. Plants set out in cool zones will benefit from late transplanting and raised beds, where soil temperatures warm up more quickly than ground level planting

22. SOIL REACTION (pH)

♣ Definition, pH scale and Factors affecting soil pH

♣ Buffering capacity and its importance

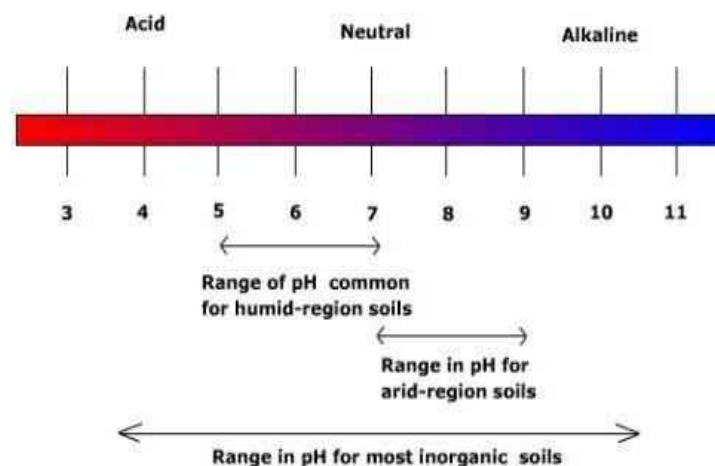
Soil pH

Soil pH is defined as the negative logarithm (to the base 10) of hydrogen ion concentration (activity)

$$\text{pH} = -\log_{10}[\text{H}]$$

The acidity or alkalinity of a solution can be expressed in terms of their hydrogen ion concentration. The scale of acidity or alkalinity is called **pH scale**. This scale ranges from 0 to 14. The neutral point in this scale is at pH 7. All the values above pH 7 represent alkalinity and below 7 denote acidity. The degree of alkalinity increases as values go above pH 7 and the degree of acidity increases as the pH decreases below 7.

pH scale and ranges in soil pH



According to the theory of electrolytic dissociation when liquids have number of H ions just equal to the number of OH ions the solution is **neutral**. While, H ion exceeds OH ion the solution is **acidic**. Conversely if OH ions are in excess the solution is **alkaline**.

- This ionic product of water has been measured by electrical conductivity measurements and is found to be of the order of 1×10^{-14} ions per litre at 22°C.
- Since in pure water, the number of H ions is equal to the number of OH ions, the concentration of each ion type must be 1×10^{-7} g ions per litre.
- If the concentration of H ions is more than 10^{-7} , the solution is **acidic**; if less solution is **alkaline**.
- This method of expressing the H ions concentration is very inconvenient and therefore

Sorenson (1969) suggested that H ions concentration be generally expressed as the numerical value of the negative power to which 10 must be raised in order to express the required concentration and this value be designated by the symbol pH.

- Thus, technically pH is the negative logarithm of the H ion concentration or the logarithm to the base ten of the reciprocal of hydrogen ion concentration

$$\text{i.e. } H^+ = 10^{-pH}$$

$$\log [H^+] = -pH \log 10$$

$$-pH = \log [H^+] / \log 10$$

$$pH = \log [H^+] / \log 10$$

Since $\log 10 = 1$, therefore $pH = -\log [H^+]$ (OR) $pH = \log 1 / [H^+]$

Soil reaction:

- Soil reaction is one of the most important physiological characteristics of the soil solution.
- The presence and development of micro - organisms and higher plants depend upon the chemical environment of soil. Therefore study of soil reaction is important in soil science.
- The reaction of a solution represents the degree of acidity or basicity caused by the relative concentration of H ions (acidity) or OH ions present in it.
- **Acidity** is due to the excess of H ions over OH ions, and **alkalinity** is due to the excess of OH ions over H ions.
- A neutral reaction is produced by an equal activity of H and OH ions. According to the theory of dissociation, the activity is due to the dissociation or ionization of compounds into ions.

Factors affecting Soil Reaction (pH)

Soil reaction varies due to following factors

1. Nature of soil colloids:

The colloidal particles of the soil influence soil reaction to a very greatest extent. When hydrogen (H^+) ion forms the predominant adsorbed cation, the soil reaction becomes acid.

2. Soil solution:

- The soil solution carries a number of salts dissolved in capillary water. Under field conditions, the concentration of salts varies with the moisture content of the soil.
- The more dilute the solution, the higher the pH value. Hence the pH tends to drop as the

soil gets progressively dry. Soil pH is also influenced by the presence of CO₂ in soil air.

- As the CO₂ concentration increases, the soil pH falls and increases the availability of the nutrients. Under field conditions, plant roots and micro-organism liberate enough CO₂, which results in lowering the pH appreciably.
- This principle of increasing the concentration of CO₂ in soil air is also used in the reclamation of alkali soils.

3. Climate:

Rainfall plays important role in determining the reaction of soil. In general, soils formed in regions of high rainfall are acidic (low pH value), while those formed in regions of low rainfall are alkaline (high pH value).

4. Soil management:

- Cultural operations in general tend to increase soil acidity. They make an acid soil more acidic, and an alkaline soil less alkaline.
- As a result of constant cultivation, basic elements are lost from the soil through leaching and crop removal. This leads to change the soil reaction to the acid side.

5. Parent materials:

- Soils developed from parent material of basic rocks generally have higher pH than those formed from acid rocks (*e.g.* granite).
- The influence of parent material is not very important as it is completely masked by the climatic conditions under which the soil is developed.

6. Precipitation:

- As water from rainfall passes through the soil, basic nutrients such as calcium (Ca) and magnesium (Mg) are leached.
- They are replaced by acidic elements including Al, H and manganese (Mn). Therefore, soils formed under high rainfall conditions are more acid than those formed under arid conditions.

7. Decomposition of organic matter:

- Soil organic matter is continuously being decomposed by micro-organisms into organic acids, carbon dioxide (CO₂) and water, forming carbonic acid.
- Carbonic acid, in turn, reacts with the Ca and Mg carbonates in the soil to form more soluble bicarbonates, which are leached away, leaving the soil more acid.

8. Native vegetation:

- Soils often become more acid when crops are harvested because of removal of bases.

- Type of crop determines the relative amounts of removal. For example, legumes generally contain higher levels of bases than do grasses.
- Calcium and Mg contents also vary according to the portion (s) of the plant harvested. Many legumes release H ions into their rhizosphere when actively fixing atmospheric N₂.
- The acidity generated can vary from 0.2 to 0.7 pH units per mole of fixed N.

9. Soil depth:

- Except in low rainfall areas, acidity generally increases with depth, so the loss of topsoil by erosion can lead to a more acid pH in the plough layer.
- The reason is that more subsoil is included in the plow layer as topsoil is lost. There are areas, however, where subsoil pH is higher than that of the topsoil.

10. Nitrogen fertilization:

- Nitrogen from fertilizer, organic matter, manure and legume N fixation produces acidity.
- Nitrogen fertilization speeds up the rate at which acidity develops. At lower N rates, acidification rate is slow, but is accelerated as N fertilizer rates increase.

11. Flooding:

- The overall effect of submergence is an increase of pH in acid soils and a decrease in basic soils.
- Regardless of their original pH values, most soils reach pH of 6.5 to 7.2 within one month after flooding and remain at the level until dried.
- Consequently, liming is of little value in flooded rice production. Further, it can induce deficiencies of micronutrients such as zinc (Zn).

Buffering capacity

- Buffering refers to resistance to a change in pH.
- If 1 ml HCl (of 0.1 N) is added to one liter of pure distilled water of pH 7.0, the resulting solution would have a pH of about 5.0.
- If on the other hand, the same amount of acid is added to a liter of soil suspension the resulting change in pH would be very small.
- There is a distinct resistance to change in pH. This power to resist a change in pH is called buffer action.
- A buffer solution is one which contains reserve acidity and alkalinity and does not change pH with small additions of acids or alkalis. Buffer capacity:
- The colloidal complex acts as a powerful buffer in the soil and does not allow rapid and

sudden changes in soil reaction.

- Buffering depends upon the amount of colloidal material present in soil. Clay soils rich in organic matter are more highly buffered than sandy soils.
- Buffer capacity of the soil varies with its cation exchange capacity (C.E.C.). The greater the C.E.C. the greater will be its buffer capacity.
- Thus, heavier the texture and the greater the organic matter content of a soil, the greater is the amount of acid or alkaline material required to change its pH.

Importance of buffering in agriculture

- Changes in soil reaction (pH) have a direct influence on the plants and it also affects the availabilities of plant nutrients.
- Deficiency of certain plant nutrients and excess availability of others in toxic amounts would seriously upset the nutritional balance in the soil.
- Buffering prevents sudden changes and fluctuation in soil pH. So it regulates the availability of nutrients and also checks direct toxic effect to plants.

Interpretation of Soil pH

pH Effect	pH Range
Acid	0.0-6.0
Normal	6.1-8.5
Moderately alkaline	8.6-9.0
Alkaline	>9.0

23. SOIL ELECTRICAL CONDUCTIVITY

Soil electrical conductivity, which is known as EC, is the ability of soil to conduct electrical current. EC is expressed in milliSiemens per meter (mS/m). Traditionally, soil scientists used EC to measure soil salinity.

- ◆ Soil salinity is a measure of the total amount of soluble salt in soil. As salinity levels increase, plants extract water less easily from soil, aggravating water stress conditions.
- ◆ High soil salinity can also cause nutrient imbalances, result in the accumulation of elements toxic to plants, and reduce water infiltration if the level of one salt element – sodium – is high.

Factors affecting EC

The conduction of electricity in soil takes place through the moisture- filled pores that occur between individual soil particles. Therefore, the EC of soil is determined by the following soil properties.

1. Porosity:

The greater soil porosity, the more easily electricity is conducted. Soil with high clay content has higher porosity than sandier soil. Compaction normally increases soil EC.

2. Water content:

Dry soil is much lower in conductivity than moist soil.

3. Salinity level:

Increasing concentration of electrolytes (salts) in soil water will dramatically increase soil EC. The salinity level in most Corn Belt soils is very low.

4. Cation exchange capacity (CEC):

- ◆ Mineral soil containing high levels of organic matter (humus) and / or 2:1 clay minerals such as montmorillonite, illite or vermiculate have a much higher ability to retain positively charged ions (such as Ca, Mg, K, Na, NH₄ or H) than soil lacking these constituents.
- ◆ The presence of these ions in the moisture filled soil pores will enhance soil EC in the same way that salinity does.

5. Temperature:

- ◆ As temperature decreases toward the freezing point of water, soil EC decreases slightly.

Below freezing, soil pores become increasingly insulated from each other and overall soil EC declines rapidly.

- ◆ Salinity tolerance is influenced by many plant, soil and environmental factors and their interrelationships.
- ◆ Generally, fruits, vegetables and ornamentals are more salt sensitive than forage or field crops. In addition, certain varieties, cultivars or rootstalks may tolerate higher salt levels than others.
- ◆ Plants are more sensitive to high salinity during seedling stages, immediately after transplanting and when subject to other (e.g. disease, insect, nutrient) stresses.
- ◆ Climate and irrigation also influence salinity tolerance. As soil dries, salt become concentrated in the soil solution, increasing salt stress.
- ◆ Therefore, salt problems are more severe under hot, dry conditions than under cool, humid conditions.
- ◆ Increasing irrigation frequency and applying water in excess of plant demand may be required during hot, dry periods to minimize salinity stress.

Interpretation based on EC measurement

EC (dSm⁻¹)	Salinity Effect	Interpretation
0.0 – 1.0	Harmless	Salinity effect negligible
1.1-3.0	Critical	Limit yield of sensitive crops
3.1 and above	Injurious	Limit yield of many crops

24. SOIL COLLOIDS

♣ Properties and types of soil colloids

♣ Layer silicate clays-Genesis and classification

♣ Sources of charges in clays

Soil colloids

The colloidal state refers to a two-phase system in which one material in a very finely divided state is dispersed through second phase. The examples are: Solid in liquid (Dispersion of clay in water) and Liquid in gas (Fog or clouds in atmosphere). The clay fraction of the soil contains particles less than 0.002 mm in size. Particles less than 0.001 mm size possess colloidal properties and are known as soil colloids.

General Properties of Soil Colloids

1. Size

- ❖ The inorganic and organic colloids are extremely small size - smaller than 2 micrometers in diameter.
- ❖ These particles cannot be seen using an ordinary light microscope but can be seen only with an electron microscope.

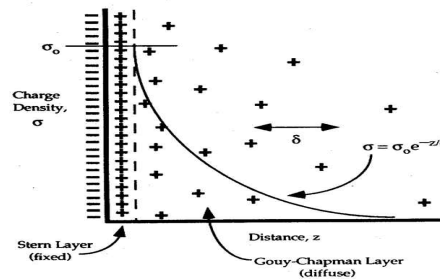
2. Surface area

- Because of their small size, all soil colloids have a larger external surface area per unit mass. The external surface area of 1 g of colloidal clay is 1000 times that of 1 g of coarse sand.
- Certain silicate clays have extensive internal surfaces occurring between plate like crystal units that make up each particle and often greatly exceed the external surface area.
- The total surface area of soil colloids ranges from 10 m²/g for clays with only external surfaces to more than 800 m²/g for clays with extensive internal surfaces. The colloid surface area in the upper 15 cm of a hectare of a clay soil could be as high as 700,000 km² g⁻¹

3. Surface charges

- Both external and internal surfaces of soil colloids carry negative and/or positive charges. Most of the organic and inorganic soil colloids carry a negative charge.

- When an electric current is passed through a suspension of soil colloidal particles they migrate to anode, the positive electrode indicating that they carry a negative charge.
- The magnitude of the charge is known as ***zeta potential***. The presence and intensity of the particle charge influence the attraction and repulsion of the particles towards each other, thereby influencing both physical and chemical properties.



4. Adsorption of cations:

- As soil colloids possess negative charge they attract and attach the ions of positive charge on the colloidal surfaces.
- They attract cations like H^+ , Al^{3+} , Ca^{2+} and Mg^{2+} . This gives rise to an ionic double layer.
- The Isomorphous substitution in the colloidal particle makes the external and internal layers of clay minerals negatively charged and these surfaces act as huge anions, which form the inner layer of the double layer.
- The outer layer is made up of a swarm of loosely held (adsorbed) cations attracted to the negatively charged surfaces.

5. Adsorption of water:

- A large number of water molecules are associated with soil colloidal particles. Some water molecules are attracted to the adsorbed cations and the cation is said to be in hydrated state.
- Others water molecules are held in the internal surfaces of the colloidal clay particles.
- These water molecules play a critical role in determining both the physical and chemical properties of soil.

6. Cohesion:

- (Attractive force between similar molecules or materials). Cohesion indicates the tendency of clay particles to stick together.
- This tendency is due to the attraction of clay particles for water molecules held between

them.

- When colloidal substances are wetted, water first adheres to individual clay particles and then brings about cohesion between two or more adjacent colloidal particles.

7. Adhesion:

- (Attractive force between different molecules or materials). Adhesion refers to the attraction of colloidal materials to the surface of any other body or substance with which it comes in contact.

8. Swelling and shrinkage:

- Some soil clay colloids belonging to smectite group like Montmorillonite swell when wet and shrink when dry.
- After a prolonged dry spell, soils high in smectite clay (e.g. Black soil -Vertisols) often show crises-cross wide and deep cracks.
- These cracks first allow rain to penetrate rapidly. Later, because of swelling, the cracks will close and become impervious.
- But soils dominated by kaolinite, chlorite, or fine grained micas do not swell or shrink. Vermiculite is intermediate in its swelling and shrinking characteristics.

9. Dispersion and flocculation:

- As long as the colloidal particles remain negatively charged, they repel each other and the suspension remains stable.
- If on any account they lose their charge, or if the magnitude of the charge is reduced, the particles coalesce, form flock or loose aggregates, and settle down.
- This phenomenon of coalescence and formation of flocks is known as flocculation. The reverse process of the breaking up of flocks into individual particles is known as de-flocculation or dispersion.

10. Brownian movement:

- When a suspension of colloidal particles is examined under a microscope the particles seem to oscillate.
- The oscillation is due to the collision of colloidal particles or molecules with those of the liquid in which they are suspended.
- Soil colloidal particles with those of water in which they are suspended are always in a constant state of motion. The smaller the particle, the more rapid is its movement.

11. Non permeability:

- Colloids, as opposed to crystalloids, are unable to pass through a semi-permeable

membrane.

- Even though the colloidal particles are extremely small, they are bigger than molecules of crystalloid dissolved in water.
- The membrane allows the passage of water and of the dissolved substance through its pores, but retains the colloidal particles.

Types of soil colloids

There are four major types of colloids present in soil

1. Layer silicate clays
2. Iron and aluminum oxide clays (sesquioxide clays)
3. Allophane and associated amorphous clays
4. Humus

Layer silicate clays, iron and aluminum oxide clays, allophane and associated amorphous clays are inorganic colloids while humus is an organic colloid.

- The soil colloids are the most active portion of the soil and largely determine the physical and chemical properties of a soil.
- **Inorganic colloids** (clay minerals, hydrous oxides) usually make up the bulk of soil colloids. Colloids are particles less than 0.001 mm in size, and the clay fraction includes particles less than 0.002 mm in size. Therefore, all clay minerals are not strictly colloidal.
- The **organic colloids** include highly decomposed organic matter generally called humus. Organic colloids are more reactive chemically and generally have a greater influence on soil properties per unit weight than the inorganic colloids.
- Humus is amorphous and its chemical and physical characteristics are not well defined.
- Both inorganic and organic colloids are intimately mixed with other soil solids. Thus, the bulk of the soil solids are essentially inert and the majority of the soil's physical and chemical character is a result of the colloids present.

Humus (Organic Colloid)

- Humus is amorphous, dark brown to black, nearly insoluble in water, but mostly soluble in dilute alkali (NaOH or KOH) solutions. It is a temporary intermediate product left after considerable decomposition of plant and animal remains.

- They are temporary intermediate because the organic substances remain continue to decompose slowly. The humus is often referred to as an organic colloid and consists of various chains and loops of linked carbon atoms.
- The humus colloids are not crystalline. They are composed basically of carbon, hydrogen, and oxygen rather than of silicon, aluminum, iron, oxygen, and hydroxyl groups.
- The organic colloidal particles vary in size, but they may be at least as small as the silicate clay particles.
- The negative charges of humus are associated with partially dissociated enolic (-OH), carboxyl (-COOH), and phenolic groups; these groups in turn are associated with central units of varying size and complexity.

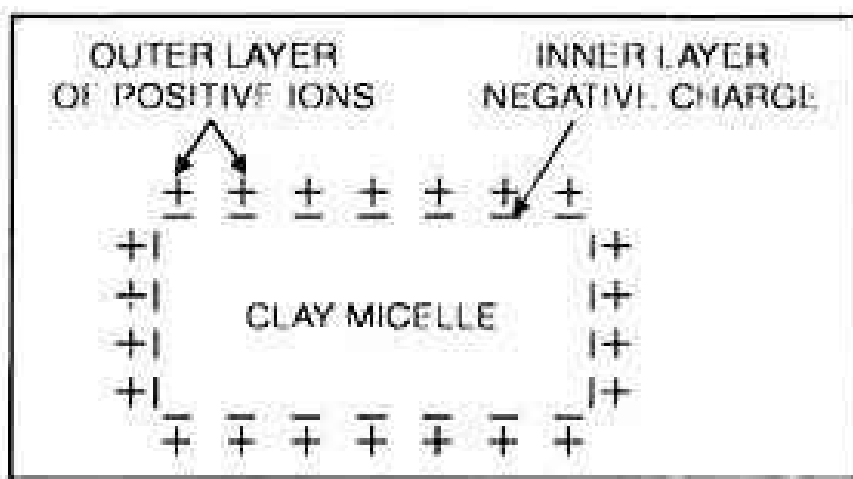


Fig. 5.1 Diagrammatic representation of colloidal clay crystals.

25. SILICATE CLAYS

♣ Genesis and Classification

♣ Amorphous minerals

These important silicate clays are also known as phyllosilicates (Phyllon - leaf) because of their leaf-like or plate like structure. These are made up of two kinds of horizontal sheets. One dominated by silicon and other by aluminum and/or magnesium.

Silica tetrahedron

- The basic building block for the silica-dominated sheet is a unit composed of one silicon atom surrounded by four oxygen atoms.
- It is called the silica tetrahedron because of its four-sided configuration.
- An interlocking array or a series of these silica tetrahedra tied together horizontally by shared oxygen anions gives a tetrahedral sheet.

Alumina octahedron

- Aluminium and/or magnesium ions are the key cations surrounded by six oxygen atoms or hydroxyl group giving an eight sided building block termed octahedron.
- Numerous octahedra linked together horizontally comprise the octahedral sheet. An aluminum-dominated sheet is known as a di-octahedral sheet, whereas one dominated by magnesium is called a tri-octahedral sheet.
- The distinction is due to the fact that two aluminum ions in a di-octahedral sheet satisfy the same negative charge from surrounding oxygen and hydroxyls as three magnesium ions in a tri-octahedral sheet.
- **The tetrahedral and octahedral sheets are the fundamental structural units of silicate clays.** These sheets are bound together within the crystals by shared oxygen atoms into different layers.
- The specific nature and combination of sheets in these layers vary from one type of clay to another and control the physical and chemical properties of each clay.

Classification of silicate clay minerals

On the basis of the number and arrangement of tetrahedral (silica) and octahedral (alumina-magnesia) sheets contained in the crystal units or layers, silicate clays are classified

into three different groups

- 1) 1 :1 Type clay minerals
- 2) 2:1 Type clay minerals
- 3) 2: 1: 1 (or) 2:2 Type clay minerals

1:1 type minerals

- The layers of the 1:1-type minerals are made up of one tetrahedral (silica) sheet combined with one octahedral (alumina) sheet.
- In soils, **kaolinite group** is the most prominent 1:1 clay mineral, which includes kaolinite, hallosite, nacrite and dickite.
- The tetrahedral and octahedral sheets in a layer of a kaolinite crystal are held together tightly by oxygen atoms, which are mutually shared by the silicon and aluminum cations in their respective sheets.
- These layers, in turn, are held together by hydrogen bonding. Consequently, the structure is fixed and no expansion ordinarily occurs between layers when the clay is wetted.
- Cations and water do not enter between the structural layers of a 1:1 type mineral particle. The effective surface of kaolinite is restricted to its outer faces or to its external surface area.



- Kaolinite crystals usually are hexagonal in shape.
- Because of the strong binding forces between their structural layers, kaolinite particles are not readily broken down into extremely thin plates.
- Kaolinite exhibits very little plasticity, cohesion, shrinkage, and swelling.

2:1-Type Minerals

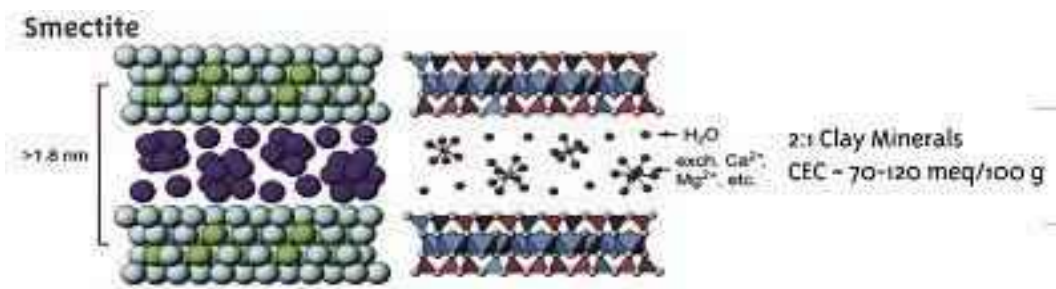
The crystal units (layers) of these minerals are characterized by an octahedral sheet sandwiched between two tetrahedral sheets. Three general groups have this basic crystal structure.

- ii) Expanding type : Smectite group and vermiculite
- iii) Non-expanding type : Mica group (illite)

Expanding Minerals:

Smectite / Montmorillonite :

- The smectite group of minerals is noted for their interlayer expansion and swelling when wetted. The water enters the interlayer space and forces the layers apart. Montmorillonite is the most prominent member of this group in soils. Beidellite, nontronite, and saponite are also found in soils.
- The flake-like crystals of smectite (e.g., **Montmorillonite**) are composed of an expanding lattice 2:1 type clay mineral. Each layer is made up of an octahedral sheet sandwiched between two tetrahedral (silica) sheets.
- There is little attraction between oxygen atoms in the bottom tetrahedral sheet of one unit and those in the top tetrahedral sheet of another.
- This permits a ready and variable space between layers, which is occupied by water and exchangeable cations.
- This internal surface exceeds the external surface of clay crystal. In montmorillonite, magnesium replaces aluminum in some sites of octahedral sheet.
- Likewise, some silicon atoms in the tetrahedral sheet may be replaced by aluminum. These substitutions give rise to a negative charge.
- These minerals show high cation exchange capacity, swelling and shrinkage properties. Wide cracks commonly form in smectite dominated soils (e.g., Vertisols) when dried. The dry aggregates or clods are very hard, making such soils difficult to till.



Vermiculites

- Vermiculites are also 2:1 type minerals in that an octahedral sheet occurs between two tetrahedral sheets.
- In the tetrahedral sheet of most vermiculite, aluminum is substituted by silicon in most of the sites. This accounts for most of the very high net negative charge associated with these minerals.
- Water molecules, along with magnesium and other ions, are strongly adsorbed in the

interlayer space of vermiculites. They act primarily as bridges holding the units together rather than as wedges driving them apart.

- The degree of swelling is less for vermiculites than for smectite. For this reason, vermiculites are considered **limited expansion clay minerals**, expanding more than kaolinite but much less than the smectite.
- The **cation exchange capacity (CEC)** of vermiculite is **higher than all other silicate clays**, including montmorillonite and other smectite because of very high negative charge in the tetrahedral sheet. Vermiculite crystals are larger than those of the smectite but much smaller than those of kaolinite.

2:1 Non-expanding minerals:

- Micas are the type of minerals in this group. (e.g.) Muscovite and biotite.
- Weathered minerals similar in structure to these micas are found in the clay fraction of soils.
- They are called fine-grained micas. Like smectite, fine-grained micas have a 2:1 type crystal. However, the particles are much larger than those of the smectite.



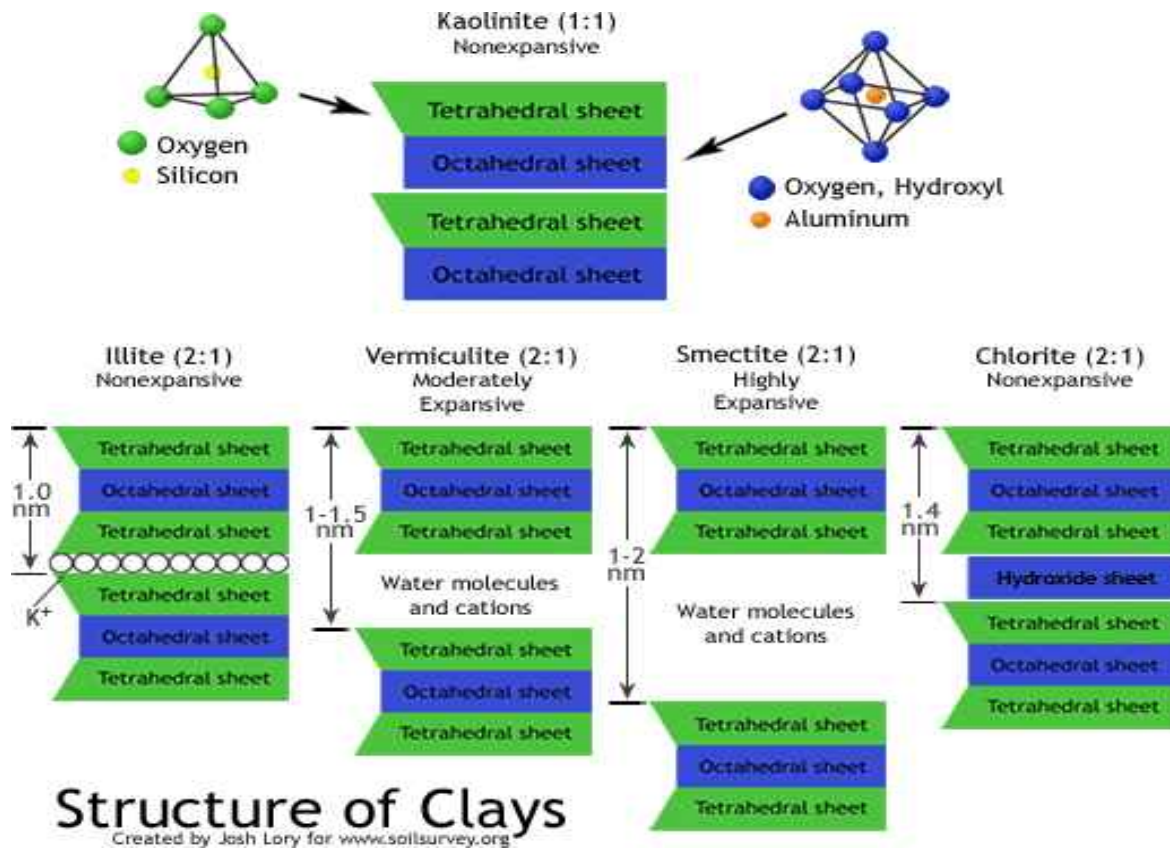
- Also, the major source of charge is in the tetrahedral sheet where aluminum atoms occupy about 20% of the silicon sites. This results in a net negative charge in the tetrahedral sheet and the charge is higher than that found in vermiculites.
- To satisfy this charge, potassium ions are strongly attracted in the interlayer space and are just the right size to fit into spaces in the adjoining tetrahedral sheets.
- The potassium thereby acts as a binding agent, preventing expansion of the crystal. Hence, fine-grained micas are quite non-expanding.
- The properties such as hydration, cation adsorption, swelling, shrinkage and plasticity are less intense in fine-grained micas. Their specific surface area varies from 70 to 100 $\text{m}^2 \text{g}^{-1}$, about one eighth that for the smectite.

2:1:1 Type Minerals:

- This silicate group is represented by **chlorites**. Chlorites are basically iron magnesium

silicates with some aluminum present.

- The crystal unit contains two silica tetrahedral sheets and two magnesium-dominated tri-octahedral sheets giving rise to the term 2:1:1 or 2:2 type structure.
- There is no water adsorption between the chlorite crystal units, which accounts for the non expanding nature of this mineral.



Other clay types

1. Mixed and interstratified layers:

- Specific groups of clay minerals do not occur independently of one another. In a given soil, it is common to find several clay minerals in an intimate mixture.
- Furthermore, some mineral colloids have properties and composition intermediate between those of any two minerals.
- Such minerals are termed mixed layer or interstratified because the individual layers within a given crystal may be of more than one type.
- Terms such as "chlorite-vermiculite" and "mica - smectite" are used to describe mixed layer minerals. In some soils, they are more common than single structured minerals such as montmorillonite.

2. Iron and aluminum oxide clays (sesquioxide clays):

- Under conditions of extensive leaching by rainfall and long time intensive weathering of minerals in humid warm climates, most of the silica and alumina in primary minerals are dissolved and slowly leached away. The remnant materials, which have lower solubility are called sesquioxides.
- Sesquioxides (metal oxides) are mixtures of aluminum hydroxide, $\text{Al}(\text{OH})_3$, and iron oxide, Fe_2O_3 , or iron hydroxide, $\text{Fe}(\text{OH})_3$.
- The Latin word *sesqui* means one and one-half times, meaning one and one-half times more oxygen than Al and Fe. These clays can grade from amorphous to crystalline.
- Examples of iron and aluminum oxides common in soils are gibbsite ($\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$) and goethite ($\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$). Less is known about these clays than about the layer silicates. These clays do not swell, not sticky and have high phosphorus adsorption capacity.

3. Allophane and other amorphous minerals:

- These silicate clays are mixtures of silica and alumina. They are amorphous in nature. Even mixture of other weathered oxides (iron oxide) may be a part of the mixture.
- Typically, these clays occur where large amount of weathered products existed. These clays are common in soils forming from volcanic ash (e.g., Allophane).
- These clays have high anion exchange capacity or even high cation exchange capacity.
- Almost all of their charge is from accessible hydroxyl ions (OH^-), which can attract a positive ion or lose the H^+ attached. These clays have a variable charge that depends on H^+ in solution (the soil acidity).

Comparative properties of silicate clay minerals

Property	Kaolinite	Mont-morillonite	Vermiculite	Mica or Illite
Structure and Size (microns)	1:1 lattice 0.5 to 5.0	2:1 0.01 to 1.0	2:1 0.1 to 5.0	2:1 0.1 to 2.0
Shape	Hexagonal crystals	Irregular flakes	Plates; flakes	Flakes
Surface area (m^2/g)	5-20	700-800	500-600	100-200
Cohesion and plasticity swelling capacity	Low	High	High	Medium
Cation exchange capacity CEC ($\text{C.mol}(\text{P}^+) \text{ k}^{-1}$)	3-15	80-120	100-180	15-40
External and internal surface	Low	Very High	High	Medium
Anion exchange capacity (AEC) ($\text{C.mol}(\text{P}^+) \text{ k}^{-1}$)	High	Low	Low	Medium

26. SOURCES OF CHARGES ON CLAYS

♣ Positive and negative charges

♣ Isomorphous substitution

♣ pH dependant charges

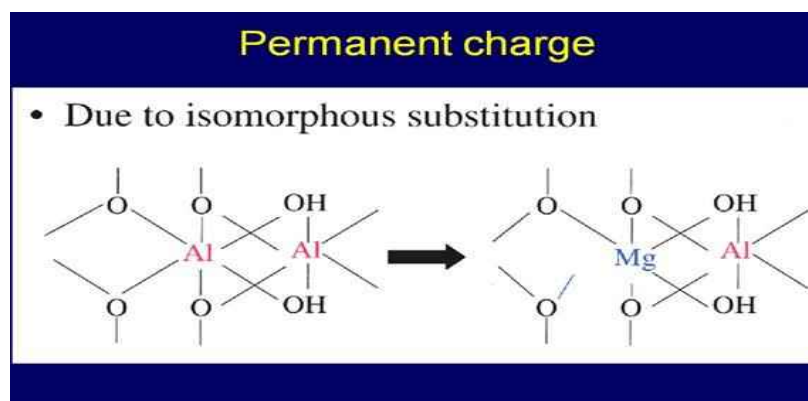
The sources of negative charge on clays are i) ionizable hydrogen ions and ii) isomorphous substitution.

Ionizable hydrogen ions:

- These are hydrogen from hydroxyl (OH) ions on clay surfaces. The Al-OH or -Si-OH portion of the clay ionizes the H and leaves an un-neutralized negative charge on the oxygen (-AlO^- or -SiO^-).
- The extent of ionized hydrogen depends on solution pH and hence these negative charges are **pH dependent charges**. More ionization occurs in alkaline (basic) solutions.

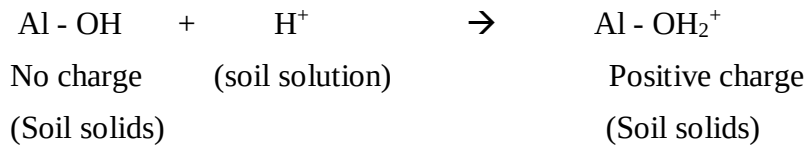
Isomorphous substitution:

- This is due to the substitution of a cation of higher valence with another cation of lower valence but similar size in the clay crystal structure.
- In clay crystals some ions fit exactly into mineral lattice sites because of their convenient size and charge.
- Dominantly, clays have Si^{4+} in tetrahedral sites and Al^{3+} in octahedral sites. Other ions present in large amounts during clay crystallization can replace some of the Al^{3+} and Si^{4+} cations.
- Common substitutions are the Si^{4+} replaced by Al^{3+} , and replacement of Al^{3+} by Fe^{3+} , Fe^{2+} , Mg^{2+} or Zn^{2+} .
- As the total negative charge from the anions (oxygen) remains unchanged, the lower positive charge of the substituted cations result in excess negative charges on clay crystals.

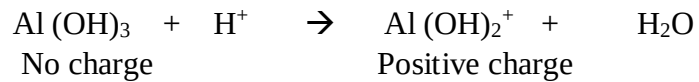


Source of positive charges

1. Isomorphous substitution: Low valence cations replaced by high valence cations.
2. Surface and exposed broken bonds of clay lattice: OH group in certain acid soils.

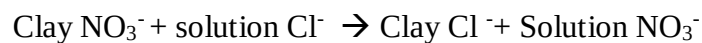


3. Complex aluminium and iron hydroxy ions in acid soils.



4. pH dependent charges are important for anion exchange of organic matter

The basic principles of cation exchange apply as well to anion exchange, except that the charges on the colloids are positive and the exchange is among negatively charged anions.



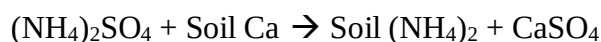
Anion exchange is an important mechanism for interactions in the soil and between the soil and plant. Together with cation exchange it largely determines the ability of soil to provide nutrients to plants promptly.

27. ION EXCHANGE REACTIONS

♣ Cation exchange and anion exchange capacity

♣ Base saturation

The **Ion Exchange phenomenon** was first identified by Harry Stephen Thompson in England during 1850. When soil was leached (washed) with ammonium sulphate, calcium sulphate was detected in the leachate. The ammonium ion in the solution replaced calcium in the soil.



- Soil colloids are responsible for most of the chemical reactions in soil. Inorganic soil colloids are clay minerals derived from different rocks and minerals during weathering and organic colloids are formed during organic matter decomposition.
- Each colloid (inorganic and organic) has a net negative (-) charge developed during the formation process (refer to sources of positive and negative charges on colloids in the previous chapter).
- The negatively charged colloids can attract and hold positively (+) charged particles or elements (ions).
- An element with a net electrical charge (positive or negative) is called an ion. Potassium (K^+), sodium (Na^+), hydrogen (H^+), Calcium (Ca^{++}), Magnesium (Mg^{++}) *etc.* all have positive charges. These ions are attracted towards cathodes and are called cations.
- Ions such as chloride (Cl^-), nitrate (NO_3^-), sulfate (SO_4^-), Phosphate (H_2PO_4^-), *etc.* are called anions.
- Positively charged ions (cations) are attracted towards the negatively charged colloids. These ions are held on the surface of the colloids. As most of the soil colloids are negatively charged, cations are the most adsorbed ions.
- The adsorbed anions are present in smaller quantities than the cations because of the predominant negative charges on the soil colloids.

Cation Exchange Capacity (CEC)

The CEC is the capacity of soil to hold and exchange cations. The cation exchange capacity is defined simply as the sum total of the exchangeable cations that a soil can adsorb. The higher the CEC of soil the more cations it can retain. Soils differ in their capacities to hold exchangeable cations.

Unit of expression

CEC is expressed as milliequivalents of cations per 100 grams of soil (meq /100g soil). After 1982, in the metric system the term equivalent is not used but moles are the accepted chemical unit. The recent unit of expression of CEC is *centi moles of protons per kilo gram soil* [**cmol (p⁺) kg⁻¹ soil**]. One meq/100 g is equal to one cmol (p⁺) kg⁻¹ soil.

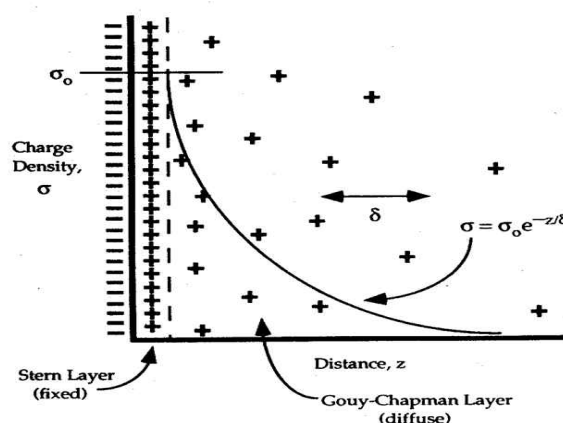
Factors affecting Cation Exchange Capacity

1. Amount and kinds of clay
 2. Amount and kind of organic matter
- A high-clay soil can have higher CEC than a low-clay soil. High organic matter content increases CEC. The CEC of clay minerals range from 10 to 150 [**cmol (p⁺) kg⁻¹**] and that of organic matter ranges from 200 to 400 [**cmol (p⁺) kg⁻¹**].
 - Clay soils with high CEC can retain large amounts of cations and reduce the loss of cations by leaching. Sandy soils, with low CEC, retain smaller quantities of cations and therefore cations are removed from soil by leaching.
 - So, nutrients in cationic form like potassium must be applied frequently in smaller quantities (split applications) in sandy soil to reduce the loss of nutrients by leaching whereas in clay soils, it can be applied in one or two splits as K is retained due to high CEC and not leached.

Importance of Cation Exchange

- ❖ Cation Exchange Reaction is considered as the second most important reaction next to photosynthesis.
- ❖ Cation exchange is an important reaction in soil fertility, in causing and correcting soil acidity and basicity, in changes altering soil physical properties, and as a mechanism in purifying or altering percolating waters.
- ❖ The plant nutrients like calcium, magnesium, and potassium are supplied to plants in large measure from exchangeable forms.
- ❖ The exchangeable K is a major source of plant K.
- ❖ The exchangeable Mg is often a major source of plant Mg.

- ❖ The amount of lime required to raise the pH of an acidic soil is greater as the CEC is greater.
- ❖ Cation exchange sites hold Ca^+ , Mg^+ , K^+ , Na^+ , and NH_4^+ ions and slow down their losses by leaching.
- ❖ Cation exchange sites hold fertilizer K^+ and NH_4^+ and greatly reduce their mobility in soils.
- ❖ Cation exchange sites adsorb many metals (Cd^{2+} , Zn^{2+} , Ni^{2+} , and Pb^{2+}) that might be present in wastewater adsorption removes them from the percolating water, thereby cleansing the water that drains into groundwater.



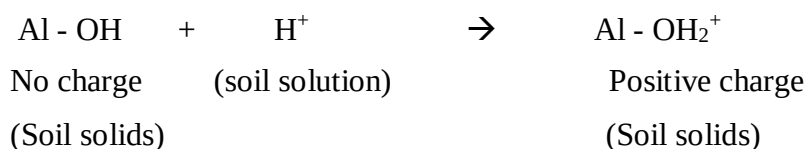
Anion Exchange

Anion adsorption:

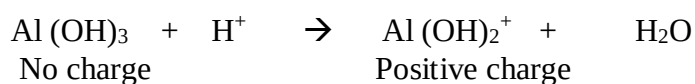
Adsorption of negative ion (anions) *e.g.* Cl^- , NO_3^- , SO_4^{2-} , and H_2PO_4^- on positively charged sites of clay and organic matter is known as anion adsorption.

Source of positive charge

1. Isomorphous substitution: Low valence cations replaced by high valence cations.
2. Surface and exposed broken bonds of clay lattice: OH group in certain acid soils.



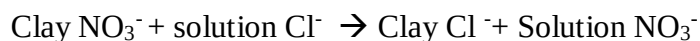
3. Complex aluminium and iron hydroxy ions in acid soils.



4. pH dependent charges are important for anion exchange of organic matter

The basic principles of cation exchange apply as well to anion exchange, except that the

charges on the colloids are positive and the exchange is among negatively charged anions.



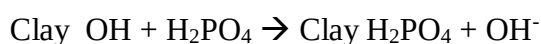
Anion exchange is an important mechanism for interactions in the soil and between the soil and plant. Together with cation exchange it largely determines the ability of soil to provide nutrients to plants promptly.

Anion exchange capacity

- “The sum total of exchangeable anions a soil can adsorb is known as anion exchange capacity”. It is expressed as cmol / kg or m.eq./ 100 g soil.
- The capacity of adsorption and exchange of anions varies with the type of clay mineral, soil reaction, and the nature of anion.
- Kaolinitic minerals have a greater anion adsorbing and exchange capacity than montmorillonitic and illitic clays because the exchange is located at only a few broken bonds.
- The capacity for holding anions increases with the increase in acidity.
- The lower the pH the greater is the adsorption. All anions are not adsorbed equally readily.
- Some anions such as H_2PO_4 are adsorbed very readily at all pH values in the acid as well as alkaline range.
- Cl and SO_4 ions are adsorbed slightly at low pH but none at neutrality, while NO_3 ions are not adsorbed at all.
- The affinity for adsorption of some of the anions commonly present in soil is of the order: $\text{NO}_3 < \text{Cl} < \text{SO}_4 < \text{PO}_4$.
- Hence at the pH commonly prevailing in cultivated soils, nitrate, chloride and sulphate ions are easily lost by leaching.

Importance of anion exchange

- The phenomenon of anion exchange assumes importance in relation to phosphate ions and their fixation. The exchange is brought about mainly by the replacement of OH ions of the clay mineral. The reaction is very similar to cation exchange.



The adsorption of phosphate ions by clay particles from soil solution reduces its availability to plants. This is known as phosphate fixation. As the reaction is reversible, the

phosphate ions again become available when they are replaced by OH ions released by substances like lime applied to soil to correct soil acidity.

Percent Base Saturation

- Percent base saturation tells what percent of the exchange sites are occupied by the basic cations. If calcium has a base saturation value of 50% and magnesium has a base saturation value of 20% as shown above, then calcium occupies half of the total exchange sites (CEC) and magnesium occupies one-fifth of the total exchange sites (CEC).
- In this example, where the soil has a CEC of 5 meq/100g, 2.5 meq/100g of the CEC is occupied by calcium and 1 meq/100g of the CEC is occupied by magnesium. If all the exchangeable bases (Ca, Mg, K and Na) total 100%, then there is no exchangeable acidity.
- The relative percentage of base forming cations in the total CEC is termed as per cent base saturation.
- The percentage of sodium in the total CEC is called Exchangeable sodium percentage (ESP) $[(Na/CEC) \times 100]$. These parameters are considered while deriving fertilizer prescription and amendments for problem soils.

28. SOIL ORGANIC MATTER

♣ Sources and Decomposition of SOM

♣ Role of SOM

♣ Mineralization and Immobilization

Soil Organic Matter (SOM) comprises an accumulation of

- i) Partially disintegrated and decomposed plant and animal residues
- ii) Other organic compounds synthesized by the soil microbes upon decay.

SOM is frequently said to consist of **humic substances** and **nonhumic substances**.

The term **SOM** is generally used to represent the organic constituents in the soil, including undecayed plant and animal tissues, their partial decomposition products, and the soil biomass

Sources of Soil Organic Matter

Primary sources

Plant tissues - the tops and roots of trees, shrubs, grasses, crop residues and soil organisms.

Secondary sources

Animals - waste products of animals, remains of animals after completion of life cycle.

Decomposition of soil organic matter

The organic materials (plant and animal residues) incorporated in the soil are attacked by a variety of microbes, worms and insects in the soil if the soil is moist. Some of the constituents are decomposed very rapidly, some less readily, and others very slowly. The constituents in terms of ease of decomposition are:

1. Sugars, starches and simple proteins - easy to decompose
2. Crude proteins
3. Hemicelluloses
4. Cellulose
5. Fats, waxes, resins
6. Lignins - Very difficult to decompose

The organic matter is also classified on the basis of their rate of decomposition

1. Rapidly decomposed : Sugars, starches, proteins etc.
2. Less rapidly decomposed : Hemicelluloses, celluloses etc.
3. Very slowly decomposed : Fats, waxes, resins, lignins etc.

Simple decomposition products under aerobic decomposition are CO₂, H₂O, NO₃, SO₄

When organic material is added to soil, three general reactions take place

- ♦ The bulk of the material undergoes enzymatic oxidation with CO₂, water, energy and heat as the major products.
- ♦ The essential elements such as N, P and S are released and / or immobilized by a series of reactions.

Molecules very resistant to microbial action is formed either through modification of compounds or by microbial synthesis.

A. Decomposition of soluble substances:

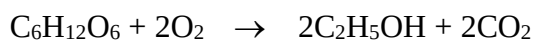
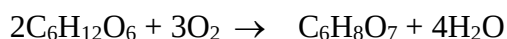
When glucose is decomposed under aerobic conditions the reaction is as under:



Under partially oxidized conditions,

Sugar + Oxygen → Aliphatic acids (Acetic, formic *etc.*) or Hydroxy acids (Citric, lactic *etc.*) or Alcohols (ethyl alcohol *etc.*)

Some of the reactions invoiced may be represented as under:

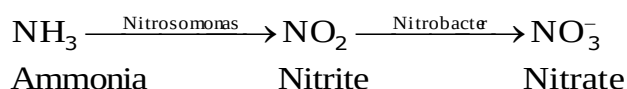


i) Ammonification :

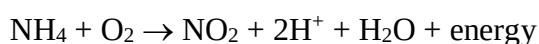
The transformation of organic nitrogenous compounds (amino acids, amides, ammonium compounds, nitrates *etc.*) into ammonia is called ammonification. This process occurs as a result of hydrolytic and oxidative enzymatic reaction under aerobic conditions by heterotrophic microbes.

ii) Nitrification:

The process of conversion of ammonia to nitrites (NO₂) and then to nitrate (NO₃⁻) is known as nitrification. It is an aerobic process by autotrophic bacteria.

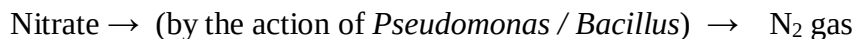


The net reactions are as follows:



iii) Denitrification:

The process, which involves conversion of soil nitrate into gaseous nitrogen or nitrous oxide, is called Denitrification. Water logging and high pH will increase N loss by Denitrification.



B. Decomposition of Insoluble Substances

i) Breakdown of Protein:

During the course of decomposition of plant materials, the proteins are first hydrolyzed to a number of intermediate products.

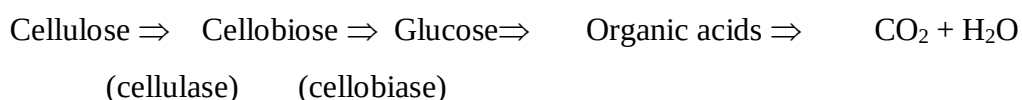
Aminization : The process of conversion of proteins to aminoacids.

Ammonification: The process of conversion of aminoacids and amides to ammonia.

ii) Breakdown of cellulose:

The decomposition of the most abundant carbohydrates.

Hydrolysis hydrolysis oxidation



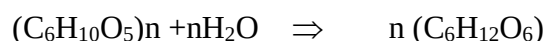
This reaction proceeds more slowly in acid soils than in neutral and alkaline soils. It is quite rapid in well aerated soils and comparatively slow in poorly aerated soils.

iii) Breakdown of Hemicellulose:

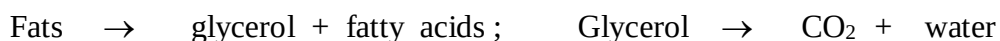
Decompose faster than cellulose and are first hydrolyzed to their components sugars and uronic acids. Sugars are attacked by microbes and are converted to organic acids, alcohols, carbon dioxide and water. The uronic acids are broken down to pentose and CO₂. The newly synthesized hemicelluloses thus form a part of the humus.

iv) Breakdown of Starch:

It is chemically a glucose polymer and is first hydrolyzed to maltose by the action of amylases. Maltose is next converted to glucose by maltase. The process is represented as under:



C. Decomposition of ether soluble substances:



D. Decomposition of lignin:

Lignin decomposes slowly, much slower than cellulose. Complete oxidation gives rise to CO₂ and H₂O.

Role of organic matter in Soils

1. Organic matter creates a granular condition of soil which maintains favorable condition of aeration and permeability.
2. Water holding capacity of soil is increased and surface runoff, erosion etc., are reduced as there is good infiltration due to the addition of organic matter.
3. Surface mulching with coarse organic matter lowers wind erosion and lowers soil temperatures in the summer and keeps the soil warmer in winter.
4. Organic matter serves as a source of energy for the microbes and as a reservoir of nutrients that are essential for plant growth and also hormones, antibiotics.
5. Fresh Organic matter supplies food for earthworms, ants and rodents and makes soil P readily available in acid soils.
6. Organic acids released from decomposing organic matter help to reduce alkalinity in soils; organic acids along with released CO₂ dissolve minerals and make them more available.
7. Humus (a highly decomposed organic matter) provides a storehouse for the exchangeable and available cations.
8. It acts as a buffering agent which checks rapid chemical changes in pH and soil reaction.

C: N ratio

- When microorganism grow and multiply on organic debris, carbon is utilized for building the cellular material of microbial cells with the release of CO₂, CH₄ and other volatile substances.
- In this process, microorganism also assimilate N, P, K, S which get bound in the cell protoplasm;
- Therefore, the C/N, C/P, C/K and C/S ratios in soil are governed by the extent of organic matter utilized by soil microorganisms depending on the O₂ content and microbial biomass at a particular stage in decomposition.

The three parallel processes going on during decomposition are :

1. Degradation of plant and animal remains by celluloses and other microbial enzymes.
2. The increase in the biomass of microorganism which comprises polysaccharides and protein
3. The accumulation and liberation of end products

Mineralization:

Mineralization is a process where microbes digest and reduce the organic complexes to inorganic materials. Inorganic materials released during this process are the essential plant nutrients (N, P and K), macronutrients and micronutrients, salts, and heavy metals.

Immobilization:

Microbial uptake of nutrients such as N, P and S which is opposite in magnitude to mineralization. Immobilization reduces the availability of nutrients to plant growth, intensity of which is related to the total microbial biomass at a given time.

Immobilization is the opposite of mineralization. Inorganic forms of N are converted into organic forms as microbes and plants die. N that enters the soil in mineral form, not from a plant or animal source, is inorganic

- The origin of soil organic matter is plant material. The rate of decomposition of plant residues depends on their composition and soil environmental conditions.
- Decomposition proceeds with the liberation of $\text{CO}_2 + \text{H}_2\text{O}$; release of N, P and S in inorganic forms (mineralization) or synthesis into organic combinations (immobilization); and the formation of new compounds that are resistant to decomposition.
- Another aspect of residue composition that affects the rate of decomposition is the C / N ratio.
- Since the organisms that decompose residues need N (and other essential elements) as well as C, if there is little N in the residue, decomposition is slow
- Also, if there is little N in the residue, microorganisms will utilize inorganic N in the soil to satisfy their N requirement, thereby competing with plants for N and reducing the amount of soil N available for plant growth.
- The C / N ratio in soil is relatively constant and = 12. In plant residues, it is highly variable and increases with maturity.
- The C / N is lower in microorganisms and = 8. Since microbes incorporate only about 1/3 of the C metabolized into biomass, the substrate material must have C / N = 24 to satisfy the N requirement of microbes.
- If the C / N ration of residue > 24, available soil N is consumed by microbes and plant-available N decreases.

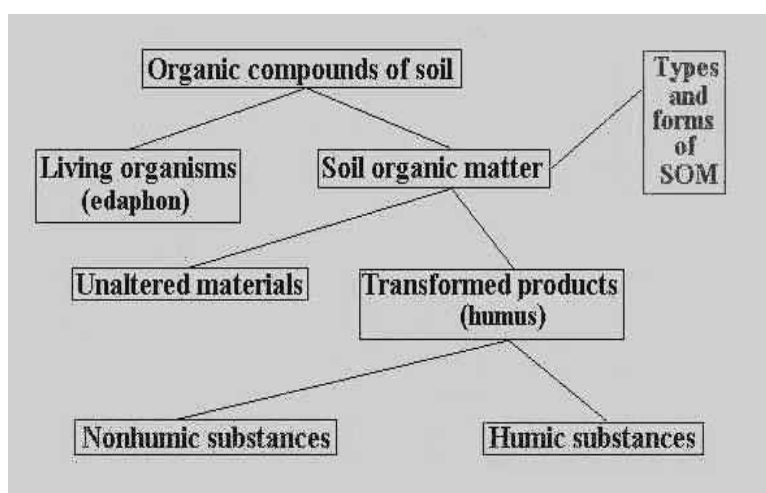
Additional reading

What is Organic Matter?

Many times we think of organic matter as the plant and animal residues we incorporate into the soil. We see a pile of leaves, manure, or plant parts which is actually organic material, not organic matter.

- ♦ What's the difference between organic material and organic matter? Organic material is anything that was alive and is now in the soil. For it to become organic matter, it must be decomposed into humus.
- ♦ Humus is organic material that has been converted by microorganisms to a resistant state of decomposition. Organic material is unstable in the soil, changing form and mass readily as it decomposes. As much as 90 percent of it disappears quickly because of decomposition.
- ♦ Organic matter is stable in the soil. It has been decomposed until it is resistant to further decomposition.
- ♦ Usually, only about 5 percent of it mineralizes yearly. That rate increases if temperature, oxygen, and moisture conditions become favorable for decomposition, which often occurs with excessive tillage. It is the stable organic matter that is analyzed in the soil test.

The Distribution of SOM is shown on the picture :



Organic compounds of soil:

Live organisms and their undecomposed, partly decomposed and completely decomposed remains as well as products of their transformation.

Living organisms: alive - edaphon.

Soil organic matter:

- ♦ Non-living components which are a heterogeneous mixture composed largely of products resulting from microbial and chemical transformations of organic debris.
- ♦ Soil organic matter can exist in different morphological patterns, which are the bases of the classification of so called forms and types of humus.

Unaltered materials : fresh and non-transformed components of older debris.

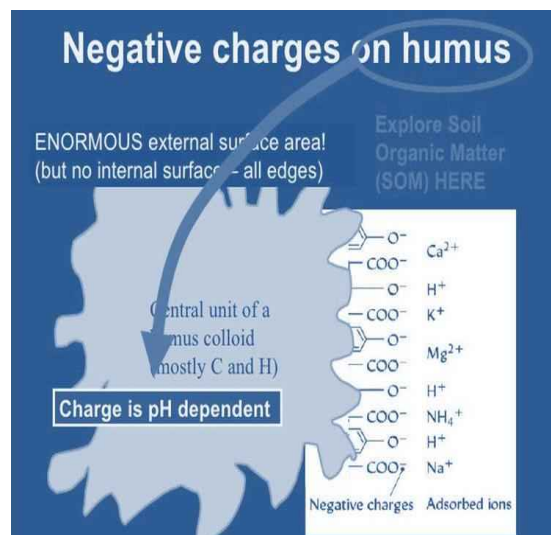
Transformed products - (humus):

- ◆ Bearing no morphological resemblance to the structures from which they were derived.
- ◆ These transformed components are referred to as the humification process products.

Humic substances:

A series of relatively high-molecular-weight, brown to black colored substances formed by secondary synthesis reactions.

- The term is used as a generic name to describe to colored material or its fractions obtained on the basis of solubility characteristics:
 - humic acids (HA)
 - fulvic acids (FA)
 - humins



Nonhumic substances:

Compounds belonging to known classes of biochemistry, such as :

- * carbohydrates
- * lipids
- * amino acids

- ✚ The chemical and colloidal properties of SOM can be studied only in the free state, that is, when freed of inorganic soil components.
- ✚ Thus the first task of the researcher is to separate organic matter from the inorganic matrix of sand, silt, and clay.

Functions of soil organic matter

Physical functions	Chemical functions	Biological functions
• Bind soil particles together in stable aggregates • Influence water holding and aeration • Influence soil temperature	• Major source of cation exchange capacity • Source of pH buffering • Binding site for heavy metals and pesticides	• Food source for microbes and small animals • Major reservoir of plant nutrients

Source: Sparrow *et al.* (1999)

29. HUMUS

♣ Theories of Formation and properties

♣ Fractionation

Humus is a complex and rather resistant mixture of brown or dark brown amorphous and colloidal organic substance that results from microbial decomposition and synthesis and has chemical and physical properties of great significance to soils and plants.

Humus Formation

The humus compounds have resulted from two general types of biochemical reactions: Decomposition and Synthesis.

1. Decomposition:

- a) Chemicals in the plant residues are broken down by soil microbes including lignin.
- b) Other simpler organic compounds that result from the breakdown take part immediately in the biochemical synthesis.
- c) These simpler chemicals are metabolized into new compounds in the body tissue of soil microbes.
- d) The new compounds are subject to further modification and synthesis as the microbial tissue is subsequently attacked by other soil microbes.

2. Synthesis:

Involves such breakdown products of lignins, phenols and quinones.

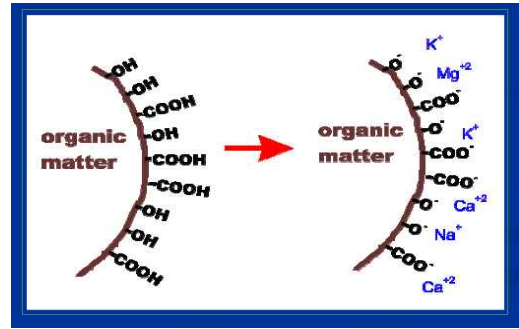
- a) These monomers undergo polymerization by which polyphenols and polyquinones are formed.
- b) These high molecular weight compounds interact with N-containing amino compounds and forms a significant component of resistant humus.
- c) Colloidal clays encourage formation of these polymers.
- d) Generally two groups of compounds that collectively make up humus are the humic group and the nonhumic group.

Properties of Humus

1. The tiny colloidal particles are composed of C, H, and O.
2. The colloidal particles are negatively charged (-OH, -COOH or phenolic groups), has

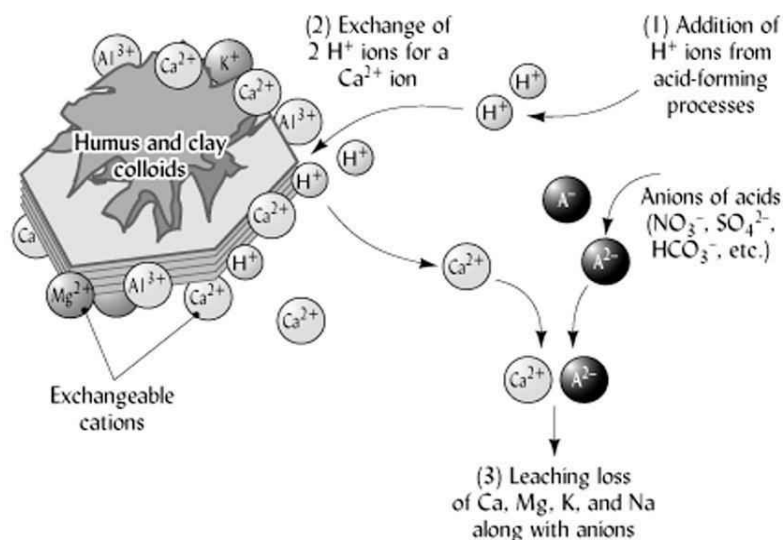
very high surface area, higher CEC ($150 - 300 \text{ cmol(p}^+) \text{ kg}^{-1}$), 4 - 5 times higher water holding capacity than that of silicate clays.

3. Humus has a very favorable effect on aggregate formation and stability.
4. Impart black colour to soils.
5. Cation exchange reactions are similar to those occurring with silicate clays.



Clay – Humus Complex

- ❖ Humus, the organic amorphous colloid supplies both basic and acidic ions which is transitory and ultimately disappears from soil.
- ❖ Clay, the inorganic crystalline colloid supplies chiefly the basic nutrient ions is more or less stable.
- ❖ Both these colloids form the soil colloidal complex and are extremely active and form important sources of plant nutrients.
- ❖ It is believed that humus and clay exist in the soil as clay – humus complex, the two being held together by cations like Ca, Fe, etc.
- ❖ Depending upon the nature of binding cation, two types of Clay – humus complex have been recognized.
- ❖ The colloidal complex bound by Ca ions is more stable and is responsible for the favorable physical condition of the soil, particularly its structure.
- ❖ The other type where Fe acts as the binding agent creates a poor physical condition of the soils.



30. SOIL BIOLOGY - SOIL ORGANISMS

♣ Soil flora and fauna

♣ Beneficial and harmful roles

♣ Microorganisms and role

♣ Soil Enzymes

Soil flora and **Soil fauna** are terms used to describe the organisms that live within soils. The flora is composed primarily of microorganisms such as bacteria, actinomycetes, algae and fungi (though some fungi are visible to the unaided eye). The fauna is composed of microfauna such as protozoans, mesofauna such as nematodes and arthropods such as mites, centipedes, springtails, and macrofauna such as insects and worms, mice, moles et cetera

The soil flora and fauna play important roles in agriculture and horticulture, performing such roles as decomposing organic materials, moving nutrients through soil layers, fixing atmospheric nitrogen, and improving soil structure.

Some parts of the soil flora and fauna are plant pathogens, while others are antagonists. Without the activities of soil organisms, organic materials would accumulate and litter the soil surface, and there would be no food for floricitas. The soil biota includes

- **Megafauna:** size range 20 mm upwards, e.g. moles, rabbits, and rodents.
- **Macrofauna:** size range 2-20 mm, e.g. woodlice, arañas, earthworms, beetles, centipedes, slugs, snails, ants.
- **Mesofauna:** size range 100 micrometre-2 mm, e.g. mites and springtails.
- **Microfauna and Microflora:** size range 1-100 micrometres, e.g. yeasts, bacteria, fungi, protozoa, roundworms, and rotifers.

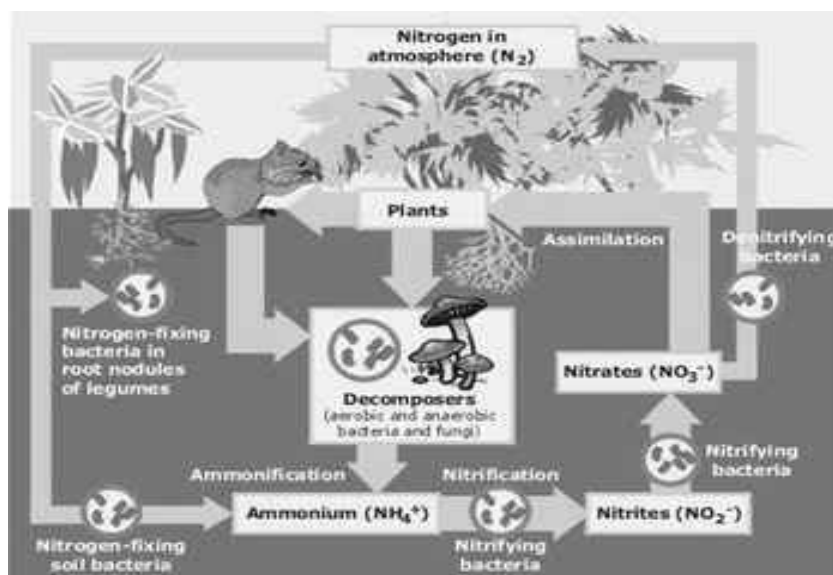
Of these, bacteria and fungi play key roles in maintaining a healthy soil. They act as decomposers that break down organic materials to produce detritus and other breakdown products. Soil detritivores, like earthworms, ingest detritus and decompose it. Saprotrophs, well represented by fungi and bacteria, extract soluble nutrients from delitro

Bacteria

- Bacteria are single-celled organisms, and are the most numerous members of the soil, with populations ranging from 100 million to 3 billion in a gram.
- They are capable of very rapid reproduction by binary fission (dividing into two) in favourable conditions.
- One bacterium is capable of producing 16 million more in just 24 hours. Most soil

bacteria live in close contact with to plant roots and are often referred to as rhizobacteria.

- Bacteria live in soil water, including the film of moisture surrounding soil particles, and some are able to swim by means of flagella.
- The majority of the beneficial soil-dwelling bacteria need oxygen (and are thus termed aerobic bacteria) and those that do not require air are referred to as anaerobic, and tend to cause putrefaction of dead organic matter.
- Aerobic bacteria are most active in a soil that is moist and neutral soil pH, and where there is plenty of food (carbohydrates and micronutrients from organic matter) is available.



The nitrogen cycle

The important roles that bacteria play are:

Nitrification

- ❖ Nitrification is important part of the nitrogen cycle wherein certain bacteria decomposition of proteins are able to transform nitrogen in the form of ammonium and then into nitrates, which are available to growing plants

Nitrogen fixation

- ❖ In another part of the cycle, the process of nitrogen fixation constantly puts additional nitrogen into biological circulation.
- ❖ This is carried out by free-living nitrogen-fixing bacteria in the soil or water such as *Azotobacter*, or by those which live in close symbiosis with leguminous plants, such as rhizobia.
- ❖ These bacteria form colonies in nodules on the roots of peas, beans, and related species.
- ❖ These are able to convert nitrogen from the atmosphere into nitrogen-containing organic substances.

Denitrification

- ❖ While nitrogen fixation converts nitrogen from the atmosphere into organic compounds, a series of processes called denitrification returns an approximately equal amount of nitrogen to the atmosphere.
- ❖ Denitrifying bacteria tend to be anaerobes, or facultatively anaerobes including *Achromobacter* and *Pseudomonas*.
- ❖ The putrefaction process caused by oxygen-free conditions converts nitrates and nitrites in soil into nitrogen gas or into gaseous compounds such as nitrous oxide or nitric oxide.

Actinobacteria

- ❖ Actinobacteria are critical in the decomposition of organic matter and in humus formation, and their presence is responsible for the sweet "earthy" aroma which is associated with a good healthy soil.
- ❖ They require plenty of air and a pH between 6.0 and 7.5, but are more tolerant of dry conditions than most other bacteria and fungi.

Fungi

- ❖ A gram of garden soil can contain around one million fungi, such as yeasts and moulds. Fungi have no chlorophyll, and are not able to photosynthesise; besides, they can't use atmospheric carbon dioxide as a source of carbon.
- ❖ Therefore they are chemo-heterotrophic, meaning that, like animals, they require a chemical source of energy rather than being able to use light as an energy source, as well as organic substrates to get carbon for growth and development.
- ❖ Many fungi are parasitic, often causing disease to their living host plant, although some have beneficial relationships with living plants.
- ❖ In terms of soil and humus creation, the most important fungi tend to be saprotrophic, that is, they live on dead or decaying organic matter, thus breaking it down and converting it to forms which are available to the higher plants.
- ❖ Some fungi are able to break down cellulose and lignins.
- ❖ Fungi spread underground by sending long thin threads known as mycelium throughout the soil; these threads can be observed throughout many soils and compost heaps.
- ❖ From the mycelia the fungi is able to throw up its fruiting bodies, the visible part above the soil (e.g., mushrooms, toadstools and puffballs) which may contain millions of spores.
- ❖ When the fruiting body bursts, these spores are dispersed through the air to settle in fresh environments, and are able to lie dormant for up to years until the right conditions for their activation arise or the right food is made available.

Mycorrhizae

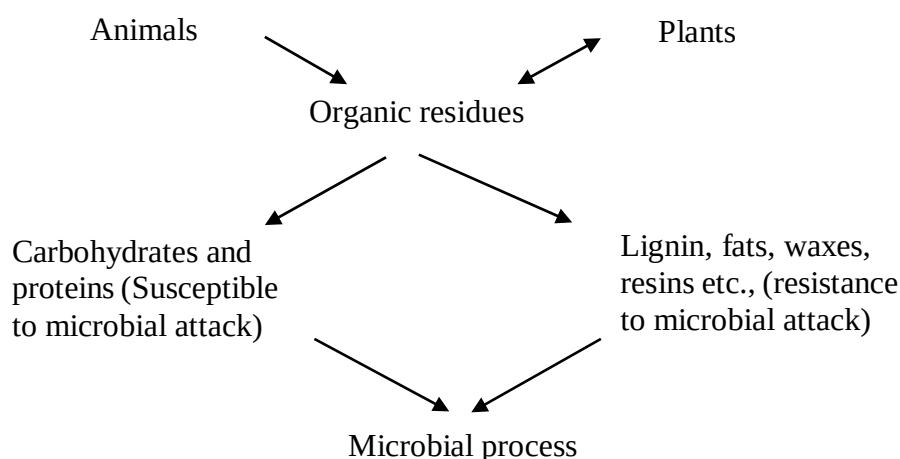
- ❖ Those fungi that are able to live symbiotically with living plants, creating a relationship that is beneficial to both, are known as Mycorrhizae (from *myco* meaning fungal and

rhiza meaning root).

- ❖ The mycorrhiza obtains the carbohydrates that it requires from the root, in return providing the plant with nutrients including nitrogen and moisture. Later the plant roots will also absorb the mycelium into its own tissues.
- ❖ Beneficial mycorrhizal associations are to be found in many of our edible and flowering crops.
- ❖ These include at least 80% of the *brassica* and *solanum* families (including tomatoes and potatoes), as well as the majority of tree species, especially in forest and woodlands

Role of soil microorganisms in the decomposition of organic matter

- ❖ Soil microorganism plays an important role in completing the carbon cycle. They convert organic matter into simple substances, liberate CO₂ and water, increase the concentration of nitrogen content and bring down the ratio between C and N in soil.
- ❖ This improves the soil fertility. The released CO₂ goes back to complete the cycle and this ensures that there is no major lock-up in organic tissues.



Mineralization into CO₂, NH₄, NO₃, NO₂

Immobilization of C, N, P, S in microbial cells

The C: N ratio is ~ 10:1. C: N ratio – is a tool to predict the rate of decomposition. If the C: N ratio is wider, decay will be slower.

Ecological significance of soil microorganisms

- Soil microorganisms are very important as almost every chemical transformation taking place in soil involves active contributions from soil microorganisms.
- In particular, they play an active role in soil fertility as a result of their involvement in the cycle of nutrients like carbon and nitrogen, which are required for plant growth.

- For example, soil microorganisms are responsible for the decomposition of the organic matter entering the soil (e.g. plant litter) and therefore in the recycling of nutrients in soil.
- Certain soil microorganisms such as mycorrhizal fungi can also increase the availability of mineral nutrients (e.g. phosphorus) to plants.
- Other soil microorganisms can increase the amount of nutrients present in the soil. For instance, nitrogen-fixing bacteria can transform nitrogen gas present in the soil atmosphere into soluble nitrogenous compounds that plant roots can utilise for growth.
- These microorganisms, which improve the fertility status of the soil and contribute to plant growth, have been termed 'bio-fertilizers' and are receiving increased attention for use as microbial inoculants in agriculture.
- Similarly, other soil microorganisms have been found to produce compounds (such as vitamins and plant hormones) that can improve plant health and contribute to higher crop yield.
- These microorganisms (called 'phytostimulators') are currently studied for possible use as microbial inoculants to improve crop yield.
- In contrast to these beneficial soil microorganisms, other soil microorganisms are pathogenic to plants and may cause considerable damage to crops.
- Large numbers of pathogenic microorganisms are routinely found in the soil and many of them can infect the plant through the roots.
- However, certain native microorganisms present in the soil are antagonistic to these pathogens and can prevent the infection of crop plants.
- Antagonism against plant pathogens usually involves competition for nutrients and/or production of inhibitory compounds such as secondary metabolites (antimicrobial metabolites and antibiotics) and extra cellular enzymes.
- Other soil microorganisms produce compounds that stimulate the natural defence mechanisms of the plant and improve its resistance to pathogens.
- Collectively, these soil microorganisms have been termed 'bio-pesticides' and represent an emerging and important alternative (i.e. biological control) to the use of chemical pesticides for the protection of crops against certain pathogens and pests.
- *Azospirillum* induces the proliferation of plant root hairs which can result in improved nutrient uptake

- **Macro-organisms** are only a small part of a soil's make up, but they are important for healthy plants and nutrient-rich soil.
- In gardening, macro-organisms are the animals, mostly invertebrate, that live in the soil, and which are generally visible to the naked eye.
- Many benefit the soil by helping to break down minerals, soil particles and nutrients. They also add other elements to the soil: egg casings, digested soil, digested plant and animal (some prey on others) materials.
- Since their life cycles are short, when they die, their bodies also add nutrients to the soil. These additional elements break down quickly in the soil, adding to the overall richness of the soil.
- Beneficial macro-organisms include earthworms, many species of nematodes, and other beneficial insects.
- Destructive macro-organisms include larvae of some insect species

Soil Enzymes

i) Sources : Sources of soil enzymes include living and dead microbes, plant roots and residues, and soil animals. Enzymes stabilized in the soil matrix accumulate or form complexes with organic matter (humus), clay, and humus-clay complexes, but are no longer associated with viable cells.

It is thought that 40 to 60% of enzyme activity can come from stabilized enzymes, so activity does not necessarily correlate highly with microbial biomass or respiration. Therefore, enzyme activity is the cumulative effect of long term microbial activity and activity of the viable population at sampling. However, an example of an enzyme that only reflects activity of viable cells is dehydrogenase, which in theory can only occur in viable cells and not in stabilized soil complexes.

ii) Significance : Soil enzymes increase the reaction rate at which plant residues decompose and release plant available nutrients. The substance acted upon by a soil enzyme is called the substrate. For example, glucosidase (soil enzyme) cleaves glucose from glucoside (substrate), a compound common in plants. Enzymes are specific to a substrate and have active sites that bind with the substrate to form a temporary complex. The enzymatic reaction releases a product, which can be a nutrient contained in the substrate.

Enzymes respond to soil management changes long before other soil quality indicator changes are detectable. Soil enzymes play an important role in organic matter decomposition and nutrient cycling (see table 1). Some enzymes only facilitate the breakdown of organic matter

(e.g., hydrolase, glucosidase), while others are involved in nutrient mineralization (e.g., amidase, urease, phosphatase, sulfates).

Role of soil enzymes				
Enzyme	Organic matter substances acted on	End product	Significance	Predictor of soil function
Beta glucosidase	carbon compounds	glucose (sugar)	energy for microorganisms	organic matter decomposition
FDA hydrolysis	organic matter	carbon and various nutrients	energy and nutrients for microorganisms, measure microbial biomass	organic matter decomposition nutrient cycling
Amidase	carbon and nitrogen compounds	ammonium (NH ₄)	plant available NH ₄	nutrient cycling
Urease	nitrogen (urea)	ammonia (NH ₃) and carbon dioxide (CO ₂)	plant available NH ₄	nutrient cycling
Phosphatase	phosphorus	phosphate (PO ₄)	plant available P	nutrient cycling
Sulfatase	sulfur	sulfate (SO ₄)	plant available S	nutrient cycling

Specific problems that might be caused by poor function of Soil Enzymes :

- ❖ Absence or suppression of soil enzymes prevents or reduces processes that can affect plant nutrition.
- ❖ Poor enzyme activity (e.g., pesticide degrading enzymes) can result in an accumulation of chemicals that are harmful to the environment; some of these chemicals may further inhibit soil enzyme activity.

Management :

- ❖ Organic amendment applications, crop rotation, and cover crops have been shown to enhance enzyme activity (figures 1 and 2).
- ❖ The positive effect of pasture (figure 2) is associated with the input of animal manure and less soil disturbance.
- ❖ Agricultural methods that modify soil pH (e.g., liming) can also change enzyme activity.

Measuring soil enzyme activity :

Enzymes are measured indirectly by determining their activity in the laboratory using biochemical assays. Enzyme assays reflect potential activity and do not represent true in situ activity levels and must be viewed as an index.

Factors affecting enzyme activity

i) Inherent :

- ❖ Soil enzymes have varying optimum pH and temperature values at which they function most effectively. For example, the activity of phosphatase, aryl sulfatase, and amidase involved in phosphorus, sulfur, and nitrogen cycling, respectively, is strongly correlated to variations in soil pH. Since enzyme structure and substrate binding can be altered by heat and extreme cold temperature, enzyme activity decreases above and below the optimum temperature.
- ❖ The activity of many enzymes often correlates with soil moisture content, as well. Drought may suppress enzyme activity.
- ❖ Soil texture influences enzyme activity, and normally enzyme activities are significantly and positively correlated with clay content. Clayey soils have greater ability to store organic matter that promotes microbial communities, and clay forms clay-enzyme complexes as against the low activity in sandy soils because low organic matter content and poor water holding capacity.

ii) Dynamic :

- ❖ Addition of organic amendments and adoption of management practices that increase soil organic matter lead to increased enzyme activity. Plant roots stimulate enzyme activity because of their positive effect on microbial activity and production of exudates rich in substrates acted on by enzymes.
- ❖ Elevated soil concentrations of chemical compounds that are end products of enzymatic reactions can inhibit enzyme activity by feedback inhibition. For example, phosphatase activity increases in phosphorus deficient soil, but its activity decreases in soil with high phosphorus concentration. Similarly, urease activity may be suppressed by ammonia-based nitrogen fertilizer because ammonium is the product of urease activity.
- ❖ Compaction may limit the activity of enzymes involved in nutrient mineralization because of decreased oxygen in the soil for those reactions or organisms requiring an aerobic environment. Conversely, anaerobic conditions from compaction or water saturation increase enzymatic reaction rates related to denitrification.
- ❖ Application of materials containing heavy metals can reduce enzyme activity (e.g., amidase) due to their toxic effect on soil organisms and roots or direct inhibition of enzyme reactions.

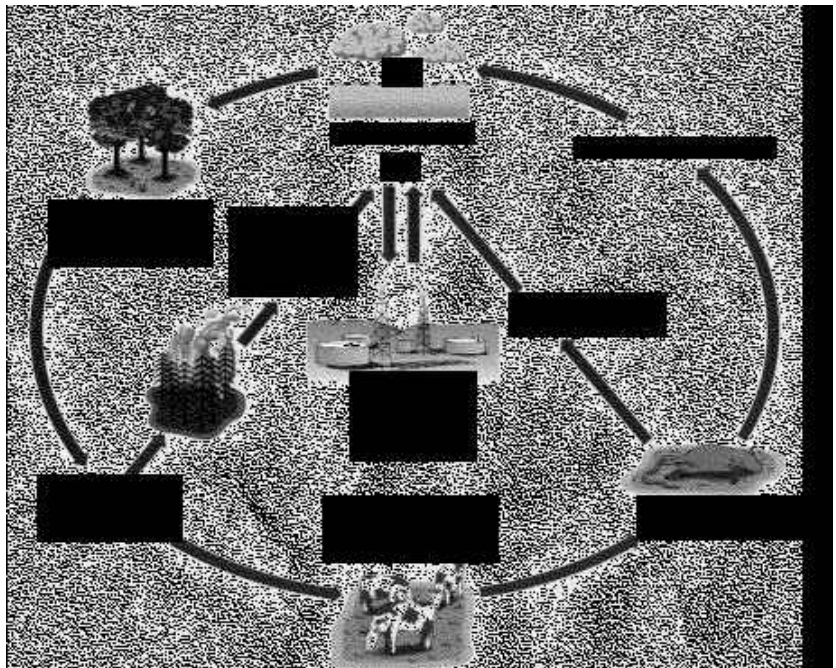
31. THE CARBON CYCLE

♣ Terrastrial carbon sesquestration

♣ Carbon trading

✚ The concentration of carbon in living matter (18%) is almost 100 times greater than its concentration in the earth (0.19%).

✚ So living things extract carbon from their nonliving environment. For life to continue, this carbon must be recycled. That is our topic.



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✚ So living things extract carbon from their nonliving environment. For life to continue, this carbon must be recycled. That is our topic.

Carbon exists in the nonliving environment as:

- Carbon dioxide (CO_2) in the atmosphere and dissolved in water (forming HCO_3^-)
- Carbonate rocks (limestone and coral = CaCO_3)
- Deposits of coal, petroleum, and natural gas derived from once-living things
- Dead organic matter, e.g., humus in the soil

Carbon enters the biotic world through the action of autotrophs:

- Primarily **photoautotrophs**, like plants and algae, that use the energy of light to convert carbon dioxide to organic matter.
- And to a small extent, **chemoautotrophs** — bacteria and archaea that do the same but use the energy derived from an oxidation of molecules in their substrate.

Carbon returns to the atmosphere and water by

- Respiration (as CO_2)
- Burning
- Decay (producing CO_2 if oxygen is present, methane (CH_4) if it is not).

The uptake and return of CO_2 are not in balance.

The carbon dioxide content of the atmosphere is gradually and steadily increasing. The increase in CO_2 probably began with the start of the industrial revolution. This increase is surely "anthropogenic"; that is, caused by human activities:

- Burning fossil fuels (coal, oil, natural gas) returns to the atmosphere carbon that has been locked within the earth for millions of years.
- Clearing and burning of forests, especially in the tropics. In recent decades, large areas of the Amazon rain forest have been cleared for agriculture and cattle grazing.

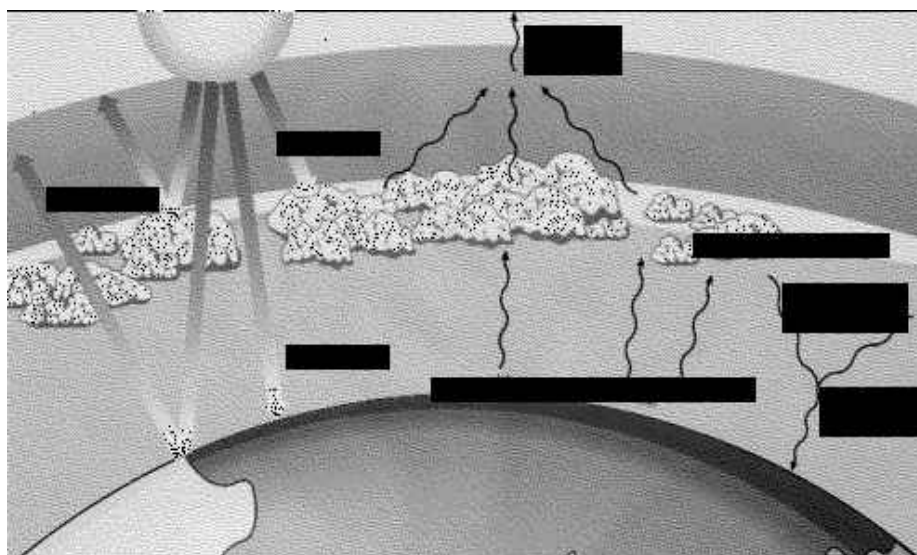
Where is the missing carbon?

Curiously, the increase in atmospheric CO_2 is only about one-half of what would have been expected from the amount of fossil fuel consumption and forest burning.

Where has the rest gone?

Research has shown that increased CO_2 levels lead to increased net production by photoautotroph. There is some evidence that the missing CO_2 has been incorporated by

- Increased growth of forests, especially in North America
- Increased amounts of phytoplankton in the oceans



The Greenhouse Effect and Global Warming

Despite these "sinks" for our greatly increased CO₂ production, the concentration of atmospheric CO₂ continues to rise.

Carbon dioxide is transparent to light but rather opaque to heat rays. Therefore, CO₂ in the atmosphere retards the radiation of heat from the earth back into space — the "greenhouse effect".

Average temperatures do seem to have increased slightly (~0.6°C) in the last century.

Some evidence:

- Careful monitoring of both ocean and land temperatures.
- Many glaciers and ice sheets are receding.
- Woody shrubs are now growing in areas of northern Alaska that 50 years ago were barren tundra.
- Many angiosperms in temperate climates are flowering earlier in the spring than they used to.
- Many species of birds and butterflies are moving north and breeding earlier in the spring

Carbon storage in the biosphere is influenced by a number of processes on different time-scales: while net primary productivity follows a diurnal and seasonal cycle, carbon can be stored up to several hundreds of years in trees and up to thousands of years in soils. Changes in those long term carbon pools (e.g. through de- or afforestation or through temperature-related changes in soil respiration) will thus directly affect global warming.

Terrestrial Carbon Sequestration

- ◆ Terrestrial carbon sequestration is the process through which carbon dioxide (CO₂) from the atmosphere is absorbed by trees, plants and crops through photosynthesis, and stored as carbon in biomass (tree trunks, branches, foliage and roots) and soils.
- ◆ The term "sinks" is also used to refer to forests, croplands, and grazing lands, and their ability to sequester carbon. Agriculture and forestry activities can also release CO₂ to the atmosphere.
- ◆ Therefore, a carbon sink occurs when carbon sequestration is greater than carbon releases over some time period.

Agricultural and forestry sequestration activities

- Forests and soils have a large influence on atmospheric levels of carbon dioxide (CO₂)—the most important global warming gas emitted by human activities.
- Tropical deforestation is responsible for about 20% of the world's annual CO₂ emissions on a global scale, however, these emissions are more than offset by the uptake of atmospheric CO₂ by forests and agriculture.
- Therefore, agricultural and forestry activities can both contribute to the accumulation of greenhouse gases in our atmosphere, as well as be used to help prevent climate change, by avoiding further emissions and by sequestering additional carbon.
- Sequestrations activities can be carried out immediately, appear to present relatively cost-effective emission reduction opportunities, and may generate environmental co-benefits.
- At the same time, it is important to recognize that carbon sequestered in trees and soils can be released back to the atmosphere, and that there is a finite amount of carbon that can ultimately be sequestered.

There are three general means by which agricultural and forestry practices can reduce greenhouse gases:

1. Avoiding emissions by maintaining existing carbon storage in trees and soils.
2. Increasing carbon storage by, e.g., tree planting, conversion from conventional to conservation tillage practices on agricultural lands.
3. Substituting bio-based fuels and products for fossil fuels, such as coal and oil, and energy-intensive products that generate greater quantities of CO₂ when used.

Carbon sequestered by agricultural and forestry practices

- Carbon sequestration rates vary by tree species, soil type, regional climate, topography and management practice.
- Soil carbon sequestration rates vary by soil type and cropping practice and are less well documented but information and research in this area is growing rapidly.
- Carbon accumulation in forests and soils eventually reaches a saturation point, beyond which additional sequestration is no longer possible.

- This happens, for example, when trees reach maturity, or when the organic matter in soils builds back up to original levels before losses occurred.
- Even after saturation, the trees or agricultural practices would need to be sustained to maintain the accumulated carbon and prevent subsequent losses of carbon back to the atmosphere

Carbon Trading

- According to the Kyoto Protocol of 1997 all countries are required to reduce their greenhouse gas emissions by 5% --from 1990 levels-- in the next ten years, ie 2012—or pay a price to those that do.
- “Let’s rewind to the Kyoto Protocol of 1997 by which all the idea is, that if a country is a consumer of an environmental value like clean air, it must pay a producer of an equivalent
- Since developing countries can start with clean technologies, they will be rewarded by those stuck with ‘dirty’ ones. Say a company in India can prove it has prevented the emission of x-tonnes of carbon, it can sell this good carbon-karma to a company in say, the US which has a bad karma.
- An environment-fundamentalist may say it’s all a bit like an indulgent epicure paying someone else to diet for him, but then that’s another story. Right now, there is a market opportunity for India—but only till 2012.
- Closer to that clean-up date prices of carbon credits will rise and in the years leading up to it there will be a scramble to buy credits cheap.
- The World Bank has built itself a role in this market as a referee, broker and macro-manager of international fund flows.
- The scheme has been entitled Clean Development Mechanism [CDM] in 2000. Or more commonly CARBON TRADING

Emissions trading (or **cap and trade**) is an administrative approach used to control pollution by providing economic incentives for achieving reductions in the emissions of pollutants

- In such a plan, a central authority (usually a government agency) sets a limit or *cap* on the amount of a pollutant that can be emitted. Companies or other groups that emit the pollutant are given *credits* or *allowances* which represent the right to emit a specific amount.

- The total amount of credits cannot exceed the cap, limiting total emissions to that level. Companies that pollute beyond their allowances must buy credits from those who pollute less than their allowances or face heavy penalties. This transfer is referred to as a trade.
- In effect, the buyer is being fined for polluting, while the seller is being rewarded for having reduced emissions.
- Thus companies that can easily reduce emissions will do so and those for which it is harder will buy a credit which reduces greenhouse gases at the lowest possible cost to society.
- There are currently several trading systems in place with the largest being the European Union's.
- The carbon market makes up the bulk of these and is growing in popularity. Many businesses have welcomed emissions trading as the best way to mitigate climate change.
- Enforcement of the caps is a problem, but unlike traditional regulation, emissions trading markets can be easier to enforce because the government overseeing the market does not need to regulate specific practices of each pollution source. However, monitoring (or estimating) and verifying of actual emissions is still required, which can be costly.
- Critics doubt whether these trading schemes can work as there may be too many credits given by the government, such as in the first phase of the European Union's scheme. Once a large surplus was discovered the price for credits bottomed out and effectively collapsed, with no noticeable reduction of emissions.

Additional Reading

Indian villages in global carbon trading

- About two years ago when Good News India reported that an Indian company was ready to export carbon credits, it had seemed a very savvy, avant-garde move.
- Now Indian villages have hurried along, ahead of many Indian corporates. A most appropriately named village, Powerguda in Adilabad district Andhra Pradesh has pioneered a sale to the World Bank for \$645.
- The village was selling 147 tonnes equivalent of saved carbon dioxide credits. What are carbon credits and what is carbon trading? Quoting from the above cited article:
- “Let’s rewind to the Kyoto Protocol of 1997 by which all countries are required to reduce their greenhouse gas emissions by 5% --from 1990 levels-- in the next ten years, ie 2012—or pay a price to those that do.”
- The idea is, that if a country is a consumer of an environmental value like clean air, it must pay a producer of an equivalent value.

- Powerguda's claim of having saved 147 MT of CO₂ is based on the bio-diesel they extracted from 4500 Pongamia trees in their village. Using this —instead of petroleum— in oil engines would enhance air quality.
- World Bank in a behaviour model worth emulating by other corporate was buying those carbon credits to balance the aviation fuel burnt by aircraft carrying Bank officials.
- Mr Emmanuel D'Silva, a former World Bank staffer has been working extensively among Andhra Pradesh villagers, creating awareness about this market opportunity that awaits them.
- In a communication to Good News India in June, 2004 he reports that five other villages have followed Power guda and made carbon credit sales.
- Mr Nalin Kishor of the World Bank who heads the programme has kindly permitted the article in the following page, to be reproduced for Good News India's readers. It originally appeared at the profor site.

32. SOIL POLLUTION

♣ Types and behavior of Pesticides

♣ Inorganic contaminants

Soil pollution :

An undesirable change in physical, chemical and biological characteristics of soil and that may be harmful to living organisms is called soil pollution

Pollutant :

The substances, which cause pollution, are called pollutants. Pollutant is defined as any substance that is released intentionally or inadvertently by man into the environment in such a concentration that may have adverse effect on environmental health. Environment Protection Act, 1986 defines pollutant, as any solid, liquid or gaseous substance present in such a concentration as may be, or tend to be, injurious to environment.

Sources of soil pollution

- Natural sources
- Artificial sources

I. Natural Sources

Natural pollutants are of plant and animal origin. They are:

(i) Plant residues : Normally plants on death and decay contribute organic matter to the soil and thereby increase the soil fertility. Sometimes residues from crops, fields and orchards carry plant pathogens and pests. These on death and decay cause soil pollution. Naturally occurring or manmade forest fires and crops add unwanted organic matter to the soil and destroy soil micro flora which create unfavorable conditions leading to pollution. Burning of crop yields residues with CO (83 per cent), NO (1.5 per cent), hydrocarbons (5.3 per cent) and particulate matter (8.5 per cent)

(ii) Animal wastes: Animal wastes, such as faecal matter, urine, blood, slaughter house waste in the form of liquid or particulate matter, bodies of dead animals etc., are all indiscriminately dumped in to the soil. These create unhealthy conditions detrimental to the growth of plants and soil organisms. Excessive organic contents are harmful for healthy growth of roots as they create hypertonic conditions in the soil causing wilting or stunted growth of the plants.

(iii) Natural calamities:

a. Floods, cyclones, earthquakes and volcanic eruptions : Sediments, garbage and dumps of waste from one place to another place leads the pollution and cause non point sources of pollution

II. Artificial Sources

(i) Industrial wastes : Rapid industrialization has given birth to the problem of waste disposal. Most Indian industries release their effluents directly into surrounding fields. These effluents are often enriched with inorganic chemicals that enhance the salt content of the soil. Addition of excessive salts into the soil changes its pH, which in turn adversely affects the soil organisms. Continuous inflow of such effluents into the soil gradually decreases the fertility of the soil rendering it unfit to support plant life. At places, slow and steady percolation or seepage of toxic heavy metals deep into the soil pollute the groundwater.

(ii) Solid waste : Solid wastes include all substances in a solid or semisolid state disposed off in the soil. These are often garbages from homes and commercial places. Major sources of solid wastes in our country come from domestic wastes, shops, offices, restaurants, hospitals, and small scale industries.

(iii) Radioactive wastes : Pollution resulting from radioactive wastes is perhaps the most dangerous and farfetched process. Sources of such pollutants are nuclear explosions, nuclear testing's, nuclear reactors etc., Radio-active dust particles produced after an atomic explosion slowly settles down on the soil surface and gradually penetrate into the soil strata. When absorbed by the plants, these enter the human body via the food chain and cause serious damages. These also disrupt the growth and multiplication of soil microorganisms.

(iv) Detergents : Wastewater from domestic sources is usually rich in detergents. It is allowed to flow unchecked into the ground. Presence of detergents in the soil increases its alkalinity and phosphate contents which affect root growth of the plants and depress the growth of soil microorganisms.

(v) Construction activities: Due to urbanization, large amount of construction activities are taking place which has resulted in large waste articles like wood, metal, bricks, plastic that can be seen by naked eyes outside any building or office which is under construction.

(vi) Mining activities: During extraction and mining activities, several land spaces are created beneath the surface. We constant hear about land caving in; this is nothing but nature's way of filling the spaces left out after mining or extraction activity.

(vii) Overcrowded landfills: Each household produces tonnes of garbage each year. Garbage like aluminum, plastic, paper, cloth, wood is collected and sent to the local recycling unit. Items that can not be recycled become a part of the landfills that hampers the beauty of the city and cause land pollution.

(viii) Deforestation and soil erosion: Deforestation carried out to create dry lands is one of the major concerns. Land that is once converted into a dry or barren land, can never be made fertile again, whatever the magnitude of measures to redeem it are. Land conversion, meaning the alteration or modification of the original properties of the land to make it use-worthy for a specific purpose is another major cause. This hampers the land immensely. Also there is a constant waste of land. Unused available land over the years turns barren; this land then cannot be used. So in search of more land, potent land is hunted and its indigenous state is compromised with.

(ix) Agrochemicals : Agrochemicals viz ., fertilizers, pesticides, herbicides used in agriculture to boost the crop productivity contributes to soil pollution considerably.

➤ **Fertilizers:** Increase in crop yield requires the use of inorganic and organic fertilizers. Common fertilizers are urea, super phosphate, bone meal etc., Accumulation of excessive fertilizers particularly inorganic types misbalances the nutrient contents of the soil. Excess phosphorus and nitrogen affects adversely the plants growth and soil flora and fauna.

➤ **Herbicides :** Herbicides like 2,4-D, 2,4,5- T, sodium arsenate, sodium chlorate etc., kill the herbs. These are poisonous in nature and their indiscriminate use completely wrecks the ecosystem of the soil.

➤ **Pesticides :** Pesticides are chemicals used to destroy pests which damage or kill crop plants. Pesticides like DDT, malathion, dieldrin etc., contains toxic heavy metals which increase the toxicity of the soil.

Effect of pollutants on soil, water and environment

- ✓ Heavy metals addition to the soil can affect soil fertility and agricultural productivity
- ✓ Industrial effluents when discharged through sewage system will poison the biological purification mechanism of sewage treatment causing several soil and water borne diseases.
- ✓ Metallic contaminants destroy bacteria and beneficial microorganisms in the soil. Heavy metals tends to precipitate phosphatic compounds and catalyze their decomposition and their accumulation in the soil for a long period will destroy the living organisms
- ✓ Severe agricultural crop damage is caused by acidity and alkalinity of the soil coming

from chemical industries. About 30% of the irrigated land of the world is now affected by salinity of soil and water logging.

- ✓ About 30 to 80 % of world's irrigated land has been degraded due to increased salinity.
- ✓ When discarded plastic material, packaging and toys of polyvinyl chloride are burnt in soil, they emit highly toxic gases like hydrochloric acid fumes, SO₂ and NO_x etc., and thereby causes acid rain.
- ✓ Chemicals either emitted into air or applied by aerosol spray ultimately reach the soil where they have significant effect on plants and animals causing a disruption in plant species.
- ✓ Contaminated soils pave the way for the entry of pathogenic bacteria which are the carriers of chronic diseases such as typhoid, cholera etc., Flies which breed or get in contact with the contaminated soil become carriers of diseases organisms

Impact on Agriculture

- Reduced soil fertility
- Reduced nitrogen fixation
- Increased erodibility
- Larger loss of soil and nutrients
- Deposition of silt in tanks and reservoirs
- Reduced crop yield
- Imbalance in soil fauna and flora

Pesticides as Pollutants

Pesticides are the potential toxic substances that are used to destroy, prevent, attract, repel or to control any pest or weed species that may cause detrimental effect to crops, animals and indirectly on human health.

Characteristics based on target pests

- | | |
|---|--|
| • Broad spectrum activity | • No adverse effect on quality of crop produce |
| • No mammalian toxicity | • Non inflammable and non corrosive |
| • Non toxic to crops and beneficial insects | • Low production cost |
| • Low persistence | • Easily available formulation |
| • Good compatibility with various agrochemicals | • Environment friendly |

Types of pesticides:

Type of pesticide	Target pest group		Type of pesticide	Target pest group
Herbicides	Plant		Miticides or acaricides	Mites
Algicides or algaecides	Algae		Molluscicides	Snails
Avicides	Birds		Nematicides	Nematodes
Bactericides	Bacteria		Rodenticides	Rodents
Fungicides	Fungi and oomycetes		Virucides	Viruses
Insecticides	Insects			

Behaviour of pesticides

Although pesticides are indispensable in modern agriculture, their use and misuse can lead to serious water quality problems. Fish kills, reproductive failure of birds, and acute illnesses in people have all been attributed to the ingestion of pesticides or exposure to pesticides. Usually this is the result of misapplication, careless storage, or careless disposal of pesticides and containers. Groundwater contamination, however, is a potential problem following normal pesticide use. Once a pesticide is applied, several things may happen. It may be taken up by plants and/or ingested by animals, insects, worms, or microorganisms in the soil; it may move downward in the soil and adhere to soil particles, or it may dissolve; it may volatilize; it may be broken down into less toxic compounds; it may be leached or moved out of the plants' root zone by rain or irrigation water; or it may be carried away by runoff water or erosion.

Factors Affecting Fate of Pesticides

There are four major factors which affect the fate of pesticides. They are as follows:

1. Properties of the pesticide.
2. Properties of the soil.
3. Conditions of the site.
4. Management practices.

(i) Pesticide Properties

Pesticide properties which affect movement to groundwater include solubility, adsorption, volatility, and degradation.

(a) Solubility : Chemicals which dissolve readily in water are said to be highly soluble. As water seeps downward through the soil, it carries with it water-soluble chemicals. This process is called leaching. Highly soluble pesticides, therefore, have a tendency to be leached from the soil to groundwater.

(b) Adsorption : Many pesticides do not leach because they are adsorbed or tightly held by soil particles. Adsorption depends not only on the chemical, but also on the soil type and amount of soil organic matter present.

(c) Volatility : Highly volatile chemicals are easily lost to the atmosphere, similar to the evaporation of water. If a pesticide is highly volatile and not very water soluble, it is likely to be lost to the atmosphere, and less will be available for leaching to groundwater. Highly volatile compounds may become groundwater contaminants, however, if they are highly soluble in water.

(d) Degradation : Another chemical property affecting leaching potential is the pesticide's rate of degradation in the soil. Pesticides are degraded, or broken down into other chemical forms, by sunlight, microorganisms in the soil, and a variety of chemical and physical properties. The longer the compound lasts before it is broken down, the longer it is subject to the forces of leaching. Many chlorinated hydrocarbons are highly persistent in soil, but they have not been found in groundwater because of their low solubility and strong adsorption to soil particles.

(ii) Soil Properties

Soil properties affecting the movement of pesticides include soil texture, soil permeability, and organic matter content.

(a) Soil Texture : Soil texture is determined by the relative proportions of sand, silt, and clay. Texture affects movement of water through soil and, therefore, affects the movement of dissolved chemicals, such as pesticides. The coarser the soil, the faster the movement of the percolating water, and the less opportunity for adsorption of dissolved chemicals. Soils with more clay and organic matter tend to hold water and dissolved chemicals longer. These soils also have far more surface area on which pesticides can be adsorbed. The coarser the texture of the soil, the greater the chance of the pesticide reaching groundwater.

(b) Soil Permeability : Soil permeability is a measure of how fast water can move downward through a particular soil. Water moves quickly through soils with high permeability. They also lose dissolved chemicals with the percolating water. In highly permeable soils, therefore, the timing and methods of pesticide application need to be carefully designed to minimize leaching losses.

(c) Organic Matter Content : Soil organic matter influences how much water a soil can hold and how well it will be able to adsorb pesticides. Increasing the soil's organic content, through practices such as application of manure or plowing under of cover crops, increases the soil's ability to hold both water and dissolved pesticides in the root zone where they will be available to plants and to eventual degradation.

(iii) Site Conditions

The conditions of the site where a pesticide is applied can also affect the movement of the pesticide. Such conditions include the depth to groundwater, geologic conditions, and climate.

(a) Depth To Groundwater : The shallower the depth to groundwater, the less soil there will be to act as a filter. Also, there will be fewer opportunities for degradation or adsorption of pesticides. Therefore, extra precautions need to be taken to protect groundwater in areas where it is close to the ground surface. In humid regions, groundwater may be only a few feet below the surface of the soil. If rainfall is high and soils are permeable, water carrying dissolved pesticides may take only a few days to percolate downward to groundwater. In arid regions, groundwater may be several hundred feet below the soil surface, and leaching of pesticides to groundwater may be a much slower process.

(b) Geologic Conditions : In addition to depth to groundwater, it is important to look at the permeability of the geologic layers between the soil and groundwater. Highly permeable materials, such as gravel deposits, allow water and dissolved pesticides to freely percolate downward to groundwater. Layers of clay, on the other hand, are much less permeable and, thus, inhibit the movement of water. Groundwater quality is most vulnerable in areas where permeability of geologic layers is rapid.

(c) Climate : Areas with high rates of rainfall or irrigation may have large amounts of water percolating through the soil and, therefore, are highly susceptible to leaching of pesticides, especially if the soils are highly permeable.

(iv) Management Practices

Management practices include the method used to apply the pesticide and the rates and timing of application.

(a) Application Methods : Another factor determining leaching potential is the way in which a pesticide is applied. Injection or incorporation into the soil, as in the case of nematicides, makes the pesticide most readily available for leaching. Most of the pesticides which have been detected in groundwater are ones which are incorporated into the soil rather than being sprayed onto growing crops.

(b) Pesticide Rates And Timing: The rate and timing of a pesticide's application also are critical in determining whether it will leach to groundwater. The larger the amount used and the closer the time of application to a time of heavy rainfall or irrigation, the more likely that some pesticide will leach to groundwater. Particular care should be taken when practicing chemigation because of the risks of back-siphoning and leaching.

33. Prevention and Mitigation of soil pollution

Prevention and Mitigation of soil pollution

- Control of pesticides by using degradable insecticides like organic phosphates , bio-control agents like parasites, predators
- Control of fertilizer pollution : Natural procedures to boost soil fertility e.g., fallowing of fields, growing leguminous crops , crop rotation , use of organic fertilizer or FYM should be encouraged
- Diluting the soil with water
- Washing with water
- Vapor extraction – expensive process- deals with pumping with vaporous fumes out of the ground, and into the atmosphere
- Phytoremediation – certain plants concentrate specific elements (for example, locoweed concentrates selenium from the soil). Over a time soil becomes depleted of the element. Each year the plants can be harvested and disposed off.
- Polluted soil is often dug up and shipped to landfills or hazardous waste sites instead of being cleaned up.

Good Management Practices to avoid Pesticide Pollution

Good management practices can help prevent unwanted pesticide movement. There are several good management practices to consider when working with pesticides. They are as follows:

1. Follow pesticide label directions.
2. Mix pesticide and calibrate equipment accurately.
3. Avoid spills and back-siphoning.
4. Dispose of pesticide wastes and containers properly.
5. Eliminate unnecessary pesticide applications and use other pest control methods if possible.
6. Consider such factors as the weather, soil type, location, timing, and application methods before applying pesticides.
7. Irrigate properly by controlling the quantity and timing of irrigation.
8. Use proper pesticide storage.
9. Always maintain records of pesticide use.

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